Computer Networks

Complete Exam Study Guide

A concise, diagram-led revision of the **entire course** — all 31 lecture decks. It follows the layer-by-layer logic of the syllabus: introduction and network models, topologies, the physical layer and signals, transmission impairments, digital and analog data transmission, modulation and spread spectrum, the data link layer, error detection and correction, data link protocols, SDLC/HDLC, the network layer and IP, routing algorithms and protocols, IP addressing and subnetting, IP multicast, LAN/MAC protocols, IEEE 802, multiplexing and switching, transmission media, network components, the application layer and the transport layer (TCP).

How to use this guide. It is organised into 24 sections plus a formula sheet. Every section opens with a navy banner naming the source decks. Exam-critical definitions are in **bold**, and every formula is isolated in an **amber box** so it is easy to memorise the night before. Every diagram from every deck is reproduced and captioned; tables summarise the comparisons that examiners love.

Scope. The decks that were already covered by the provided *Networks_Study_Guide.pdf* are included here in the same spirit and at the same or greater depth, and this guide extends that treatment to the full set of decks so that nothing in the course is left out.

Decks covered (31): 1Introduction, 2networkhardwaresoftware, 5OSI referencemodel, 6TCPIP, 4Topology, 1Physical Layer, 2Digital Signals, 3transmission_impairments, 4Digital Data Transmission, 5Digital Data Transmission1, 6Analog Data Transmission, 8spread spectrum, 1Data Link Layer, 2Error Control, 3Error Correction, 4Data Link Layer Protocols, 5SDLC & HDLC, 1Network Layer, 2INTERNET PROTOCOL (IP), 3Routing Algorithms, 4IP Addressing, 6IP Multicast, 5Routing Protocols, 6LANMAC, 7IEEE802, 7Multiplexing, 7Transmission Media, 3NETWORK COMPONENTS, 5Application Layer, 1Transmission Control Protocol (TCP), 2TCP Data Flow.

Generated study aid — read cover-to-cover in a day. Diagrams and formulas are complete and unabridged.

Contents

24 sections, grouped by layer, plus a formula & quick-reference sheet.

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10	Error Detection: Parity, Checksum, CRC	2Error Control
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19	LAN Protocols: MAC, ALOHA, CSMA, IEEE 802	6LANMAC, 7IEEE802
20	Multiplexing & Switching	7Multiplexing
21	Transmission Media	7Transmission Media
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1. Introduction to Computer Networks

Source decks: 1Introduction, 2networkhardwaresoftware

What a network is

- A **network** is a group of two or more devices interconnected so they can exchange information. Tanenbaum: "a collection of autonomous computers interconnected by a single technology."
- Goal: **resource sharing** — make programs, equipment and especially data available to anyone on the network regardless of physical location.
- Size varies from small to large; hosts (also called **nodes**) include servers, desktops, laptops, phones.
- A **node** is any device that can receive, create, store or send data: modem, hub, bridge, switch, router, printer, server, phone.
- We need (1) ways to represent all types of information as bits, and (2) ways to move lots of bits everywhere, cheaply, with quality of service.

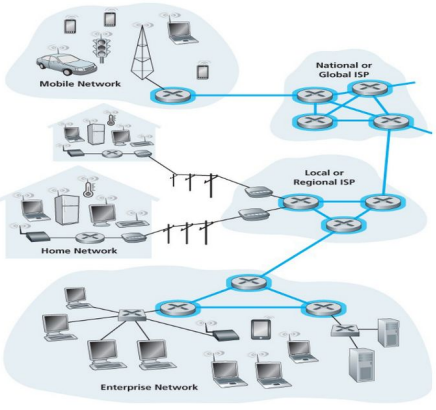
Uses of computer networks

Domain	Typical uses
Business	Resource sharing and information distribution; client-server model; communication medium (email, VoIP); desktop sharing; e-commerce
Home	Peer-to-peer file and media sharing; home networking
Mobile users	Wireless access anywhere; roaming
Social	Social networks, messaging, online communities

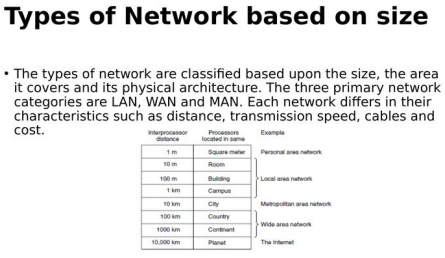
Types of networks (by size / coverage)

Type	Coverage	Key characteristics
LAN	Room, building, campus	Wired/wireless; coaxial or CAT-5; very low noise; 10–100 Mbps; e.g. a college computer lab
WLAN	Short range	High-frequency radio; base-station or ad-hoc (peer) mode; short range, high speed
MAN	5–50 km	Connects LANs; often a backbone; owned by an organization/individual; lower rate than LAN
WAN	Country / worldwide	Contains many LANs and MANs; satellites and microwave relays; best example is the Internet
PAN	Personal, ~10 m	Organized by an individual; short-range technologies (Bluetooth)
SAN	Data centre	Connects servers to storage over fibre-optic cables (backup, mirroring)
VPN	Over any network	Virtual Private Network — private tunnel over a public network

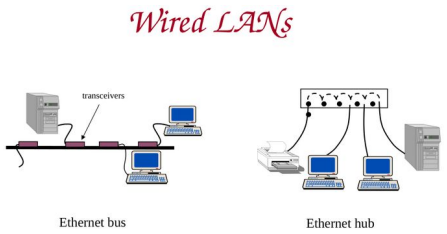
1Introduction — slide 6



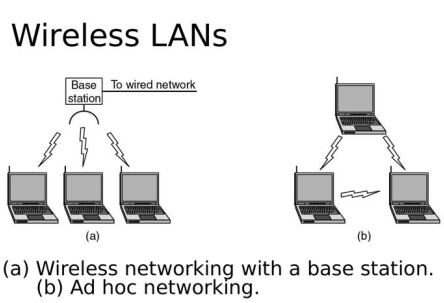
1Introduction — slide 7



Types of Network based on size

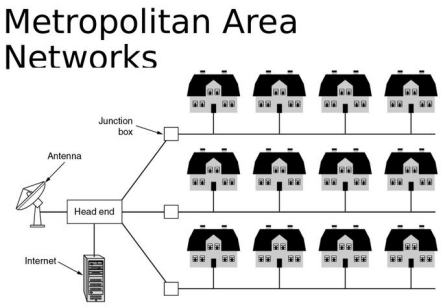


Ethernet bus



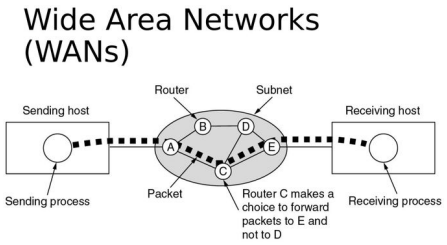
(a) Wireless networking with a base station.
(b) Ad hoc networking.

Wireless LANs



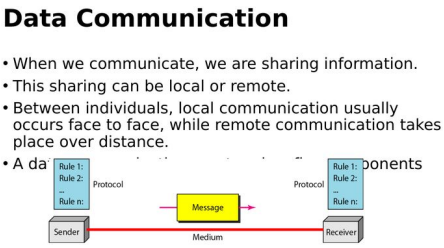
A metropolitan area network based on cable TV.

Metropolitan Area Networks

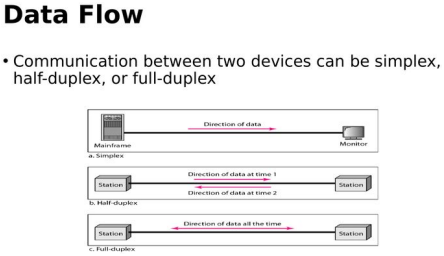


A stream of packets from sender to receiver.

Wide Area Networks (WANs)



Data Communication



Data Flow

Data communication

Communication is the sharing of information; it may be local (face-to-face) or remote (over distance). A data communications system has **five components**:

- **Message** — the information (data) to be communicated.
- **Sender** — the device that sends the message.
- **Receiver** — the device that receives the message.
- **Transmission medium** — the physical path a message travels from sender to receiver.
- **Protocol** — a set of rules governing communication; an agreement between devices. Without a protocol two devices may be connected but not communicating.

Data representation

Form	Representation
Text	Bit pattern (sequence of 0s/1s). A set of patterns is a **code**. **Unicode** (32-bit, universal) is prevalent; **ASCII** is the first 127 characters of Unicode ("Basic Latin").
Numbers	Not coded with ASCII — the number is converted directly to binary to simplify arithmetic.
Images	A matrix of **pixels** (picture elements); pixel size depends on resolution (e.g. 1920×1080).
Audio	Recording/broadcasting of sound; *continuous*, not discrete.
Video	Recording/broadcasting of a picture or movie — continuous (TV camera) or a sequence of discrete images conveying motion.

Communication media — features

- **Transmission speed** — measured in bits per second (bps).
- **Bandwidth** — capacity of the medium; for analog signals in hertz (Hz), for digital signals in bps. Measures how much information can be carried in a given time.
- **Transmission mode** — asynchronous or synchronous.
- **Transmission direction** — simplex, half-duplex, full-duplex.

Transmission mode	How it works
Asynchronous	Data sent one character at a time, each 5–8 bits; typical rates 1200–19200 bps; used for medium quantities over long distances.
Synchronous	Data transferred as a block of bits at once, with no start/stop bits; more efficient; used for large quantities (e.g. assembled disk contents).

Data flow (direction)

Mode	Description	Example
Simplex	Unidirectional, like a one-way street; only one device transmits.	Keyboard, traditional monitor
Half-duplex	Both can transmit and receive but not at the same time.	Walkie-talkie, CB radio
Full-duplex	Both transmit and receive simultaneously.	Telephone network

Network hardware and software

- Hardware = the physical connection between machines: adaptors, cables, routers, bridges, etc.
- Software = a set of **protocols** — a formal language among components that makes hardware usable by applications.

Physical structures — type of connection

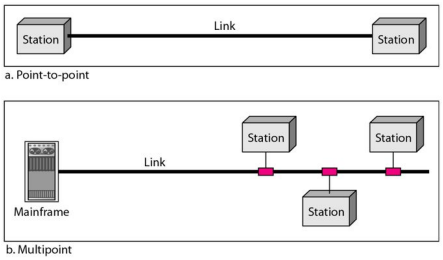
Connection	Meaning
Point-to-point	A dedicated link between two devices; the entire capacity of the link is reserved for those two. E.g. infrared remote → TV; microwave or satellite links.
Multipoint (multi-drop)	More than two devices share a single link; capacity is shared spatially or temporally (timeshared). Broadcast = one channel shared by all machines; multicast = communication to a specified group.

Network software

- Software is now highly structured. Topics: **protocol hierarchies**, **design issues for the layers**, **connection-oriented vs connectionless services**, **service primitives**, and the **relationship of services to protocols**.
- **Protocol hierarchy**: to reduce design complexity, networks are organized as layers, each built on the one below. Layer n on one machine converses with layer n on another; the rules of that conversation are the layer-n **protocol**.
- Between adjacent layers is an **interface**. A set of layers and protocols is a **network architecture**; a list of protocols (one per layer) is a **protocol stack**.

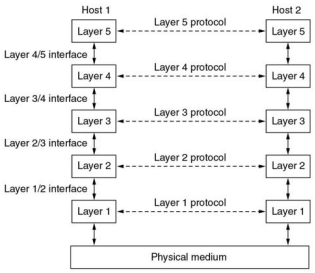
Design issues for the layers

- **Addressing** — identify senders and receivers at each layer.
- **Rules of data transfer** — simplex, half-duplex, full-duplex.
- **Error control** — error detection and correction.
- **Flow control** — preserve message order (disassemble, transmit, reassemble).
- **Multiplexing** — avoid setting up a connection for each process pair.
- **Routing** — choose a route when multiple paths exist.



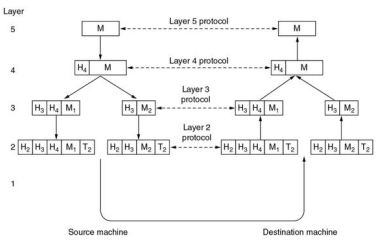
2networkhardwaresoftware — slide 5

The layering Principle



The layering Principle

The layering Principle



The layering Principle

Connection-Oriented vs. Connectionless

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Connection-Oriented vs. Connectionless

Service Primitives

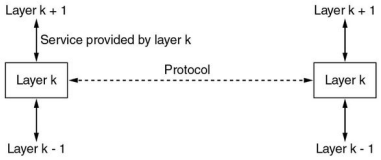
- A service is formally specified by a set of primitives (basic operations) available to a user or other entity to access the service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Example: five service primitives for implementing a simple connection-oriented service.

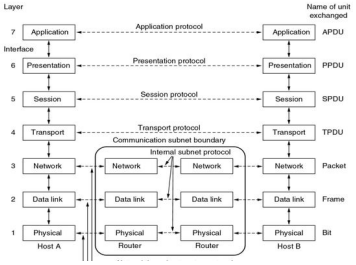
Service Primitives

The relationship of Service



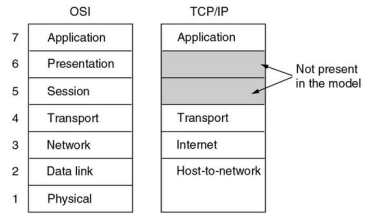
The relationship of Service

The OSI Reference Model



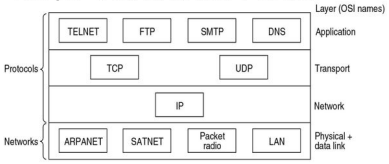
The OSI Reference Model

The TCP/IP Reference Model



The TCP/IP Reference Model

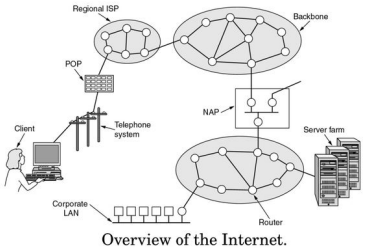
The TCP/IP Reference Model



Protocols and networks in the TCP/IP model initially.

The TCP/IP Reference Model

The TCP/IP Reference Model



Overview of the Internet.

The TCP/IP Reference Model

Services vs protocols

- A **service** is a set of primitives (operations) that a layer provides to the layer above it.
- A **protocol** is a set of rules governing the format and meaning of the frames/packets/messages exchanged by peer entities within a layer.

2. Network Models: OSI and TCP/IP

Source decks: 5OSI referencemodel, 6TCPIP

Why reference models

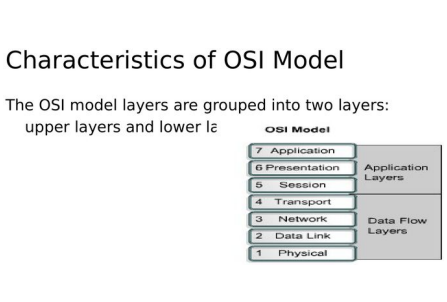
- A **communication architecture** is a strategy for connecting hosts and other equipment; it defines a standard for communicating hosts.
- A **reference model** is an abstract framework of clearly defined concepts that lets a community communicate ideas clearly. In networking it standardizes communication between heterogeneous networks.
- Two popular reference models: the **OSI model** (ISO, 1984) and the **TCP/IP protocol suite**.

The OSI reference model

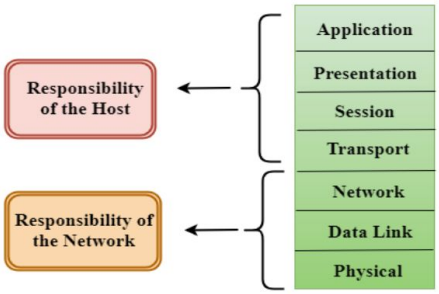
- OSI = **Open System Interconnection**; it describes how information moves from an application on one computer through a physical medium to an application on another.
- Seven layers, each performing a well-defined, self-contained function; the whole task is divided into smaller manageable tasks.
- Principles: layers created for different abstractions; each performs a well-defined function; function chosen with international standard protocols in mind; minimise information flow across interfaces; number of layers optimum.

Upper vs lower layers

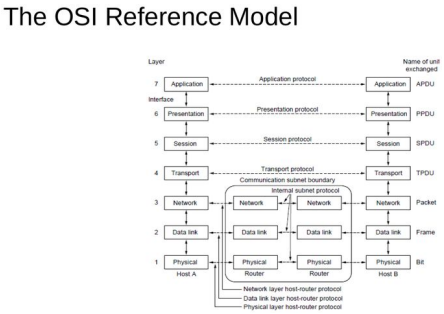
- **Upper layers** (session, presentation, application) deal with application-related issues; implemented only in software; the application layer is closest to the user.
- **Lower layers** (physical, data link, network, transport) deal with data-transport issues; the data link and physical layers are implemented in hardware and software; the physical layer is closest to the medium and places information on it.



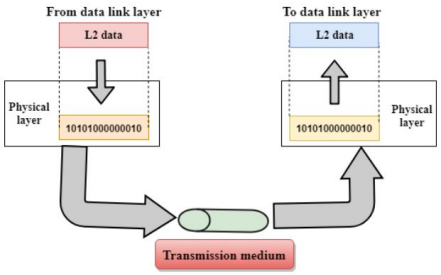
Characteristics of OSI Model



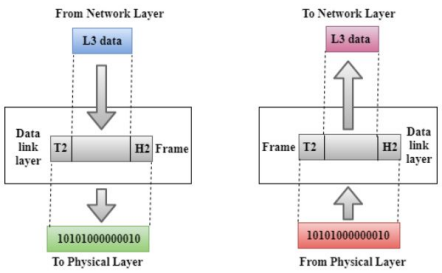
5OSI referencemodel — slide 9



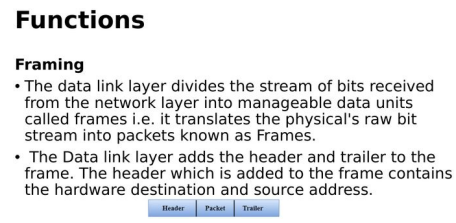
The OSI Reference Model



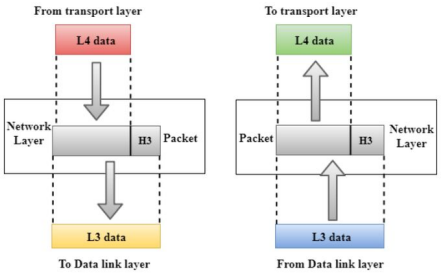
5OSI referencemodel — slide 15

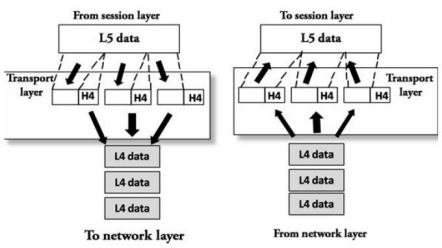


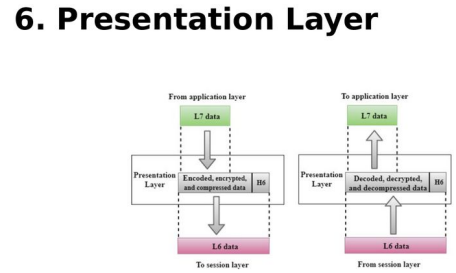
5OSI referencemodel — slide 18



5OSI referencemodel — slide 22



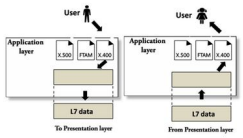




5OSI referencemodel — slide 29 *5OSI referencemodel — slide 34* *6. Presentation Layer*

7. Application layer

Layer 7 of the Open Systems Interconnection (OSI) reference model, in which network-aware, user-controlled software is implemented—for example, e-mail, file transfer utilities, and terminal access. The application layer represents the window between the user and the network.



7. Application layer

The seven layers

#	Layer	Function
7	Application	Provides services to the user (email, file transfer, terminal access).
6	Presentation	Translation, compression and encryption of the data syntax/semantics.
5	Session	Establishes, manages and terminates sessions (dialog control, synchronization).
4	Transport	Reliable process-to-process message delivery; end-to-end.
3	Network	Moves packets from source to destination (addressing, routing, packetizing).
2	Data link	Error-free transfer of frames; framing, physical addressing, flow control.
1	Physical	Transmits individual bits; establishes, maintains, deactivates the physical connection.

Physical layer (layer 1)

- Transmits individual bits from one node to another; establishes, maintains and deactivates the physical connection; specifies mechanical, electrical and procedural interface specifications.
- Four characteristics: **Mechanical** (physical properties of the interface/connector), **Electrical** (bit representation, voltage, data rate, synchronisation), **Functional** (functions of individual circuits), **Procedural** (sequence of events by which bit streams are exchanged).
- Functions: line configuration, data transmission (simplex/half/full duplex), topology, and signals.

Data link layer (layer 2)

- Turns the physical layer's raw bit-stream into a reliable link; responsible for error-free transfer of frames; defines the data format; uniquely identifies each device on a LAN.
- Functions: **framing**, **physical addressing**, **flow control**, **error control** (CRC in the trailer, retransmission), and **access control** (which device controls the link).
- IEEE Project 802 splits layer 2 into **LLC** (flow control, error correction, resequencing; hands packets to the network layer; identifies the network-layer protocol) and **MAC** (method to gain access to the medium; bridge between LLC and physical).
- LAN examples: IEEE 802.3 (CSMA/CD Ethernet) and IEEE 802.5 (token-passing token ring). WAN encapsulations: PPP, HDLC (point-to-point); Frame Relay, ATM, SMDS, X.25 (multipoint).

Network layer (layer 3)

- Logical addressing and routing of packets; establishing/releasing connections and paths; transferring data, generating and confirming receipts, resetting connections.
- Manages device addressing and tracks device locations; determines the best path based on network conditions and priority; routers are layer-3 devices.
- Functions: **internetworking** (logical connection between different devices), **addressing** (source/destination in header), **routing** (best path among multiple), **packetizing** (converting upper-layer data into packets via IP).
- Protocols: IP, IPv6; an OSI example is X.25 (on top of the X.21 physical protocol).

Transport layer (layer 4)

- End-to-end layer; ensures messages are delivered in order with no duplication; converts data into **segments**.
- Services: flow control, packet sequencing (segmentation and remote reassembly), error handling and acknowledgements, multiplexing, and virtual circuits.

Session, presentation and application layers

Layer	Function
Session (5)	Establishes and terminates sessions. Dialog control (half- or full-duplex) and synchronization (checkpoints in a data stream).
Presentation (6)	Syntax and semantics: **translation** between encoding systems, **encryption/decryption**, and **compression**.
Application (7)	Window between the user and the network: file transfer/access/management, mail services, directory services (FTP, HTTP, telnet).

The TCP/IP reference model

- The **Internet Protocol Suite** — general design guidelines and implementations of protocols that let computers communicate over a network; provides end-to-end connectivity specifying how data is formatted, addressed, transmitted, routed and received.
- Created in the 1970s by **DARPA** (US DoD); evolved from **ARPANET**, the first WAN and predecessor of the Internet; specified and widely used before OSI.
- RFC 1122 defines a four-layer model: **Application**, **Transport**, **Internet**, **Link/Network-access**.

Design goals

- Must keep working during war: fully **decentralized** (no central installation) and fully **redundant**; flexible across applications (file transfer to time-sensitive voice).
- Later goals: hardware independence, software independence (heterogeneous networks), failure recovery and high-error-rate handling, low overhead, ability to add networks without disruption, and routable data (single meaningful addressing scheme).
- **End-to-End principle**: originally, keep state and intelligence at the edges; the network stays stateless, fast and simple (changed in practice by firewalls, NAT, caches).
- **Robustness principle**: be conservative in what you send, liberal in what you accept.

The four layers

Layer	Function	Covers OSI
Application	High-level protocols, representation; create user data and pass to other processes (peers).	Application + Presentation + Session
Transport	Uniform host-to-host interface; reliability, flow control, correction; opens/maintains connections (TCP, UDP).	Transport (+ part of Session)
Internet	Exchanges datagrams across network boundaries; defines addressing and routing (IP, ARP, ICMP).	Network (+ part of LLC)
Link / Network-access	Networking methods within the local link; encapsulate IP datagrams into frames; map IP to physical addresses (Ethernet, token ring, FDDI, X.25, Frame Relay).	Physical + Data link (MAC)

Application-layer protocols

Protocol	Purpose
HTTP	HyperText Transfer Protocol — access data over the WWW (plain text, audio, video).
SNMP	Simple Network Management Protocol — manage devices on the Internet.
SMTP	Simple Mail Transfer Protocol — send email.
DNS	Domain Name System — map names to IP addresses.
TELNET	Terminal Network — local terminal appears as a terminal at a remote system.
FTP	File Transfer Protocol — transfer files between computers.

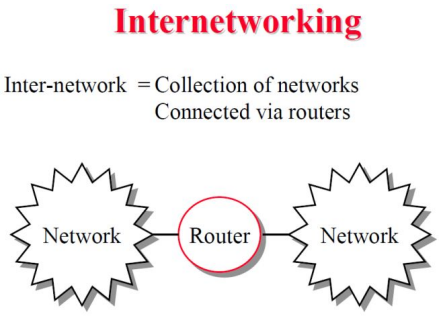
Transport-layer protocols: UDP vs TCP

Aspect	UDP	TCP
Service	Connectionless, end-to-end delivery	Full transport service; virtual circuit for the transmission duration
Reliability	Unreliable — discovers errors but does not specify/recover them (ICMP reports to sender)	Reliable — detects errors and retransmits; all segments acknowledged

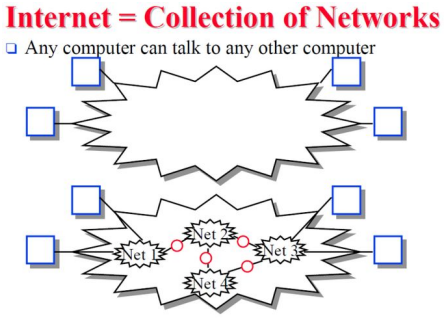
Aspect	UDP	TCP
Fields / mechanism	Source port, destination port, total length, 16-bit checksum (no segment ID)	Divides the message into segments, each with a sequence number; reorders at the receiver

Internet layer protocols

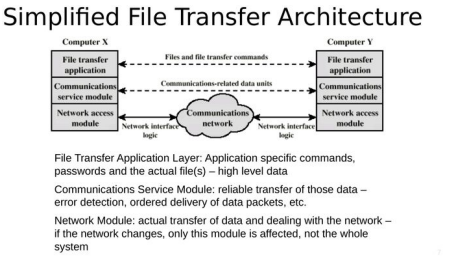
- **IP**: defines IP addresses; responsibilities are IP addressing (logical host addresses), host-to-host communication (determines the path), data encapsulation and formatting (builds the IP datagram), fragmentation and reassembly (MTU), and routing (direct or indirect delivery through routers).
- **ARP** (Address Resolution Protocol): finds the physical address from an IP address. Sender *broadcasts* an ARP request; only the device that recognises the IP address replies with an ARP reply and caches the mapping.
- **ICMP** (Internet Control Message Protocol): hosts/routers send notifications about datagram problems back to the sender (disabled links, congestion). It reports problems, it does not correct them; messages go only to the source, not to intermediate routers.



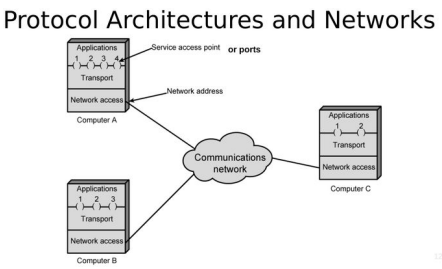
6TCPIP — slide 3



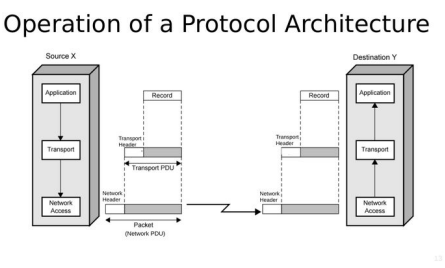
6TCPIP — slide 4



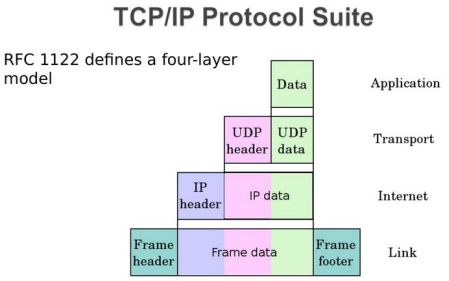
Simplified File Transfer Architecture



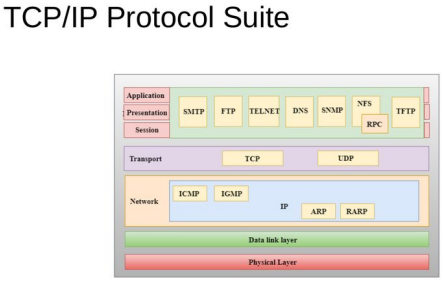
Protocol Architectures and Networks



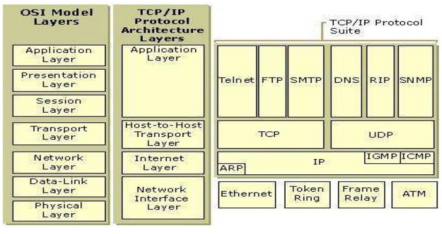
Operation of a Protocol Architecture



RFC 1122 defines a four-layer model



TCP/IP Protocol Suite



6TCPIP — slide 40

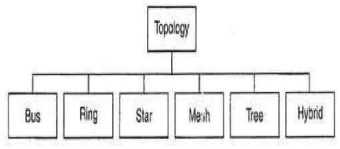
3. Network Topologies

Source deck: 4Topology

What topology means

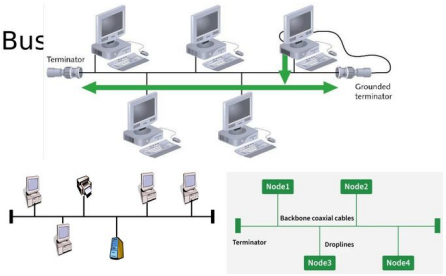
- Topology = the logical/physical layout of wires and equipment; the shape or physical connectivity of the network; it defines how nodes are connected.
- Three goals: maximum reliability (alternate routing), least-cost routing between endpoints, and the best response time and throughput for the user.
- Physical topology = how nodes are physically connected (cables, computers, peripherals). Logical topology = dedicated connections between selected source-destination pairs over the physical topology. Choose the logical topology so that average hop distance, packet delay, or maximum flow is minimised, and routing stays simple.
- Node relationships: **peer-to-peer** (all devices equal — ring, mesh) and **primary-secondary** (one device controls traffic — star).

Taxonomy



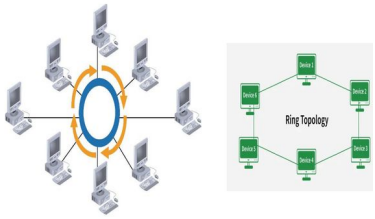
Taxonomy

Bus



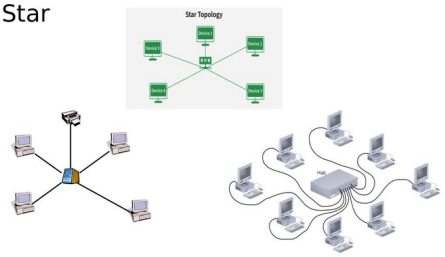
Bus

Ring



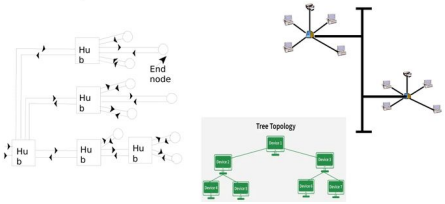
Ring

Star



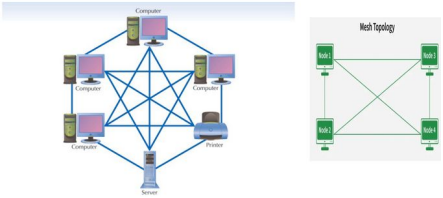
Star

A Hub/Tree Network



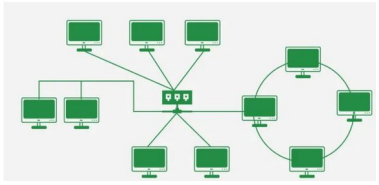
A Hub/Tree Network

Mesh Topology



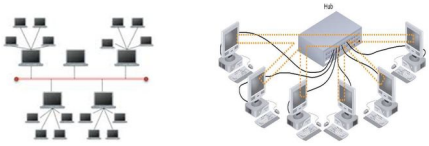
Mesh Topology

Hybrid Physical Topologies



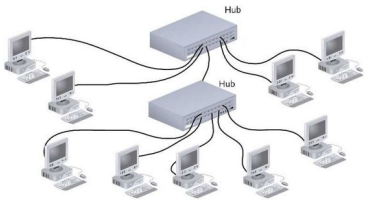
Hybrid Physical Topologies

Star-Wired Ring



Star-Wired Ring

Star-Wired Bus



Star-Wired Bus

Bus topology

- A linear bus is a main run of cable with a **terminator** at each end; all servers, workstations and peripherals attach to the cable; a single cable connects all nodes with no intervening devices.
- Terminators stop signals and prevent reflection (signal bounce). Useful in LANs; does not rely on a central host.
- Devices share responsibility for delivery: a message is broadcast in both directions; stations monitor for messages meant for them and ignore others. Access control is needed to avoid collisions.

Advantages	Disadvantages
Easy to set up; small amount of wire; keeps working if one computer fails; quick response; inexpensive; works well for small networks.	Slow; easy to crash; not scalable; hard to troubleshoot; not fault-tolerant (a bad connection can short and bring down the network); collisions; high management cost; congestion.

Ring topology

- All workstations connected in a continuous loop; no central server; each node connects to its two nearest neighbours; the signal is regenerated at each node.
- Shared-access, broadcast-capable; needs access control (e.g. token passing).
- **Single ring**: one cable, data travels one direction; each device waits its turn. **Dual ring**: two rings, each in a different direction, allowing more packets.

Advantages	Disadvantages
Point-to-point connections; easy to insert/delete nodes; scalable and modular; reliable; easy fault isolation and recovery; good for long distances; handles high traffic.	Expensive; more cable and equipment initially; less widely used than bus; fewer equipment options; limited high-speed expansion; only two alternate routes (handles a single fault); large diameter → high attenuation and blocking probability.

Star topology

- Each station connects via a point-to-point link to a central point — a **hub**, multiport repeater, or concentrator. Useful where processing must be centralised and some can be done locally.
- The central point may be **passive** (just connects the arms, no regeneration), **active** (regenerates signals), or **intelligent** (regenerates and performs path selection and management).
- Any single cable connects only two devices; cabling problems affect at most two nodes; more fault-tolerant; easy to move/isolate/interconnect; supports up to 1024 addressable nodes on a logical network.

Advantages	Disadvantages
Easy to troubleshoot (all information passes a central point); good for modern networks; low startup cost; easy to manage; room for expansion; most popular topology.	The hub is a single point of failure (all communication passes through it); requires more cable than a bus.

Tree / hub topology

- A tree or hybrid topology combines linear bus and star (and/or ring): groups of star-configured workstations connected to a linear bus backbone.
- Wires connecting nodes collapse into a central unit (hub); the hub does not switch — it consists of repeaters that retransmit signals to all nodes in the same way.

Advantages	Disadvantages
Supported by many vendors; allows more devices on a single central hub and increases distance; can isolate and prioritise communication from different computers (priority by secondary hub).	Segment length limited by cabling; a broken backbone brings down the segment; harder to configure and wire; higher cabling cost; difficult to determine time slots; signal-balancing problems (adjust transmitter strength to meet minimum signal and SNR).

Mesh topology

- Uses redundant connections between nodes (fault-tolerance strategy); each node connects to the rest of the nodes.
- Robust: failure of one node/link does not collapse the system; no traffic congestion (dedicated links); faster transmission; data privacy/security; easy fault identification.
- Expensive and difficult to install/reconfigure; high cabling cost — **n(n−1)/2** links are required to connect n nodes.

```
Mesh links for n nodes: L = n(n − 1) / 2
```

Hybrid physical topologies

- **Star-wired ring** and **star-wired bus** combine a physical star (central wiring concentrator) with a logical ring or bus.

4. Physical Layer & Signals

Source decks: 1Physical Layer, 2Digital Signals

Role of the physical layer

- Deals with the transfer of data between two *directly connected* devices (linked by a single transmission path, not a network); the process of transmitting signals across a communications link.
- Both analog and digital transmission are used; a signal consists of a spectrum of components across a range of electromagnetic frequencies. Transmission properties depend on the frequencies involved, and impairments (e.g. attenuation) depend on frequency.
- One major task is to provide services for the data link layer: the stream of 0s and 1s in frames must be converted into **signals**.

Transmission terminology

- Transmission media are **guided** (waves follow a physical path — twisted pair, coaxial cable, optical fibre) or **unguided** / wireless (waves are not guided — air, vacuum, seawater).
- **Analog data** are continuous and take continuous values; **digital data** have discrete states and discrete values.
- **Analog signals** can have an infinite number of values in a range; **digital signals** have only a limited number of values.
- A **signal** is an electric current or electromagnetic field used to convey data; it is generated by the transmitter, carried over a channel, and received. The signal is a function of time but can also be expressed as a function of frequency — the **frequency-domain view** is more important for understanding data transmission.

Analog signal parameters

Parameter	Meaning
Amplitude	Absolute value of the signal's highest intensity (peak or peak-to-peak); proportional to the energy carried; measured in volts.
Period (T)	Time to complete one cycle; expressed in seconds.
Frequency (f)	Number of periods (cycles) in one second; measured in hertz (Hz). Period is the inverse of frequency.
Phase	Position of the waveform relative to time zero; measured in degrees or radians.
Wavelength (λ)	Distance occupied by a single cycle (distance between two points of corresponding phase of consecutive cycles).

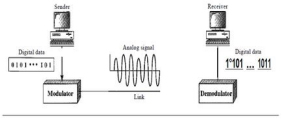
- Frequency is the rate of change with respect to time. A change in a short span of time = high frequency; over a long span = low frequency.
- Two extremes: a signal that never changes completes no cycle → frequency **0 Hz**; a signal that changes instantaneously has period 0 → frequency **infinite**.

```
Period & frequency: T = 1/f   |   Wavelength: λ = v/f = v · T   (v = propagation speed)
```

- While frequency is independent of the medium, wavelength depends on both frequency and the medium. In a vacuum light travels at 3×10^8 m/s; slower in air, slower still in cable.
- The general sine wave is fully described by **amplitude, frequency and phase**; it is the most fundamental periodic analog signal and cannot be decomposed further.

Analog Signals

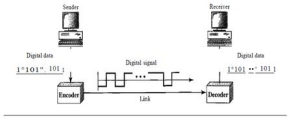
• An **analog signal** is one in which the signal intensity varies in a smooth fashion over time i.e., Analog signals are continuous, non-discrete



Analog Signals

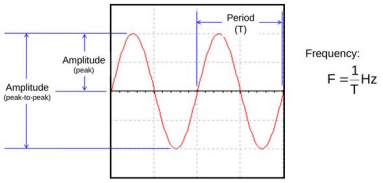
Digital Signals

• A **digital signal** is one in which the signal intensity maintains a constant level for some period of time and then abruptly changes to another constant level i.e., Digital signals are non-continuous, discrete



Digital Signals

Parts of an Analog Signal



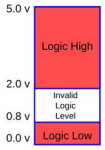
Parts of an Analog Signal

Logic Levels

A logic level is a voltage level that represents a defined digital state.

Logic HIGH: The higher of two voltages, typically 5 volts

Logic LOW: The lower of two voltages, typically 0 volts

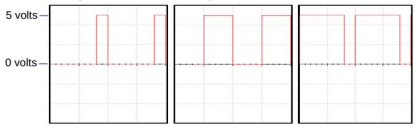


Logic Level	Voltage	True/False	On/Off	0/1
HIGH	5 volts	True	On	1
LOW	0 volts	False	Off	0

1Physical Layer — slide 13

Example of Digital Signals

- Digital signal are commonly referred to as square waves or clock signals.
- Their minimum value must be 0 volts, and their maximum value must be 5 volts.
- They can be periodic (repeating) or non-periodic.
- The time the signal is high (t_H) can vary anywhere from 1% of the period to 99% of the period.



Example of Digital Signals

Parts of a Digital Signal

Amplitude:
For digital signals, this will ALWAYS be 5 volts.

Period:
The time it takes for a periodic signal to repeat. (seconds)

Frequency:
A measure of the number of occurrences of the signal per second. (Hertz, Hz)

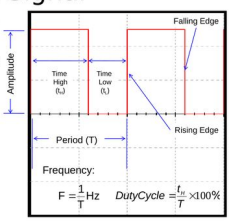
Time High (t_H):
The time the signal is at 5 v.

Time Low (t_L):
The time the signal is at 0 v.

Duty Cycle:
The ratio of t_H to the total period (T).

Rising Edge:
A 0-to-1 transition of the signal.

Falling Edge:
A 1-to-0 transition of the signal.



Parts of a Digital Signal

Periodic signals

a signal $s(t)$ is defined to be periodic if and only if

$$s(t+T) = s(t) \quad -\infty < t < +\infty$$

where the constant T is the period of the signal (T is the smallest value that satisfies the equation).

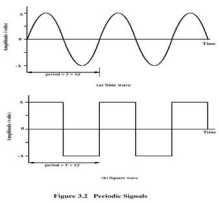


Figure 3.2 Periodic Signals

Periodic signals

Aperiodic signals

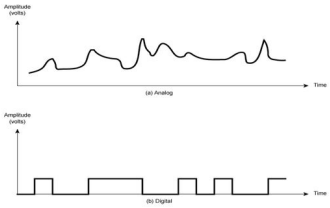
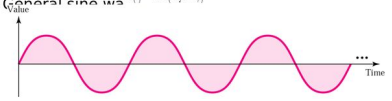


Figure 3.1 Analog and Digital Waveforms

Aperiodic signals

- Sine wave – most fundamental form of a periodic analog signal
- Fully described by: Amplitude, Frequency and Phase
- General sine wa

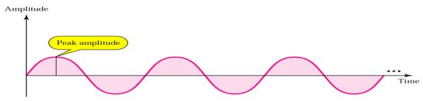
$$s(t) = A\sin[2\pi f t + \phi]$$



Sine wave – most fundamental form of a periodic analog signal Fully described by: Amplitude,...

Amplitude

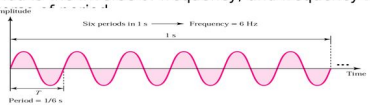
- Absolute value of a signal's highest intensity- proportional to the energy it carries.
- Normally measured in volts



Amplitude Absolute value of a signal's highest intensity- proportional to the energy it carr...

Period and Frequency

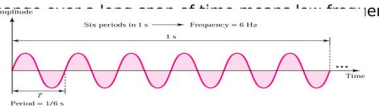
- Period - amount of time to complete one cycle, expressed in seconds
- Frequency – number of periods in one second
- Period is the inverse of frequency, and frequency is the in



Period and Frequency Period - amount of time to complete one cycle, expressed in seconds Fre...

Frequency

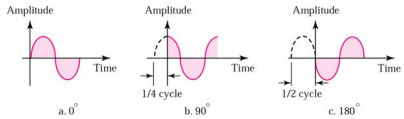
- Rate of change with respect to time, expressed in hertz (Hz)
- Change in a short span of time means high frequency
- Ch ency



Frequency Rate of change with respect to time, expressed in hertz (Hz) Change in a short spa...

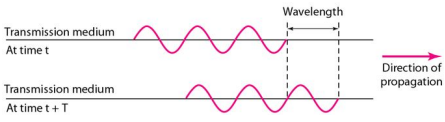
Phase

- Position of the waveform relative to time zero
- Measured in degrees or radians



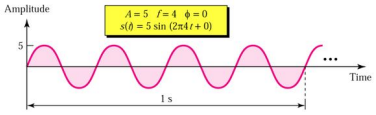
Phase Position of the waveform relative to time zero Measured in degrees or radians

Wavelength and Period

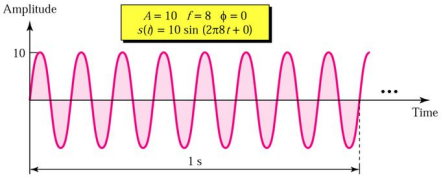


Wavelength and Period

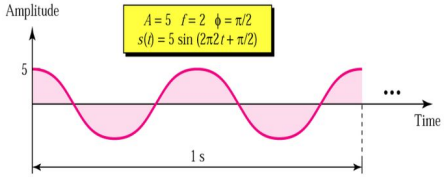
Sine Wave examples



Sine Wave examples

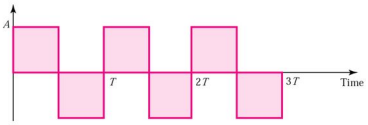


1Physical Layer — slide 30



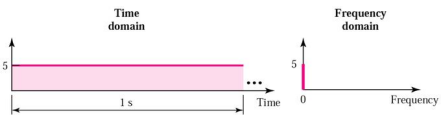
1Physical Layer — slide 31

Square Wave



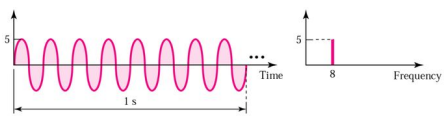
$$s(t) = \frac{4A}{\pi}\sin 2\pi f t + \frac{4A}{3\pi}\sin[2\pi(3f)t] + \frac{4A}{5\pi}\sin[2\pi(5f)t] + \ldots$$

Square Wave



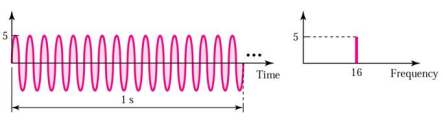
a. A signal with frequency 0

1Physical Layer — slide 34



b. A signal with frequency 8

1Physical Layer — slide 35



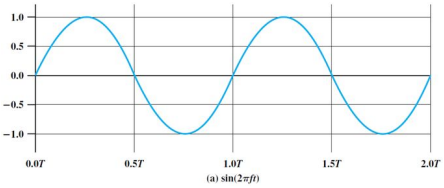
c. A signal with frequency 16

1Physical Layer — slide 36

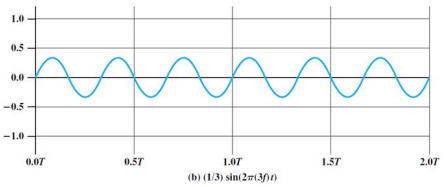
Composite Signals

- Composed of many simple sine waves of differing frequencies
- Fourier – showed any composite signal is a sum of a set of sine waves of different frequencies, phases, and amplitudes (Harmonics)
- Fourier analysis $s(t) = A \times \frac{4}{\pi} \times \sum_{k\,odd,k=1}^{\infty} \frac{\sin(2\pi k f t)}{k}$
- The scaling factor of 4/π is used to produce a wave whose peak amplitude is close to 1.

Composite Signals



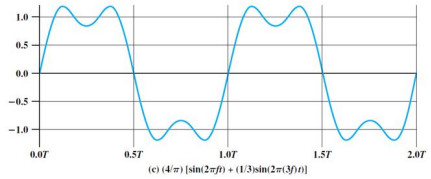
1Physical Layer — slide 38



1Physical Layer — slide 39

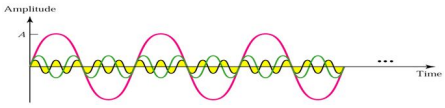
Addition of Frequency Components

T = 1/f



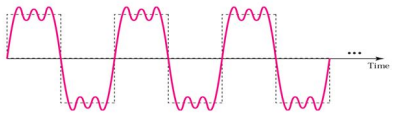
Addition of Frequency Components T = 1/ f

- Three harmonics



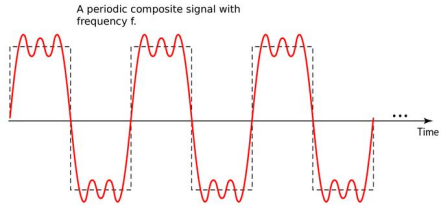
Three harmonics

- Adding first three harmonics



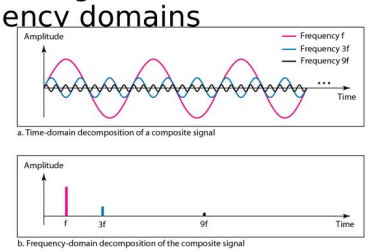
Adding first three harmonics

A composite periodic signal



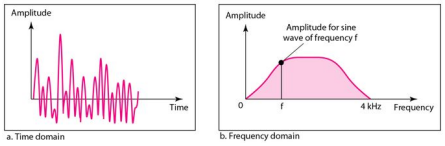
A composite periodic signal

Decomposition of a composite periodic signal in the time and frequency domains



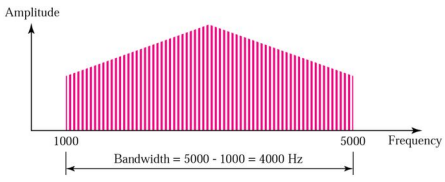
Decomposition of a composite periodic signal in the time and frequency domains

A nonperiodic composite signal



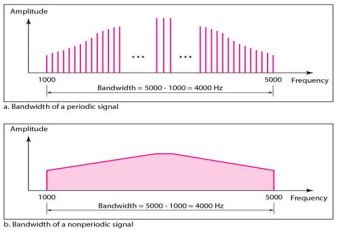
A nonperiodic composite signal

Bandwidth



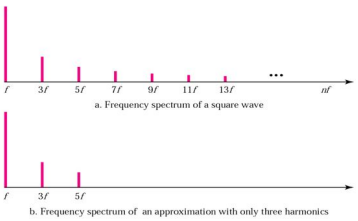
Bandwidth

The bandwidth of periodic and nonperiodic composite signals



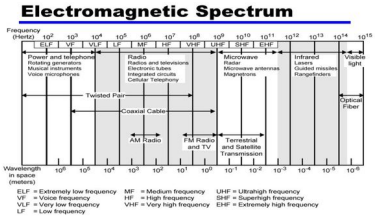
The bandwidth of periodic and nonperiodic composite signals

Frequency spectrum comparison



Frequency spectrum comparison

Frequency Spectrum



Frequency Spectrum

- no transmission medium is perfect – each medium passes some frequencies and blocks or weakens others
- composite signal sent at one end of transmission medium (comm. channel), may not be received in the same form at the other end
- passing a square wave through any medium will always deform the signal !!!

no transmission medium is perfect – each medium passes some frequencies and blocks or weaken...

Digital signal parameters

- A digital signal maintains a constant level for some time then abruptly changes — it is discrete/non-continuous, commonly a square wave (min 0 V, max 5 V). It may be periodic or non-periodic.
- A **logic level** is a voltage representing a digital state: HIGH ≈ 5 V (true/on/1) and LOW ≈ 0 V (false/off/0).

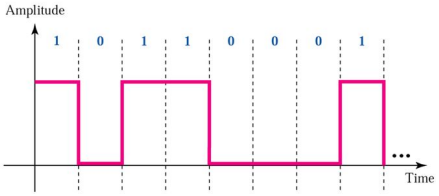
Parameter	Meaning
Amplitude	For digital signals this is always the supply level (e.g. 5 V).
Period (T)	Time for a periodic signal to repeat.
Frequency	Number of occurrences per second (Hz).
Time high (t_H)	Time the signal is at the high level; varies from 1% to 99% of the period.
Time low (t_L)	Time the signal is at the low level.
Duty cycle	Ratio of t_H to the total period (T).
Rising / falling edge	A 0→1 transition / a 1→0 transition.

Bit interval, bit rate and baud rate

- **Bit interval (bit length)** = time to send one bit; **bit rate** = number of bit intervals per second (bps) = 1 / bit interval.
- A signal with L levels carries $\log_2 L$ bits per level.
- **Baud rate** (signal rate) = rate at which the signal changes. A two-level signal has the same baud and bit rate; with more levels a symbol carries several bits.

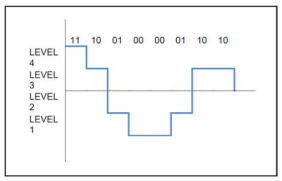
`Bit rate = 1 / bit interval  |  bits per level = log2L`

Digital signal



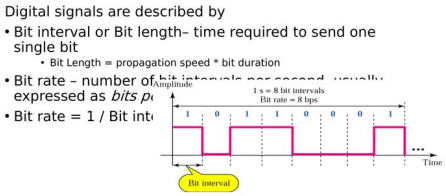
Digital signal

A digital signal with four levels



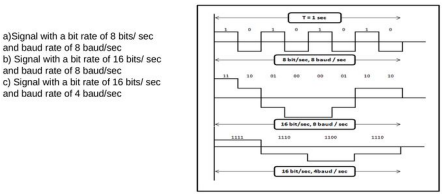
A digital signal with four levels

Bit Interval and Bit Rate



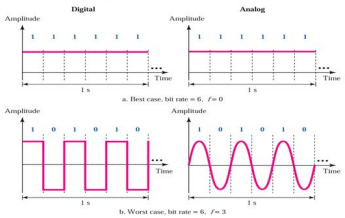
Bit Interval and Bit Rate

Three signals with different bit rates and baud rates



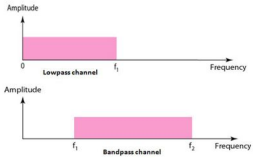
Three signals with different bit rates and baud rates

Using only one harmonic (Digital versus Analog)



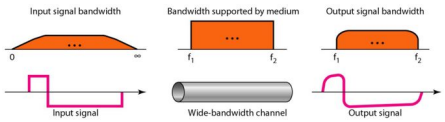
Using only one harmonic (Digital versus Analog)

Lowpass Channel & Bandpass Channel

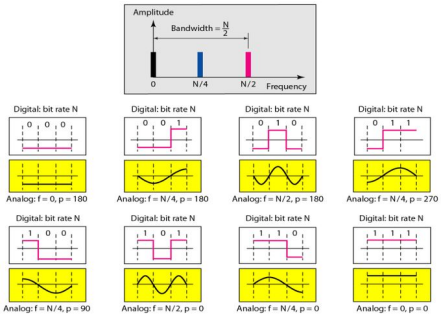


Lowpass Channel & Bandpass Channel

2Digital Signals — slide 17

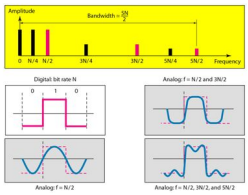


2Digital Signals — slide 19



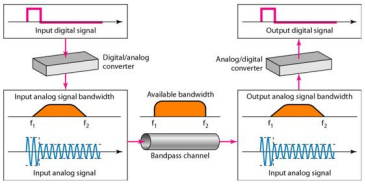
2Digital Signals — slide 21

To make the shape of the analog signal look more like that of a digital signal, we need to add more harmonics of the frequencies.
i.e. to increase the bandwidth to 3N/2, 5N/2, 7N/2, and so on.



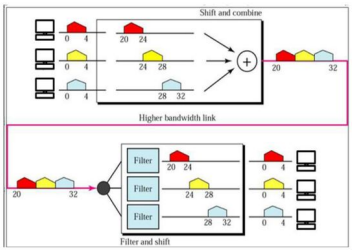
To make the shape of the analog signal look more like that of a digital signal, we need to a...

Modulation of a digital signal



Modulation of a digital signal

Broadband Transmission Involving Modulation & Demodulation



Broadband Transmission Involving Modulation & Demodulation

Periodic, aperiodic, and composite signals

- A **periodic** signal repeats a pattern over identical periods (one full pattern = a cycle); an **aperiodic** signal changes without a repeating pattern. In data communications we commonly use periodic analog signals and non-periodic digital signals.
- Periodic analog signals are **simple** (a sine wave) or **composite** (multiple sine waves).
- **Fourier** showed any composite signal is a sum of sine waves of different frequencies, phases and amplitudes (harmonics).
- A **harmonic** is a wave whose frequency is an integral multiple of a reference frequency.
- A periodic composite signal decomposes into sine waves with **discrete** frequencies (integers); a non-periodic composite signal decomposes into a continuum of sine waves with **continuous** frequencies.

Time domain and frequency domain

- A **time-domain plot** shows signal amplitude against time. A **frequency-domain plot** shows only the peak value and the frequency (amplitude vs frequency) — the best way to represent an analog signal.

Bandwidth

Term	Meaning
Bandwidth (of a medium)	Range of frequencies a medium can pass without losing one-half of the power contained in the signal; the difference between the highest and lowest frequencies the medium can pass satisfactorily.
Bandwidth (of a signal)	The portion of the electromagnetic spectrum occupied by the signal; the frequency range over which it is transmitted.
Analog signal bandwidth	Range of frequencies the composite analog signal carries = $f_{highest} - f_{lowest}$ (e.g. 90 − 30 = 60 Hz).
Digital signal bandwidth	Maximum bit rate of the signal, measured in bits per second.
Absolute bandwidth	Width of the signal spectrum: B = $f_{highest} - f_{lowest}$.
Effective bandwidth	Range of frequencies where the signal contains most of its power/energy.
Frequency spectrum	Range (set) of frequencies the signal contains; depends on the medium.

- The bandwidth of the medium must always be greater than the bandwidth of the signal, else the signal is attenuated or distorted (loss of information). The channel bandwidth determines whether an analog or digital signal can be transmitted.
- Digital signals are composite analog signals with theoretically **infinite** bandwidth; high-frequency components attenuate quickly due to trace capacitance and skin effect.

Channel types and transmission of digital signals

Channel	Range passed
Low-pass	From 0 Hz up to some non-zero frequency f_1 (passes 0…f_1).

Channel	Range passed
Band-pass	From a non-zero f_1 up to a non-zero f_2 (passes $f_1 \ldots f_2$).

- **Baseband transmission**: send the digital signal without modulation; requires a low-pass channel (bandwidth starts at zero, i.e. a dedicated medium). The signal bandwidth must be less than the channel bandwidth (wideband if the channel is wider; narrowband if narrower).
- **Broadband transmission**: with a band-pass channel, first convert the digital signal to analog using modulation (a carrier), transmit, then demodulate back to digital at the receiver.
- Approximating a digital signal with a low-pass channel: worst cases 0101…/1010… need an analog signal of frequency f = N/2 (bit rate N). Adding harmonics (to bandwidth 3N/2, 5N/2, …) makes the shape closer to the digital signal.
- The relation between bit rate (n) and required bandwidth (B): with the first harmonic B = n/2, i.e. **n ≤ 2B**.

Low-pass rough approximation: B = n / 2 → n ≤ 2B (more harmonics: B = 3n/2, 5n/2, …)

Maximum data rate of a channel

Data rate depends on three factors: the **bandwidth** available, the **levels** of signals used, and the **quality** of the channel (noise). A noiseless channel uses the Nyquist bit rate; a noisy channel uses the Shannon capacity.

$$\text{Nyquist (noiseless): BitRate} = 2 \times \text{Bandwidth} \times \log_2 L \quad (L = \text{number of signal levels})$$

$$\text{Shannon (noisy): Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

- Shannon capacity does *not* consider the number of signal levels (unlike Nyquist).

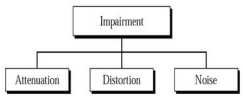
5. Transmission Impairments

Source deck: 3transmission_impairments

Why signals are impaired

- Transmission media are imperfect, so the signal at the end differs from the signal at the beginning ("what is sent is not what is received").
- Consequences: for **analog** signals, degradation of signal quality; for **digital** signals, bit errors.
- The three causes of impairment are **attenuation, distortion and noise**.

• Three causes of impairment are attenuation, distortion, and noise.



Three causes of impairment are attenuation, distortion, and noise.

• Figure shows the effect of attenuation and amplification.

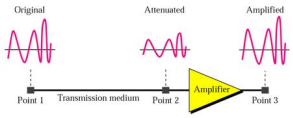


Figure shows the effect of attenuation and amplification.

Decibel

To show that a signal has lost or gained strength, engineers use the unit of the decibel.

• The decibel (dB) measures the relative strengths of two signals or one signal at two different points.

$$A_{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

• Variables P1 and P2 are the powers of a signal at points 1 and 2, respectively.
• The decibel is negative if a signal is attenuated and positive if a signal is amplified.
• Decibel in terms of voltage instead of power.
• Power is proportional to the square of the vc the formula is

$$A_{dB} = 20 \log_{10} \frac{V_2}{V_1}$$

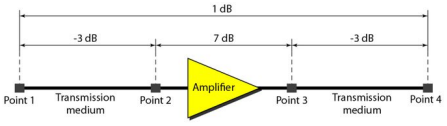
Decibel

• A signal travels through an amplifier, and its power is increased 10 times. This means that P2 = 10P1 . In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

3transmission_impairments — slide 9



3transmission_impairments — slide 11

Figure shows the effect of distortion on a composite signal.

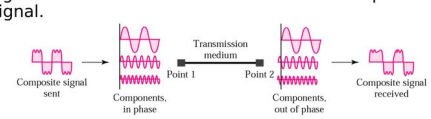


Figure shows the effect of distortion on a composite signal.

Figure shows the effect of noise on a signal.

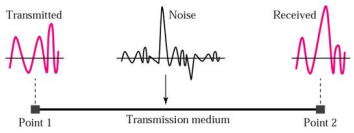


Figure shows the effect of noise on a signal.

Signal to Noise Ratio (SNR)

• The signal-to-noise ratio is defined as

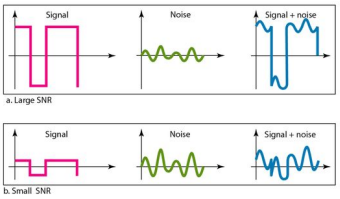
$$SNR = \frac{\text{Average signal power}}{\text{Average noise power}}$$

• We need to consider the average signal power and the average noise power because these may change with time.
• SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
• A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.
• Because SNR is the ratio of two powers, it is often described in decibel units,
• SNR dB, is defined as

$$SNR_{db} = 10 \log_{10} SNR$$

Signal to Noise Ratio (SNR)

Two cases of SNR: a high SNR and a low SNR



Two cases of SNR: a high SNR and a low SNR

• The values of SNR and SNRdB for a noiseless channel are

$$SNR = \frac{\text{signal power}}{0} = \infty$$

$$SNR_{dB} = 10 \log_{10} \infty = \infty$$

• We can never achieve this ratio in real life; it is an ideal.

The values of SNR and SNRdB for a noiseless channel are We can never achieve this ratio in r...

Attenuation

- Loss of energy; signal strength falls off with distance as energy is converted to heat overcoming the medium's resistance (a wire warms up).
- For guided media attenuation is generally exponential and expressed as a constant number of decibels per unit distance; for unguided media it is a more complex function of distance and atmosphere.
- A received signal must be strong enough for the receiver's circuitry and must stay sufficiently above noise; **amplifiers** (analog) or **repeaters** (digital) are used.

The decibel

- The **decibel (dB)** measures the relative strengths of two signals or one signal at two points. Negative = attenuated; positive = amplified.
- Because power is proportional to the square of voltage, the voltage form uses 20 $\log_{10}$.
- dB values **add** when cascading several points (attenuation then amplification then attenuation...).

$$dB = 10 \log_{10}(P_2/P_1) \quad | \quad dB = 20 \log_{10}(V_2/V_1)$$

- Example: power reduced to half ($P_2 = \frac{1}{2}P_1$) → $10 \log_{10}(0.5) = 10(-0.3) = \mathbf{-3\ dB}$ (losing half the power).
- Example: power increased 10× ($P_2 = 10P_1$) → $10 \log_{10}(10) = \mathbf{+10\ dB}$.

Distortion

- The signal changes its form or shape — happens in a composite signal of different frequencies. Each component has its own propagation speed and therefore its own delay; the phase relationships change and the composite shape is no longer the same.
- **Attenuation distortion**: attenuation is often an increasing function of frequency, so some components are attenuated more than others — noticeable for analog signals (reduces intelligibility).
- **Phase (delay) distortion**: velocity of propagation varies with frequency, so components arrive at different times → phase shifts. Critical for digital data: components of one bit spill into other bit positions, causing **intersymbol interference** — a major limit on maximum bit rate.

Noise

- Noise = unwanted signals inserted between transmission and reception; the major limiting factor in communications performance. The received signal = transmitted signal (modified by distortion) + noise.

Category	Cause / nature
Thermal (white) noise	Random motion of electrons in a wire (thermal agitation); present in all devices and media; a function of temperature; cannot be eliminated → upper bound on performance.
Intermodulation noise	Signals at different frequencies sharing a medium produce energy at the sum/difference of the originals (e.g. f_1+f_2), which can interfere with a wanted signal.
Induced noise	From sources such as motors and appliances acting as a sending antenna, with the medium as the receiving antenna.
Crosstalk	Effect of one wire on another (unwanted coupling between signal paths, e.g. nearby twisted pairs); typically ≤ thermal noise.
Impulse noise	Non-continuous irregular spikes of short duration and high amplitude (e.g. lightning); a minor annoyance for analog data but the **primary source of error in digital data**.

Signal level and signal-to-noise ratio

- Level = signal magnitude; comparative. In telephony measured in **dBm** (referenced to 1 mW); in wireless in dBW. If the level is too high, amplifiers overload (distortion); if too low, loudness/quality suffers.
- **SNR** = ratio of what is wanted (signal) to what is not wanted (noise). A high SNR means less corruption; a low SNR means more corruption. Because it is a ratio of powers it is often expressed in decibels.
- Typical requirements: voice 40 dB (45 dB for some), data ≈ 15 dB (based on a specified error rate and modulation type).
- For a noiseless channel SNR and SNR_{dB} are infinite (SNR = ∞); this ideal can never be achieved in real life.

$$SNR = \text{average signal power} / \text{average noise power} \quad | \quad SNR_{dB} = 10 \log_{10}(SNR)$$

$$SNR_{dB} = \text{level (signal in dBm)} - \text{level (noise in dBm)} \quad | \quad dBm = dBm0 + dBr$$

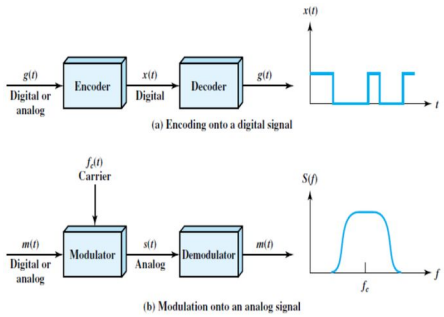
6. Digital Data Transmission (Line Coding, Block Coding, Scrambling)

Source deck: 4Digital Data Transmission

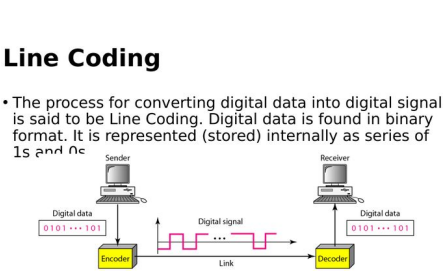
Four conversion families

Input → Output	Technique
Digital data → digital signal	Line coding (mandatory); block coding and scrambling (optional).
Digital data → analog signal	Modulation: ASK, FSK, PSK (and QAM). A modem converts digital data for an analog line.
Analog data → digital signal	Pulse Code Modulation (PCM): sample periodically, then quantize.
Analog data → analog signal	AM, FM, PM (modulate a carrier to a different band).

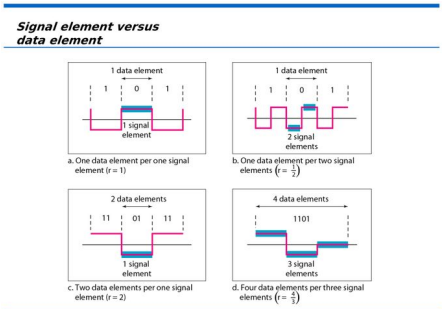
- For digital signalling, a data source g(t) is encoded into a digital signal x(t); the form of x(t) is chosen to optimise the medium (conserve bandwidth or minimise errors).
- For analog signalling, data rides a continuous constant-frequency **carrier**; modulation alters one or more of the carrier's amplitude, frequency or phase. The input is the **modulating (baseband) signal** m(t); the result is the **modulated signal** s(t), a band-limited (band-pass) signal.



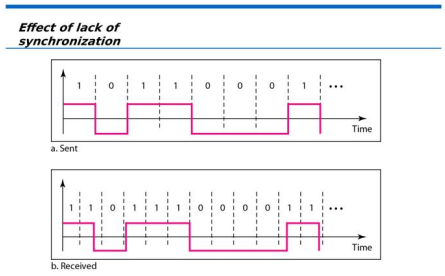
4Digital Data Transmission — slide 8



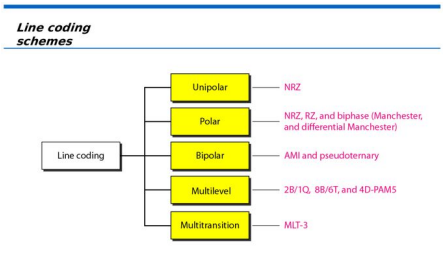
Line Coding



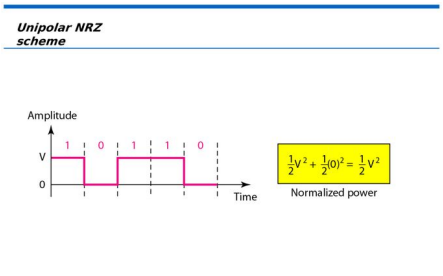
Signal element versus data element



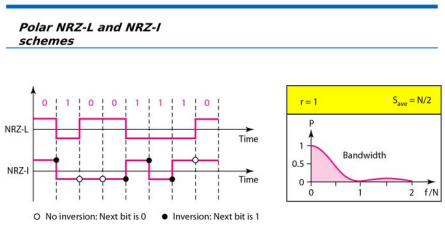
Effect of lack of synchronization



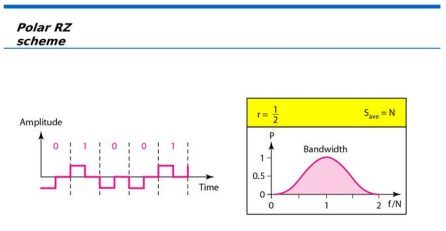
Line coding schemes



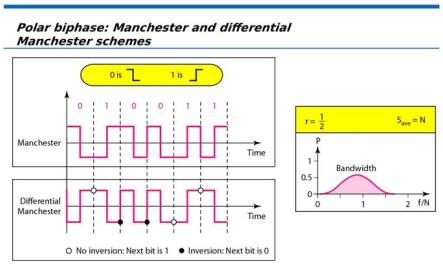
Unipolar NRZ scheme



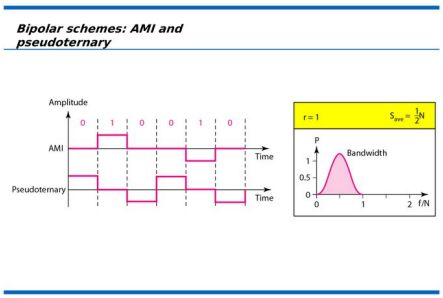
Polar NRZ-L and NRZ-I schemes



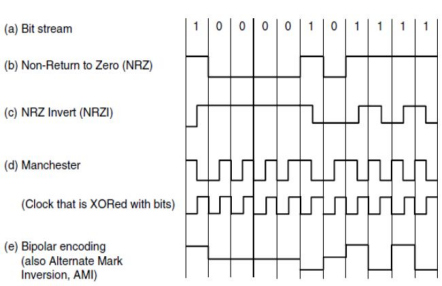
Polar RZ scheme



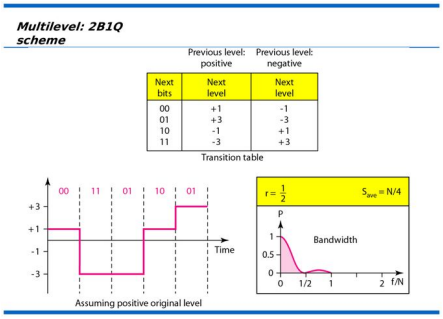
Polar biphase : Manchester and differential Manchester schemes



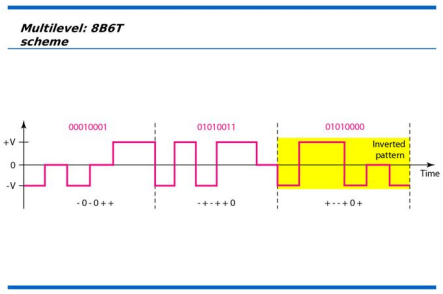
Bipolar schemes: AMI and pseudoternary



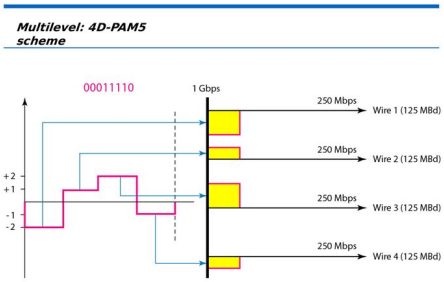
4Digital Data Transmission — slide 39



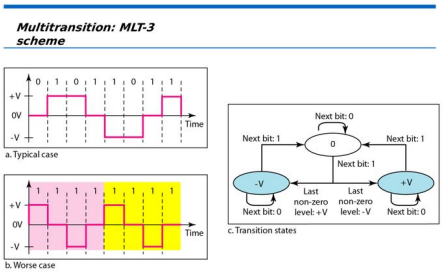
Multilevel: 2B1Q scheme



Multilevel: 8B6T scheme



Multilevel: 4D-PAM5 scheme

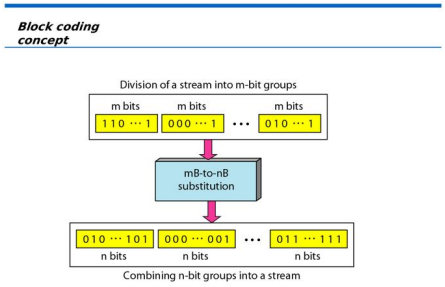


Multitransition : MLT-3 scheme

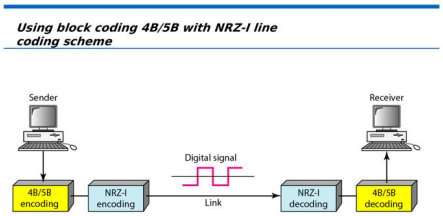
Summary of line coding schemes

Category	Scheme	Bandwidth (average)	Characteristics
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
Unipolar	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
Multilevel	8B6T	$B = 3N/4$	Self-synchronization, no DC
Multilevel	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

4Digital Data Transmission — slide 51



Block coding concept



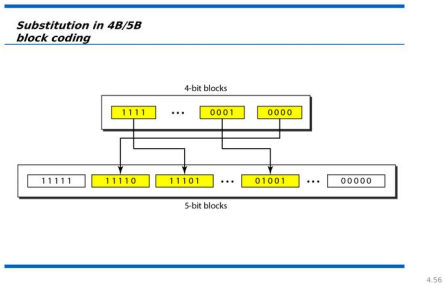
Using block coding 4B/5B with NRZ-I line coding scheme

4B/5B mapping code

Data Sequence	Encoded Sequence	Control Sequence	Encoded Sequence
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

4.55

4Digital Data Transmission — slide 55



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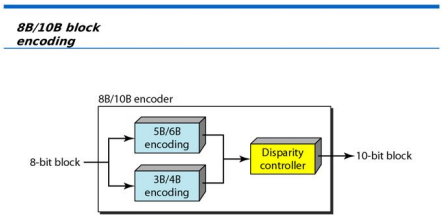
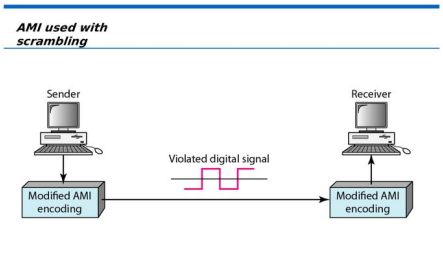
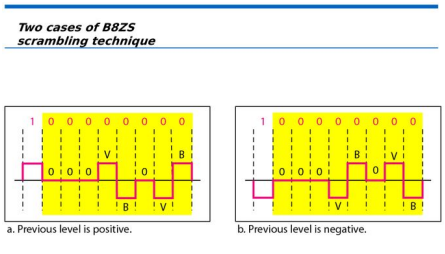


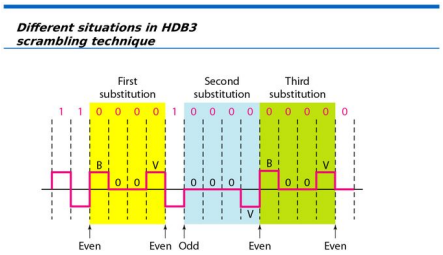
8B/10B block encoding



AMI used with scrambling



Two cases of B8ZS scrambling technique



Different situations in HDB3 scrambling technique

Line coding — signal element vs data element

- A **data element** is the smallest entity that carries information — the bit. A **signal element** is the shortest time-wise unit of the digital signal. Data elements are what we need to send; signal elements are

what we can send (the carriers).

- The ratio **r** = number of data elements carried by one signal element: r = 1 (one bit per element), r = 1/2 (two elements per bit), r = 2 (one element carries two bits), r = 4/3 (four bits carried by three elements).
- A data symbol can be coded into one or many signal elements (1 → +V, 0 → −V, or 1 → +V and −V, etc.).
- **Data rate** N = number of bits per second (bps, bit rate). **Signal rate** S = number of signal elements per second (bauds, modulation rate). Goal: increase data rate while reducing baud rate.

```
Signal (baud) rate: S = c × N × (1/r) bauds   (c = case factor: worst/best/average)
```

Considerations for choosing a line code

Criterion	Meaning
Baseline wandering	The receiver evaluates the average power (baseline) to detect bits. A long run of 0s or 1s makes the baseline drift and causes errors. Good codes prevent long runs of fixed amplitude.
DC components	A constant voltage level for long periods increases low frequencies. Most channels are band-pass and cannot pass low frequencies, so the DC component must be removed.
Self synchronization	Sender and receiver clocks must have the same bit interval; a faster/slower receiver clock misinterprets the stream. Transitions let the receiver resynchronise.
Error detection	Some codes include a transition that is not part of valid data; when it occurs the receiver knows a symbol error happened.
Noise and interference	Some encodings make the signal immune to noise/interference (stronger than error detection).
Complexity	The more robust the code, the more complex and costly (paid in baud rate or bandwidth).

Line coding schemes

Scheme	Idea / properties
Unipolar NRZ	All levels on one side of the axis; +V = 1, 0 V = 0; signal does not return to zero mid-bit. Prone to baseline wandering and DC; no sync or error detection; simple but costly in power.
Polar NRZ-L	Level determines the bit (+V = 1, −V = 0).
Polar NRZ-I	Inversion (change of polarity) = 1, no change = 0. Both NRZ-L and NRZ-I have average signal rate N/2 Bd; both have DC and baseline wandering (worse for NRZ-L); no sync or error detection.
Polar RZ	Three levels (+, 0, −); each symbol has a mid-transition. More transitions → wider bandwidth; no DC/baseline wandering; self-synchronising; more complex; no error detection.
Manchester	Combines NRZ-L and RZ: every symbol has a mid-bit transition (high→low or low→high); two levels. Used for synchronization.
Differential Manchester	Combines NRZ-I and RZ: mid-bit transition always; the level at the start of the symbol encodes the value (one symbol causes a change, the other does not).
Bipolar AMI	Three levels; 0 = zero voltage, 1 alternates between +V and −V. No DC/baseline wandering; no self-sync for long runs of 0s; no error detection.
Pseudoternary	Reverse of AMI (1 = zero, 0 alternates).

- Manchester and differential Manchester use the mid-bit transition for synchronization; their minimum bandwidth is **2× that of NRZ**; no DC, no baseline wandering; neither has error detection.

Multilevel schemes (mBnL)

- Increase the number of data bits per symbol to raise the bit rate. Combine m binary elements into 2^m symbols; with L levels and n elements we can create L^n signal elements.
- If $2^m > L^n$ we cannot represent the data; if $2^m = L^n$ there is an exact mapping; if $2^m < L^n$ we have spare signals and can pick the most distinct ones (better noise immunity and error detection).
- Notation **mBnL**: m = length of the binary pattern, B = binary, n = length of the signal pattern, L = number of levels (B = 2, T = 3 ternary, Q = 4 quaternary).

```
Multilevel condition: 2^m ≤ L^n
```

Scheme	Idea
2B1Q	Two binary bits per one quaternary signal; no redundancy so a DC component is present.
8B6T	Uses redundancy: choose only codes with a + or 0 weight and invert any code that would create a DC component (receiver inverts "−" weighted codes as they are invalid).

Scheme	Idea
4D-PAM5	Multiple channels: split the signal over x links with z-level PAM (xD-YYYz); reduces the signalling rate per link.
MLT-3	Multitransition coding: differential at the bit level, forcing transitions at bit boundaries. Signal rate same as NRZ-I, but the worst-case pattern 1111 becomes periodic, so it can be approximated as an analog signal of frequency 1/4 the bit rate.

Block coding (mB/nB)

- Adds redundancy for error detection and synchronization. Slash notation **mB/nB**: an m-bit block is substituted with an n-bit block (n > m). Three steps: division, substitution, combination. The result is then line coded for transmission.
- It prevents bit combinations that would cause DC components or poor sync quality under the line code.
- Example **4B/5B**: 16 data words → 32 code words, so 16 extra words are used for control/signalling.
- Example **8B/10B**: more redundant bits allow choosing code words that prevent long runs of a voltage level (no DC component).

Scrambling

- The best code adds no bandwidth for synchronization and has no DC component. **Scrambling** creates a bit sequence that is self-clocking with no low frequencies and no wide bandwidth; it is implemented at the same time as encoding (on the fly).
- It replaces "unfriendly" runs of bits with a violation code that is easy to recognise.
- **B8ZS** substitutes eight consecutive zeros with 000VB0VB, where V = violation of the line-encoding rule and B = bipolar (follows the AMI rule).
- **HDB3** substitutes four consecutive zeros with 000V or B00V depending on the number of non-zero pulses after the last substitution: even → B00V, odd → 000V (so the total number of non-zero pulses stays even).

7. Analog-to-Digital Conversion (PCM, Delta) & Transmission Modes

Source deck: 5Digital Data Transmission1

Why convert analog to digital

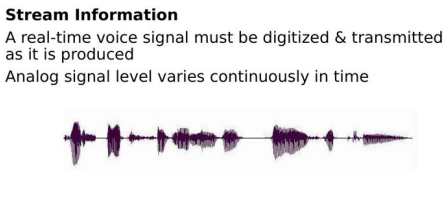
- Analog data (microphone voice, camera video) is a continuous waveform; to send it over digital signals we convert it. Two techniques: **Pulse Code Modulation (PCM)** and **delta modulation**.
- A real-time voice signal must be digitized and transmitted as it is produced; the analog level varies continuously in time.

Pulse Code Modulation (PCM)

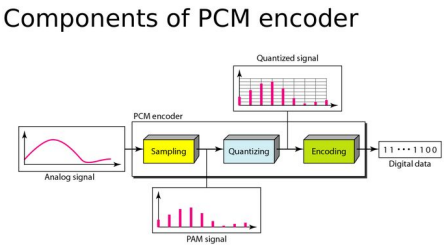
PCM digitizes an analog signal in three steps: **sampling** (obtain samples of x(t) at uniformly spaced intervals), **quantization** (map each sample to a finite-precision value) and **binary encoding/compression**. Before sampling, filter the signal to limit its maximum frequency (removing high-frequency components that distort the shape).

- Sampling every T_s seconds; sampling rate $f_s = 1/T_s$. The result (PAM) has analog, non-integer values.
- Nyquist theorem: the sampling rate must be at least twice the highest frequency in the signal.
- Three sampling methods: **ideal** (an impulse at each instant — not easily implemented), **natural** (a short pulse retaining the analog shape), and **flat-top / sample-and-hold** (constant amplitude per sample — the common method).

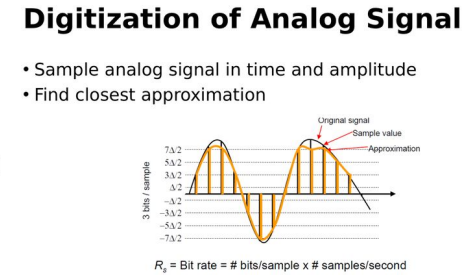
$$\text{Sampling: } f_s = 1/T_s \quad | \quad \text{Nyquist: } f_s \geq 2 \times f_{max}$$



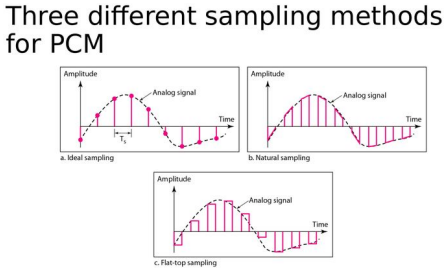
Stream Information A real-time voice signal must be digitized & transmitted as it is produce...



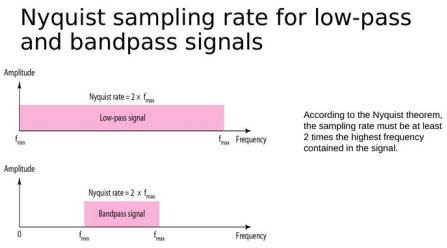
Components of PCM encoder



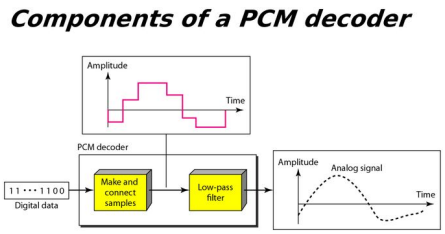
Digitization of Analog Signal



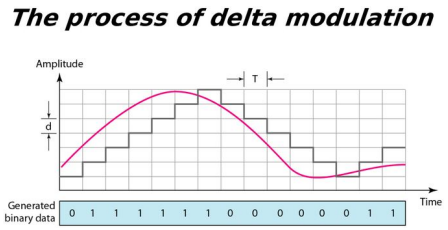
Three different sampling methods for PCM



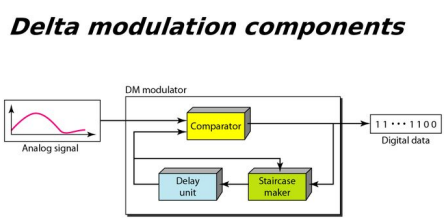
Nyquist sampling rate for low-pass and bandpass signals



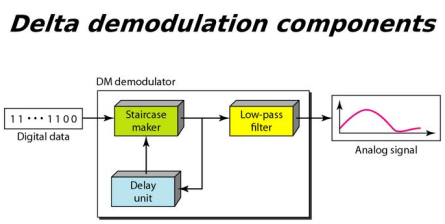
Components of a PCM decoder



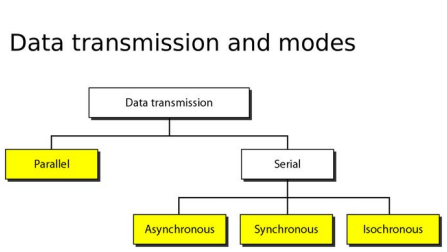
The process of delta modulation



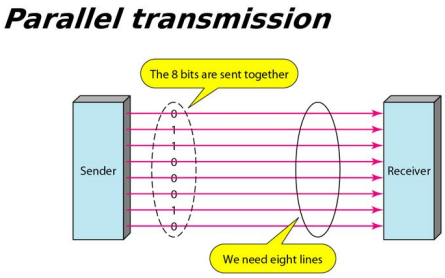
Delta modulation components



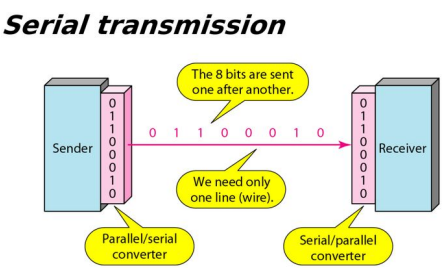
Delta demodulation components



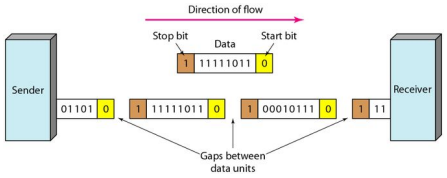
Data transmission and modes



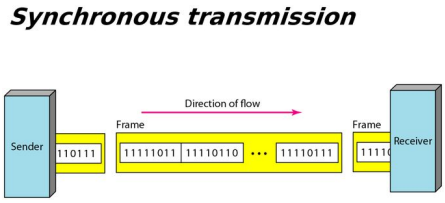
Parallel transmission



Serial transmission



5Digital Data Transmission1 — slide 34



Synchronous transmission

Quantization

- Divide the min–max amplitude range into L zones of height Δ; assign each zone the value of its **midpoint** (0 … L−1). Each sample is approximated to its zone midpoint.
- Bits per sample: $n_b = \log_2 L$. Quantization error = difference between the actual and coded (midpoint) value; more zones → smaller Δ → smaller error but higher bit rate.
- Signals with lower amplitude suffer more (the error range Δ/2 is fixed). Use **non-linear quantization** to keep the signal-to-quantization-noise ratio fixed: logarithmic levels, or **companding** (compress at the sender, expand at the receiver; fixed-height zones).

$$\text{Zone height: } \Delta = (\max - \min)/L \quad | \quad \text{bits per sample: } n_b = \log_2 L$$

- Example: $V_{min} = -20$ V, $V_{max} = +20$ V, L = 8 → Δ = 40/8 = 5; zones −20…−15, …, +15…+20 with midpoints −17.5, −12.5, …, 17.5; $n_b = 3$ bits (codes 000–111).

$$\text{PCM bit rate} = n_b \times f_s$$

- A digitized signal always needs more bandwidth than the original analog signal — the price paid for robustness.
- **PCM decoder**: a hold circuit keeps each pulse's amplitude until the next arrives; then a low-pass filter with cutoff = the highest pre-sampled frequency recovers the signal. Larger L gives less distortion.

Delta modulation

- Send only the difference between successive samples: 1 if the pulse rose, 0 if it fell. Works well for small changes; large changes cause large errors.
- **Delta PCM (DPCM)**: use more bits to quantize the difference — more bits, more levels, higher accuracy.

Transmission modes

Mode	How	Cost / note
Parallel	n wires send n bits per clock tick.	n× faster than serial but needs n lines — expensive; limited to short distances.
Serial — asynchronous	Byte-wise with a start bit (0) and stop bit(s) (1); gaps allowed.	No shared clock at byte level; timing unimportant, patterns used instead.
Serial — synchronous	Frames of many bytes, no gaps between bytes.	Faster (no extra bits/gaps); receiver must separate bits; for high-speed links.

Mode	How	Cost / note
Serial — isochronous	Guarantees a fixed data rate with equal gaps.	Needed for real-time audio/video where uneven frame delays are unacceptable.

- Parallel: computers produce/consume data in groups of bits; n bits go at once, so speed rises by a factor of n over serial, but cost rises too (n communication lines).
- Serial: one bit follows another over one channel; conversion devices (parallel-to-serial, serial-to-parallel) are needed at the interfaces.
- Asynchronous: at the byte level sender and receiver need not be synchronised; within each byte they must be (same bit duration). Synchronous: byte synchronisation is done at the data link layer; there may be uneven gaps between frames but none between characters. Isochronous: synchronisation of the entire bit stream, not just characters.

8. Analog Data Transmission (Modulation) & Spread Spectrum

Source decks: 6Analog Data Transmission, 8spread spectrum

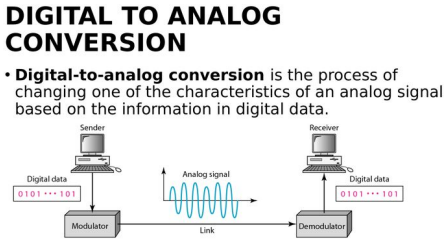
Analog transmission

- Analog transmission conveys information (voice, data, video, image) using a continuous signal whose amplitude/phase varies in proportion to a variable.
- Digital transmission is desirable but needs a low-pass channel; if only a band-pass channel is available, analog transmission is the only choice. Converting digital data to a band-pass analog signal is **digital-to-analog conversion**; converting a low-pass analog signal to a band-pass signal is **analog-to-analog conversion**.

Digital-to-analog conversion

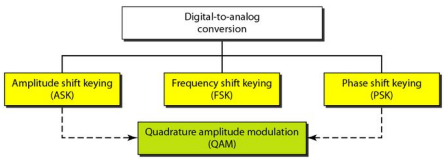
- A **carrier signal** of frequency f_c transports the digital data in an analog waveform; the carrier is manipulated to identify the digital data.
- A sine wave has three characteristics — amplitude, frequency, phase — so there are three mechanisms plus a combined one: **ASK**, **FSK**, **PSK**, and **QAM** (amplitude + phase, the most efficient and the one commonly used today).

```
Signal (baud) rate: S = N × (1/r) bauds   (r = data bits per signal element; S ≤ N)
```



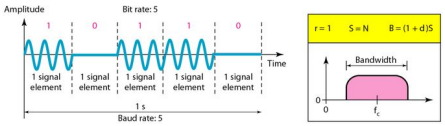
DIGITAL TO ANALOG CONVERSION

Types of digital-to-analog conversion



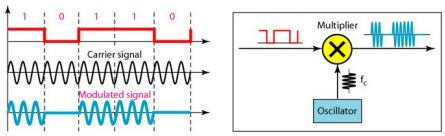
Types of digital-to-analog conversion

Binary amplitude shift keying(BASK)/ On-Off Keying(OOK)



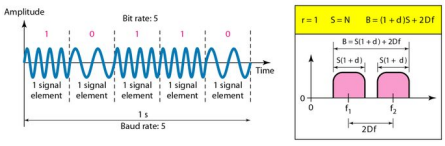
Binary amplitude shift keying(BASK)/ On-Off Keying(OOK)

Implementation of binary ASK



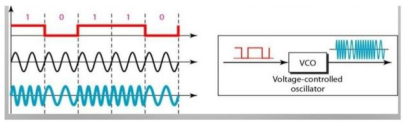
Implementation of binary ASK

Binary frequency shift keying



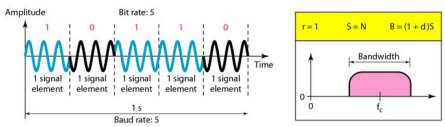
Binary frequency shift keying

Implementation of BFSK



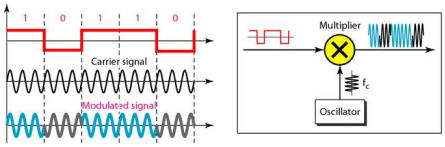
Implementation of BFSK

Binary phase shift keying



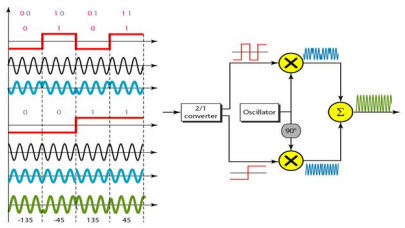
Binary phase shift keying

Implementation of BASK



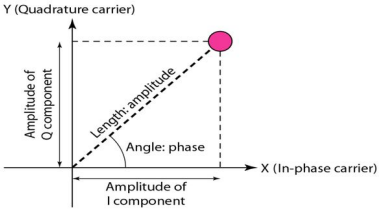
Implementation of BASK

QPSK and its implementation

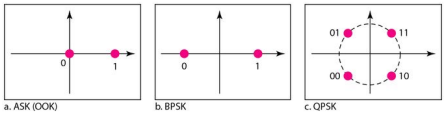


QPSK and its implementation

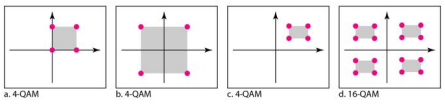
Concept of a constellation diagram



Three constellation diagrams



Constellation diagrams for some QAMs

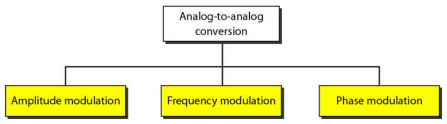


Concept of a constellation diagram

Three constellation diagrams

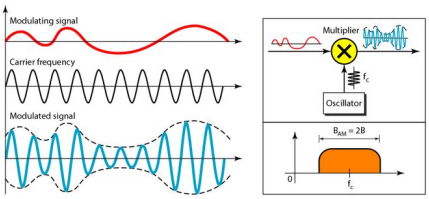
Constellation diagrams for some QAMs

Types of analog-to-analog modulation

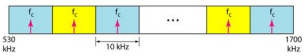


Types of analog-to-analog modulation

Amplitude modulation

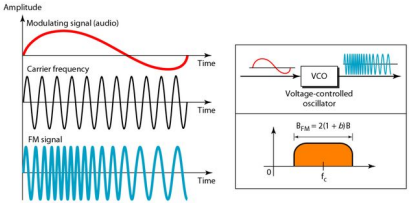


- The total bandwidth required for AM can be determined from the bandwidth of the audio signal: $B_{AM} = 2B$.
- *AM band allocation*



The total bandwidth required for AM can be determined from the bandwidth of the audio signal...

Frequency modulation

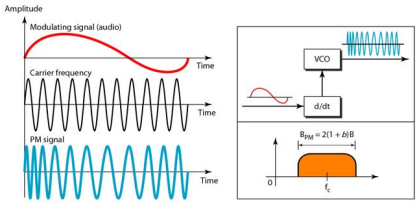


Frequency modulation

FM band allocation

FM band allocation

Phase modulation



Phase modulation

Amplitude Shift Keying (ASK)

- Change the carrier amplitude to reflect amplitude levels in the digital signal (e.g. 1 = signal present, 0 = no signal → On-Off Keying, OOK). The line encoding determines the analog waveform values.

```
Bandwidth of ASK: B = (1 + d) S   (d ∈ [0,1] due to modulation and filtering)
```

Frequency Shift Keying (FSK)

- The digital stream changes the carrier frequency: 1 → $f_1 = f_c + \Delta f$, 0 → $f_2 = f_c - \Delta f$.
- **Coherent FSK** switches frequency only at the same phase; **non-coherent FSK** does not adhere to the current phase.
- **Multi-level FSK** uses several frequencies, each representing a group of bits.

```
Bandwidth of FSK: B = (1 + d) S + 2Δf   |   Multi-level FSK: B = (1 + d)S + (L−1)/2 · Δf = L · S
```

Phase Shift Keying (PSK) and QAM

- Vary the phase of the carrier to represent data. PSK is more robust than ASK because it is not vulnerable to noise that changes amplitude.
- **QPSK** (quadrature PSK): parallelize the bit stream so every two bits modulate a carrier; one carrier is phase-shifted 90° (in quadrature) from the other; the two PSK signals are added → four signal elements (L = 4).
- **QAM** combines ASK and PSK; a constellation of amplitude/phase points.

```
Bandwidth of PSK: B = (1 + d) S
```

Constellation diagrams

- A constellation diagram defines amplitude and phase when two carriers (one in quadrature) are used: X-axis = in-phase carrier, Y-axis = quadrature carrier.
- ASK: two points on the X-axis (0 V and 1 V). BPSK: polar NRZ, two points on the X-axis (amplitudes +1 and −1, i.e. 180° apart). QPSK: four points of amplitude $2\sqrt{2}$ at phases 45°, 135°, −135°, −45°. QAM: fills a grid.

Analog-to-analog conversion

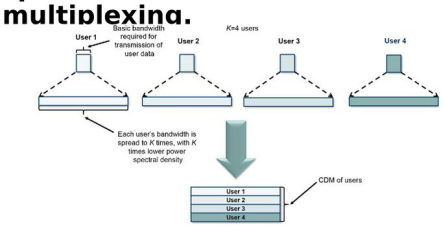
- Represents analog information with an analog signal; needed when the medium/channel is band-pass (e.g. radio, where each station's low-pass signal is shifted to a different band).

Scheme	What changes	Bandwidth
AM	Amplitude only (frequency & phase fixed); the modulating signal is the envelope.	B_{AM} = 2B (one sideband may be dropped)
FM	Frequency follows the modulating voltage (amplitude & phase fixed).	B_{FM} = 2(1 + β)B, β usually 4 (≈10× signal frequency)
PM	Phase follows the modulating voltage (amplitude & frequency fixed); the phase change appears as a frequency change proportional to the derivative of amplitude.	B_{PM} = 2(1 + β)B, β ≈ 2 (higher than AM)

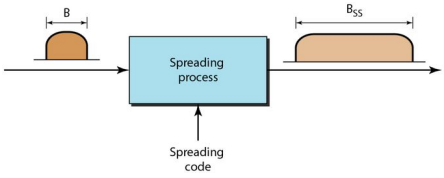
Spread spectrum

- Primary resources are **bandwidth** and **power**. Sometimes efficiency is sacrificed to meet other design objectives. Multiplexing combines sources to achieve bandwidth efficiency; spread spectrum combines sources to fit into a *larger* bandwidth for different goals.
- Designed for wireless, where concerns outweigh bandwidth efficiency: sharing the medium without eavesdropping or jamming.
- A spread-spectrum system transmits information over a bandwidth much wider than strictly necessary ($B_{ss} \gg B$), increasing robustness against narrowband interference. Two principles: (1) allocate far more bandwidth than needed (redundancy, a protective envelope); (2) the spreading must be done by a process *independent* of the original signal, using a **spreading code** (a pseudo-random sequence).

The principle of spread spectrum based user multiplexing.



The principle of spread spectrum based user multiplexing.



8spread spectrum — slide 7

Spread Spectrum

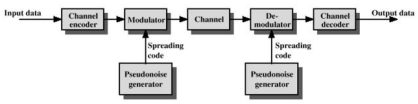


Figure 7.1 General Model of Spread Spectrum Digital Communication System

Spread Spectrum

Frequency Hoping Spread Spectrum

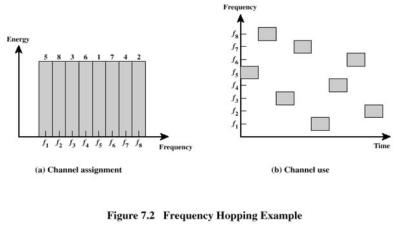
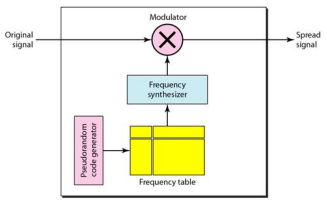


Figure 7.2 Frequency Hopping Example

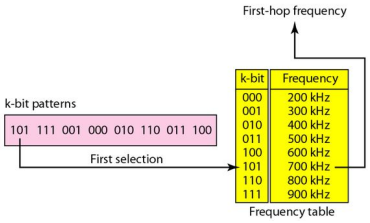
Frequency Hoping Spread Spectrum

Frequency hopping spread spectrum (FHSS)



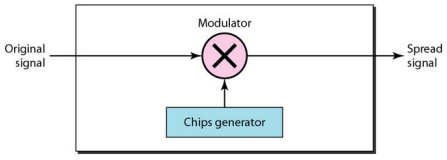
Frequency hopping spread spectrum (FHSS)

Frequency selection in FHSS



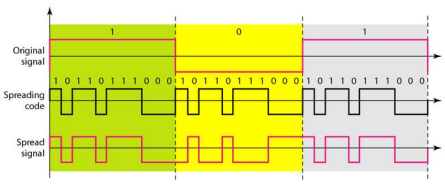
Frequency selection in FHSS

DSSS



DSSS

DSSS example



DSSS example

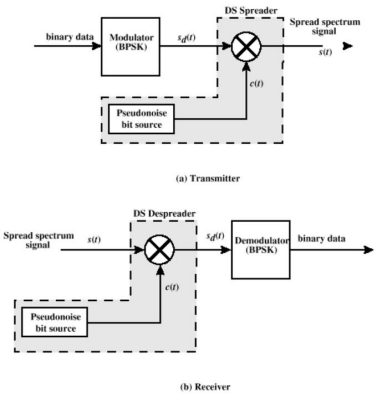


Figure 7.7 Direct Sequence Spread Spectrum System

8spread spectrum — slide 23

What is gained, and the types

- Gains: immunity from noise and multipath distortion; hiding and encrypting signals; several users can share the same higher bandwidth with little interference.

- The input is fed to a channel encoder (narrowband analog), then modulated by a spreading sequence from a pseudo-noise generator (widening the bandwidth); at the receiver the same sequence demodulates, then a channel decoder recovers the data.
- Goals: anti-jam (AJ), low probability of intercept (LPI), encryption, and multiple-access.

Type	Idea
FHSS (frequency hopping)	Rapidly hop the carrier among sub-bands in a code-controlled order; interference affects only a short interval. Enables CDMA; avoids interference/eavesdropping.
DSSS (direct sequence)	Each data bit is represented by multiple bits; the spreading code (XOR with the data stream) widens the spectrum; de-spreading at the receiver restores the information while reducing interference.
THSS (time hopping)	Pulse-position modulation; achieves AJ/LPI (2^k pulse positions to send k bits per pulse).
CSS (chirp)	Wideband linear frequency-modulated chirp pulses (frequency rises or falls over time).

- IEEE 802.11 uses either FHSS or DSSS in its radio interface. FHSS and DSSS use pseudorandom number sequences to control the spreading pattern.

9. Data Link Layer — Fundamentals

Source deck: 1Data Link Layer

Role and position

- The physical layer offers only a raw bit-stream service; the DLL turns it into a reliable link and controls activation, maintenance and deactivation of the link.
- Responsible for the error-free transfer of data frames, defines the data format on the network, and uniquely identifies each device on a LAN.
- Lowest layer that performs hop-by-hop (node-to-node) delivery; the network layer above does the end-to-end job.

IEEE 802 splits the DLL into two sublayers

Sublayer	Position	Responsibility
LLC — Logical Link Control	Upper	Flow control, error correction, resequencing for connection-oriented service; also supports connectionless. Hands packets up to the network layer and identifies the network-layer protocol from the header.
MAC — Media Access Control	Lower	A method for stations to gain access to the shared medium; the bridge between LLC and the physical layer; transfers packets over the network.

Functions of the DLL

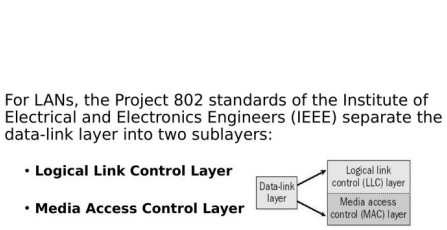
- **Framing** — group raw physical-layer bits into frames.
- **Physical addressing** — identify the sender/receiver on the link.
- **Error control** — detect and recover from transmission errors (CRC, ARQ).
- **Flow control** — keep a fast sender from drowning a slow receiver.
- **Access control** — decide which device controls the link at any instant.
- Provides a well-defined service interface to the network layer, plus general link management.

Services offered to the network layer

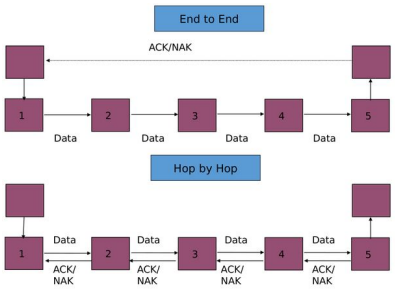
Service	Guarantees	Where it fits
Unacknowledged connectionless	No ACK; losses recovered at higher layers.	Reliable media (coax, fibre) with low error rate; suitable for voice where delay matters more than occasional bad data.
Acknowledged connectionless	Each frame acknowledged.	Unreliable media (wireless). ACKs add delay; useful as an optimisation, since resending a small frame beats resending a whole packet at the network layer.
Acknowledged connection-oriented	Most reliable: every frame arrives, exactly once, in order.	Guaranteed service; needs explicit setup/teardown and sequencing.

End-to-end vs hop-by-hop

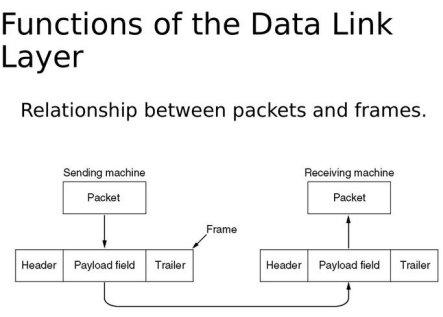
- In the **hop-by-hop** model each link acknowledges the previous node; in the **end-to-end** model only the final destination acknowledges. DLL reliability is hop-by-hop.



For LANs, the Project 802 standards of the Institute of Electrical and Electronics Engineers...

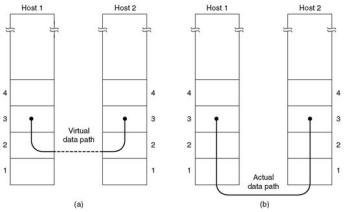


1Data Link Layer — slide 7



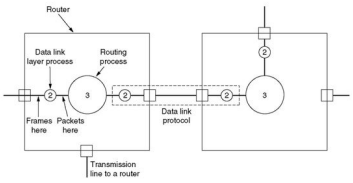
Functions of the Data Link Layer

Services Provided to Network Layer

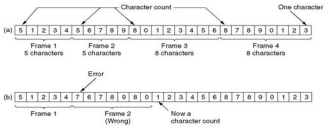


Services Provided to Network Layer

Placement of DLL

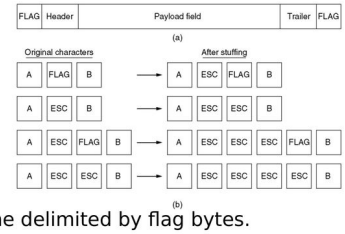


Placement of DLL



A character stream. (a) Without errors. (b) With one error.

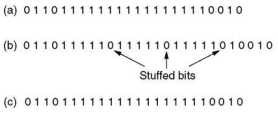
A character stream. (a) Without errors. (b) With one error.



(a) A frame delimited by flag bytes.
(b) Four examples of byte sequences before and after stuffing.

(a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after st...

Framing with bit stuffing



Bit stuffing
(a) The original data.
(b) The data as they appear on the line.
(c) The data as they are stored in receiver's memory after destuffing.

Framing with bit stuffing

Framing techniques

Framing decides how the bit stream is chopped into frames and how the receiver re-synchronises after an error.

Method	Idea	Weakness / note
Character (byte) count	A header field gives the number of bytes in the frame, so the receiver knows where it ends.	If the count is garbled by one bit flip the receiver loses sync; even with a checksum it cannot find the next frame start. No longer used.
Flag bytes with byte stuffing	Each frame begins and ends with the same special flag byte; a data byte equal to the flag is preceded by an escape byte (stuffing).	Assumes a fixed 8-bit character, so it cannot handle heterogeneous environments.
Flag bytes with bit stuffing	Flag pattern 01111110; the transmitter inserts a 0 after every five consecutive 1s of data; the receiver removes it after five 1s.	Works for arbitrary bit-oriented data; the basis of HDLC/SDLC.
Physical-layer coding violations	Reserved (redundant) signals that never occur in normal data delimit frames (e.g. high-high and low-low, since data uses high-low and low-high).	Only possible when the physical encoding has spare symbols.

- Byte stuffing example: data A ESC B FLAG C → transmitted A ESC ESC B ESC FLAG C (an escape is added before ESC and FLAG).
- With byte stuffing, two consecutive flag bytes indicate the end of one frame and the start of the next, so the receiver can always resynchronise by searching for flags.

```
General frame: Flag | Address | Control | Data (variable) | FCS | Flag   (flags delimit, FCS detects errors)
```

10. Error Detection: Parity, Checksum, CRC

Source deck: 2Error Control

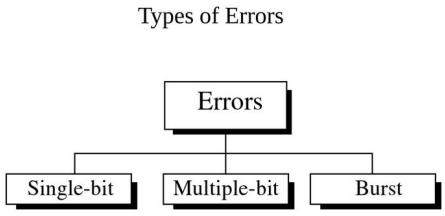
Error control basics

- Reliable delivery needs positive and negative feedback (ACK/NAK) plus timers. A timer rescues the case where a frame vanishes and neither ACK nor NAK arrives.
- Retransmission can deliver a frame more than once → give frames **sequence numbers**.
- Use error detection on reliable media (retransmission is cheap); use error correction on unreliable media (retransmission may fail again).
- Errors are handled at the data link or transport layer. Two strategies: add enough redundancy to deduce the original (error-correcting codes, i.e. **Forward Error Correction**), or add just enough to notice an error (error-detecting codes) and request retransmission.

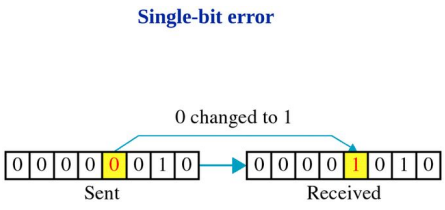
Error types

Type	Description	Notes
Single-bit	Exactly one bit flips.	Least likely in serial transmission — noise would need duration ≈ one bit time (e.g. 1 µs at 1 Mbps). Common in parallel transmission.
Multiple-bit	Two or more (non-consecutive) bits flip.	Occurs in both serial and parallel links.
Burst	A run of bits from the first to the last corrupted bit changes.	Most likely in serial links, because noise usually lasts longer than one bit. Length = distance from first to last corrupted bit; inner bits may be unharmed.

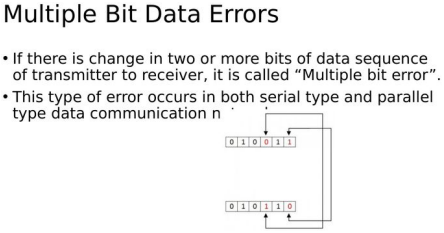
- Burst length depends on data rate: at 1 kbps a noise of 1/100 s affects 10 bits; at 1 Mbps the same noise affects 10,000 bits.



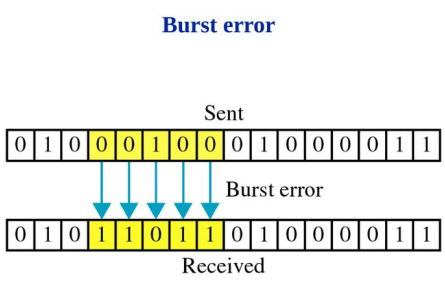
Types of Errors



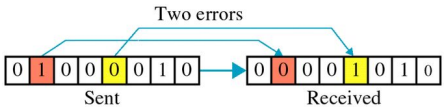
Single-bit error



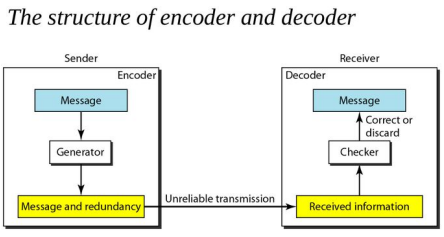
Multiple Bit Data Errors



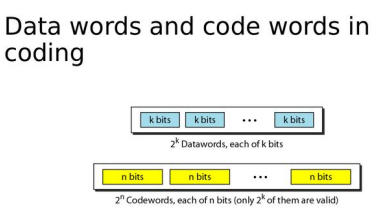
Burst error



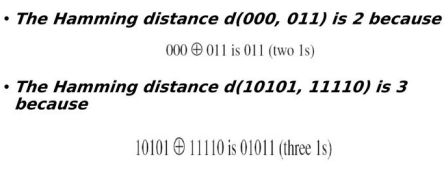
2Error Control — slide 11



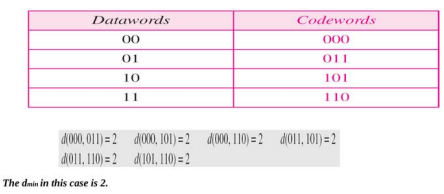
The structure of encoder and decoder



Data words and code words in block coding



The Hamming distance d(000, 011) is 2 because The Hamming distance d(10101, 11110) is 3 because



The d min in this case is 2.

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

$d(00000, 01011) = 3$ $d(00000, 10101) = 3$ $d(00000, 11110) = 4$
$d(01011, 10101) = 4$ $d(01011, 11110) = 3$ $d(10101, 11110) = 3$

The d_{min} in this case is 3.

The d min in this case is 3.

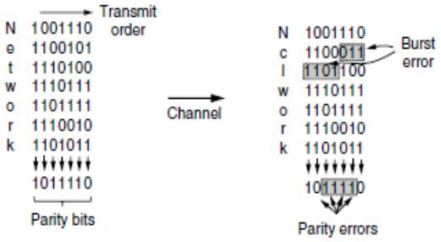


Figure 3-8. Interleaving of parity bits to detect a burst error.

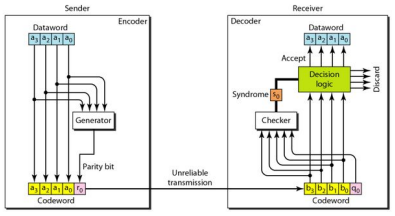
2Error Control — slide 39

Simple parity-check code C(5, 4)

Datawords	Codewords	Datawords	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

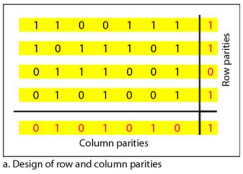
Simple parity-check code C(5, 4)

Encoder and decoder for simple parity-check code



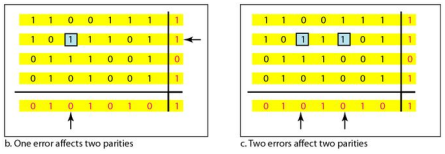
Encoder and decoder for simple parity-check code

Two-dimensional parity-check code



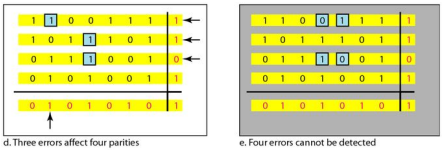
a. Design of row and column parities

Two-dimensional parity-check code



b. One error affects two parities
c. Two errors affect two parities

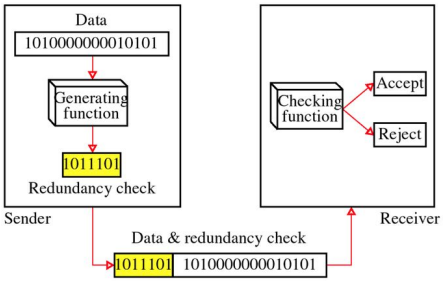
2Error Control — slide 46



d. Three errors affect four parities
e. Four errors cannot be detected

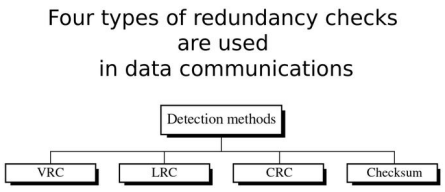
2Error Control — slide 47

Redundancy

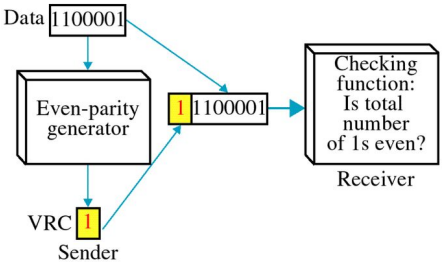


Redundancy

Four types of redundancy checks are used in data communications

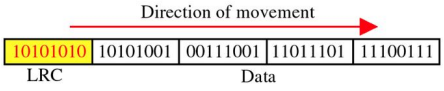


Vertical Redundancy Check VRC

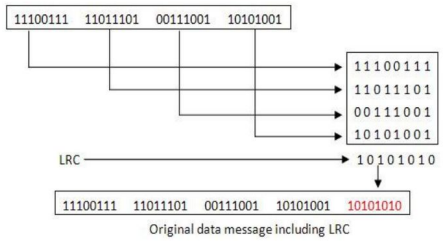


Vertical Redundancy Check VRC

Longitudinal Redundancy Check LRC

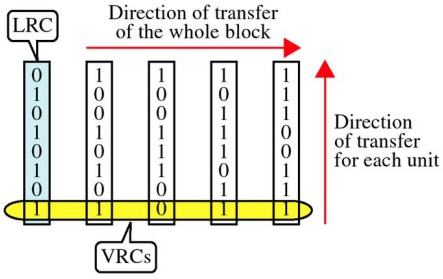


Longitudinal Redundancy Check LRC



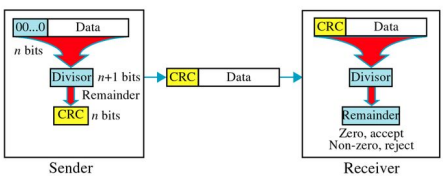
2Error Control — slide 54

VRC and LRC



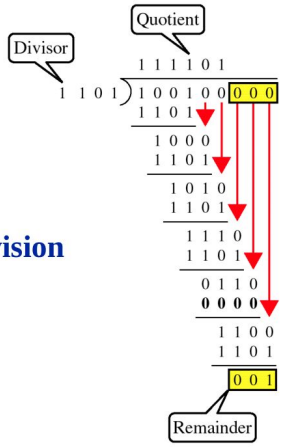
VRC and LRC

Cyclic Redundancy Check CRC

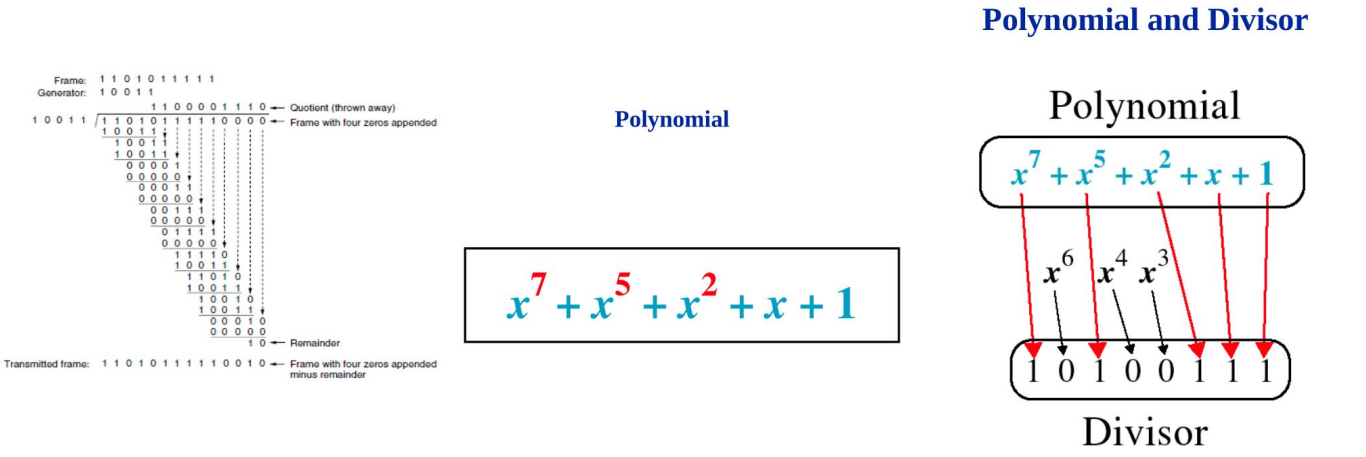


Cyclic Redundancy Check CRC

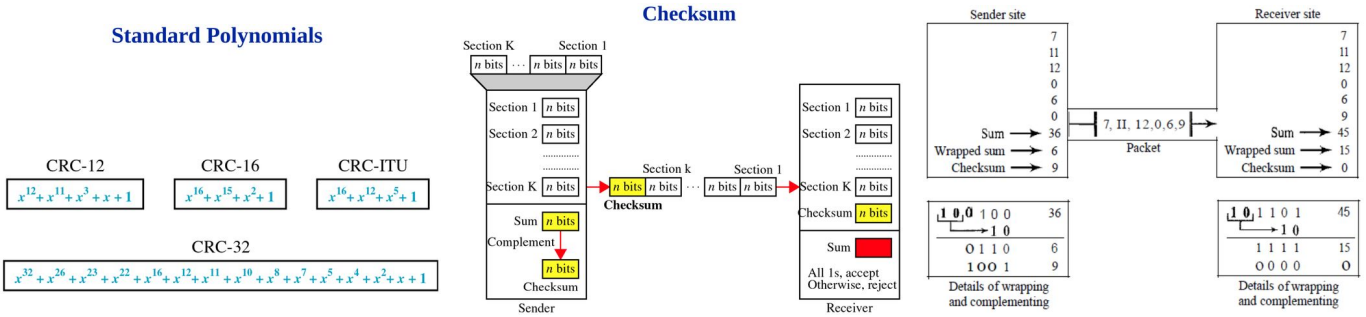
Binary Division



Binary Division



2Error Control — slide 63 *Polynomial* *Polynomial and Divisor*



Standard Polynomials *Checksum* *2Error Control — slide 70*

Ternary view: detect vs correct

- Detection only asks whether any error happened; a single-bit error is as bad as a burst.
- Correction needs to know the exact number of corrupted bits and their location — harder, so the message size and error count matter.
- Correction methods: FEC (the receiver guesses the message from redundant bits; best when errors are few) or retransmission (the receiver asks for the message again).
- Redundancy is the central idea: extra bits added by the sender, removed by the receiver. Block coding and convolutional coding are the two families; convolutional coding is more complex.

Modular arithmetic

- Work in a limited integer range 0…N−1 with modulus N; a value ≥ N is reduced to its remainder.
- Modulo-2 arithmetic uses only {0,1}: addition 0+0=0, 0+1=1, 1+0=1, 1+1=0; subtraction is identical → both are **XOR**.

```
Modulo-2: 0+0=0, 0+1=1, 1+0=1, 1+1=0   (= XOR, and subtraction identical)
```

Block coding & Hamming distance

- Split the message into k-bit data words; add r redundant bits to make n = k + r bit code words. Encoding is one-to-one, so $2^n - 2^k$ code words go unused.
- A receiver detects a change only if it holds the valid code-word list and the received word is invalid.
- Hamming distance d(x,y) = number of differing bit positions = number of 1s in x ⊕ y.
- Minimum Hamming distance d_{min} = smallest distance over all pairs. A scheme is written C(n, k) with an expression for d_{min}.
- Linear block code: XOR of any two valid code words is again valid; d_{min} = number of 1s in the lightest nonzero code word.

```
Detect up to s errors → d_min ≥ s + 1   |   correct d errors → d_min = 2d + 1
```

Error-detecting codes (all linear, systematic, block)

Code	How it works	Strength
Parity (VRC)	One extra bit makes the count of 1s even (or odd); equivalently XOR of data bits.	$d_{min} = 2$, detects all single-bit errors; a long burst is caught with only probability ½.
Two-dimensional parity	Parity per row and per column; interleaving spreads a burst across columns.	Isolates and can even correct a single bit; catches most bursts.
Longitudinal (LRC)	Arrange data in rows/columns, send parity of each column.	Detects burst and single-bit errors; fails only for 2-bit errors in the same vertical slice.
Checksum	Arithmetic sum of message words (one's complement), negated and appended.	Detects all odd-error cases and most even ones; weakest for compensating bit pairs.

Cyclic Redundancy Check (CRC)

- Treat a block of data as coefficients of a polynomial and divide by a fixed generator polynomial G(x); both high- and low-order bits of G(x) are 1.
- The transmitter generates an n-bit **frame check sequence (FCS)** so the whole (k + n)-bit frame is exactly divisible by G(x). The receiver divides again; a nonzero remainder means an error.

CRC algorithm (modulo-2 division)

- Let r = degree of G(x). Append r zero bits to the frame (now m + r bits, polynomial $x^r M(x)$).
- Divide the $x^r M(x)$ bit string by G(x) using modulo-2 division.
- Subtract the r-bit remainder (modulo-2) from $x^r M(x)$; the result is the transmitted frame.

```
Transmitted frame = x^rM(x) ⊕ remainder   →   divisible by G(x); receiver remainder = 0 ⇔ no error
```

Checksum (incl. Internet 16-bit checksum)

- The checksum is a group of check bits; placed at the end of the message as the complement of the sum. Adding the whole received codeword should give zero.
- Sender: initialise 0 → add all data items (checksum counts as one) → wrap the carry (add surplus bits back) → complement → send.
- Receiver: add all items including checksum → wrap → complement; result 0 means clean, otherwise drop the packet.
- Internet checksum uses 16-bit words and one's-complement addition; the complement of the sum is the checksum.
- Detects all errors involving an odd number of bits and most even ones; fails when compensating bit pairs cancel.

```
Sender: checksum = complement(wrapped one's-complement sum)   |   receiver accepts iff recomputed value = 0
```

11. Error Correction Codes

Source deck: 3Error Correction

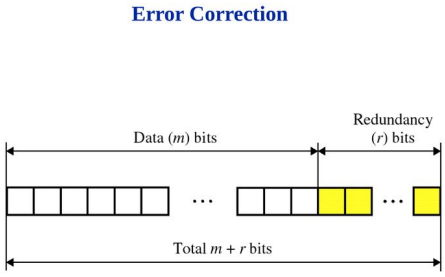
Two ways to correct

- Ask the sender to retransmit the whole data unit, or use an error-correcting code that automatically fixes certain errors.
- A frame carries m data bits + r redundant (check) bits. A **linear** code computes the r check bits as a linear (XOR) function of the m data bits; a **systematic** code sends the m data bits unchanged alongside the check bits; a **block** code computes r solely from the m bits of the same block.
- Four families of error-correcting codes: **Hamming**, **binary convolutional**, **Reed-Solomon**, **low-density parity-check (LDPC)**. All add redundancy.

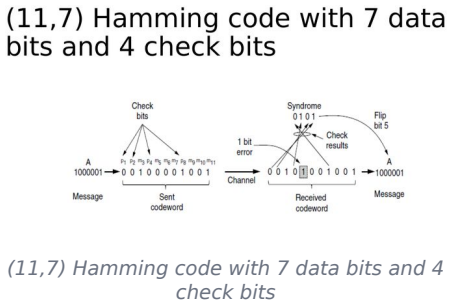
How many redundancy bits? (single-bit correction)

To correct a single error and identify which bit, m data + r check bits must satisfy:

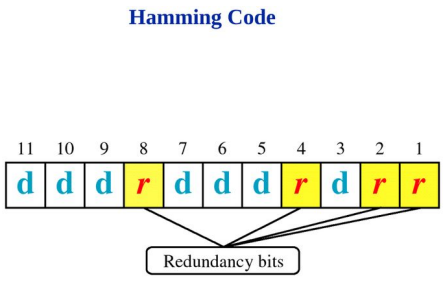
$$2^r \geq m + r + 1 \quad \text{(total message sent = } m + r\text{)}$$



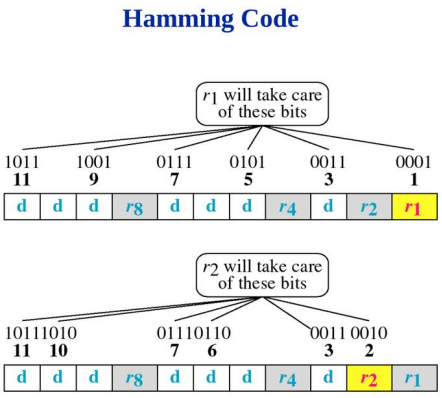
Error Correction



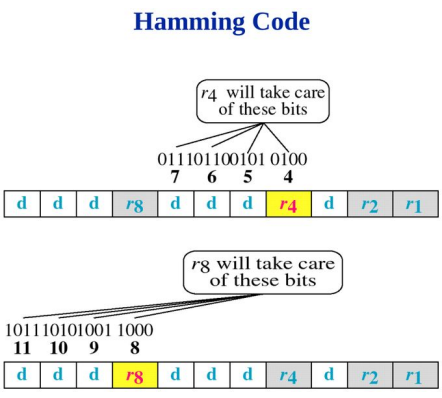
(11,7) Hamming code with 7 data bits and 4 check bits



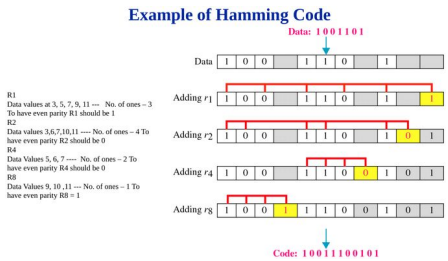
Hamming Code



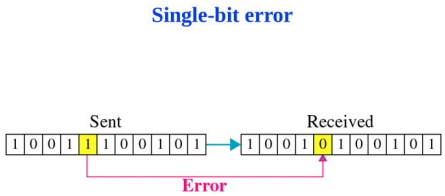
Hamming Code



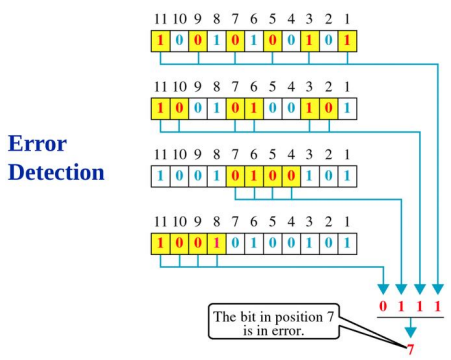
Hamming Code



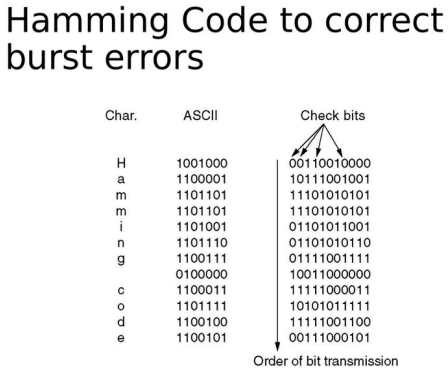
R1 Data values at 3, 5, 7, 9, 11 --- No. of ones – 3 To have even parity R1 should be 1 R2 D...



Single-bit error

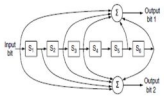


Error Detection



Hamming Code to correct burst errors

• Convolutional codes are widely used in deployed networks, for example, as part of the GSM mobile phone system, in satellite communications, and in 802.11.



The NASA binary convolutional code used in 802.11.

Convolutional codes are widely used in deployed networks, for example, as part of the GSM mo...

Hamming codes

- Number codeword bits from 1 leftward; positions that are powers of 2 (1, 2, 4, 8, 16, …) are check bits; the rest carry data bits.
- Each check bit forces the parity of some collection of bits (including itself) to be even (or odd).
- A data bit at position k is checked by exactly the check bits appearing in the power-of-2 expansion of k — e.g. 11 = 1 + 2 + 8, so bit 11 is checked by bits 1, 2, 8; 29 = 1 + 4 + 8 + 16.
- The (11, 7) Hamming code has 7 data bits and 4 check bits (m = 7, r = 4, $2^4 = 16 \geq 12$).
- Performing the checks yields a nonzero **error syndrome** that pinpoints the bad bit — e.g. check results 0,1,0,1 give syndrome 0101 = 4 + 1 = 5, so bit 5 is wrong.

Check	Data positions covered	Count of ones	To make parity even
R1	3, 5, 7, 9, 11	3	R1 = 1
R2	3, 6, 7, 10, 11	4	R2 = 0
R4	5, 6, 7	2	R4 = 0
R8	9, 10, 11	1	R8 = 1

Use of HD for correction: to correct d errors, need code list with $d_{min} \geq 2d + 1$ (e.g. HD = 5 corrects 2 single-bit errors)

Convolutional codes

- Not a block code: the encoder processes a sequence of input bits and generates a sequence of output bits. There is no natural message size or encoding boundary.
- The output depends on the current and previous input bits — the encoder has **memory**. The number of previous bits on which the output depends is the **constraint length**. Codes are specified by rate and constraint length.
- The NASA binary convolutional code has rate r = 1/2 and k = 7 (used on Voyager from 1977, and in 802.11): each input bit produces two output bits (XOR sums of input and internal state). It is not systematic (no output bit is simply the input bit).
- The internal state is kept in six memory registers; each new input shifts them right. Seven shifts flush an input completely, so k = 7.
- Decoded by the **Viterbi** algorithm: walk the observed sequence, keeping for each step and each state the input sequence producing the fewest errors; the sequence with fewest errors at the end is the most likely message.

Reed-Solomon codes

- Linear, often systematic, block codes that operate on m-bit **symbols** (not single bits).
- Based on the fact that an n-degree polynomial is fixed by n + 1 points: send data points plus check points on the same curve; if one arrives in error, fit the curve to recover it.
- For m-bit symbols, codewords are $2^m - 1$ symbols; with m = 8 (bytes) a codeword is 255 bytes. The (255, 233) code adds 32 redundant symbols to 233 data symbols.
- With 2t redundant symbols, up to t symbol errors are corrected. A single-bit error and an m-bit burst both count as one symbol error.
- Decoded by the **Berlekamp-Massey** algorithm; used in DSL, cable, satellite, CD/DVD/Blu-ray. Often nested inside a convolutional code to mop up bursts.

LDPC codes

- Linear block codes (Gallagher, 1962) where each output bit depends on only a fraction of the input bits, so the parity-check matrix has a low density of 1s.

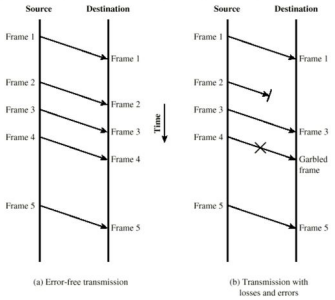
- Decoded by an iterative approximation that converges on the best-fit legal codeword.
- Practical for large blocks; excellent error-correction that beats many alternatives, so it is rapidly entering new protocols.

12. Data Link Protocols: Flow Control, Stop-and-Wait & ARQ

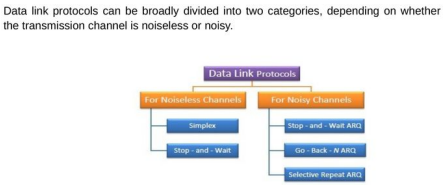
Source deck: 4Data Link Layer Protocols

- Flow control stops the sender overwhelming the receiver (buffer overflow). **Transmission time** = time to emit all bits; **propagation time** = time for a bit to traverse the link.
- Protocols split by channel: noiseless vs noisy.

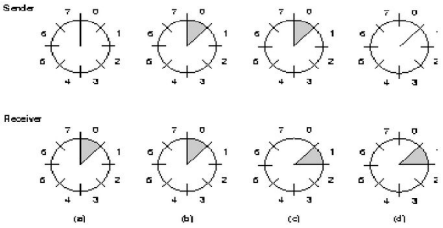
Model of Frame Transmission



Model of Frame Transmission

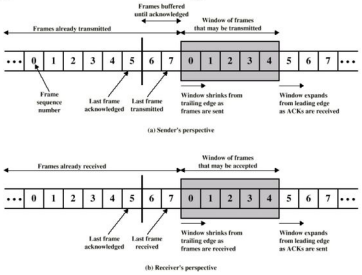


Data link protocols can be broadly divided into two categories, depending on whether the tra...



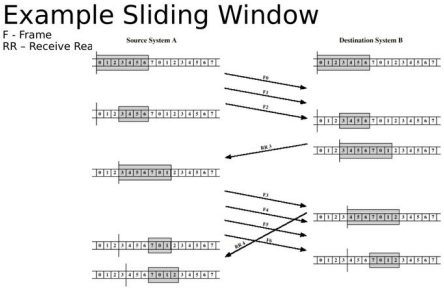
4Data Link Layer Protocols — slide 21

Sliding Window Diagram

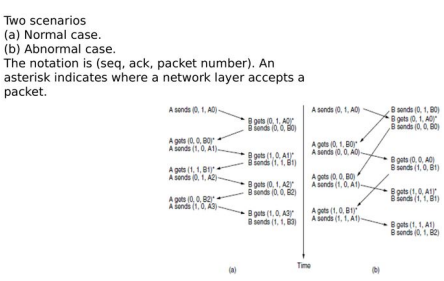


Sliding Window Diagram

Example Sliding Window



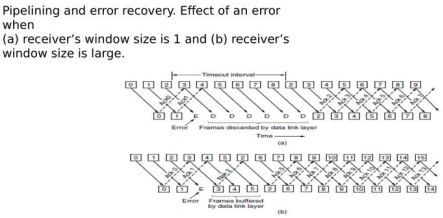
Example Sliding Window F - Frame RR – Receive Ready



Two scenarios (a) Normal case. (b) Abnormal case. The notation is (seq , ack , packet numbe...

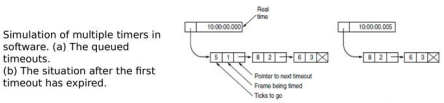
• Sliding Window Protocol(Duplex)
  • At any instant of time, the sender & receiver maintain a set of sequence numbers corresponding to frames it is permitted to send/receive
  • Piggybacking
  • Pipelining

Sliding Window Protocol(Duplex) At any instant of time, the sender & receiver maintain a set...



Pipelining and error recovery. Effect of an error when (a) receiver's window size is 1 and (...

Out-of-order buffering not mandatory at destination
• Maximum Window = Sequence Number Space - 1
• N timers or 1 timer may be used



Out-of-order buffering not mandatory at destination Maximum Window = Sequence Number Space -...

Protocols

Protocol	Assumptions	Behaviour
Unrestricted simplex (Utopia)	Infinite receiver buffer, error-free, sender always ready, one direction.	Framing only — no error or flow control. Hypothetical.
Stop-and-wait (noiseless)	Finite buffer & processing speed; still error-free.	Sender sends one frame and waits for the receiver to signal it is ready — flow control only. Good for a few large frames.
Stop-and-wait ARQ (noisy)	Finite buffer; errors/losses possible.	Positive ACK + retransmission: sender keeps a copy, waits a finite time for ACK; on timeout or NAK, retransmits; on ACK, sends the next frame.

ARQ building blocks

- Positive ACK; a 1-bit sequence number in frames (not in ACKs); a timeout to detect lost frames/ACKs; retransmission. Can fail under early timeout.
- Full-duplex works with **piggybacking**: temporarily delay an ACK so it can ride on the next outgoing data frame.
- **Fragmentation**: split large blocks into small frames to bound buffer size, detect errors sooner, retransmit less on error, and stop one station hogging the medium. (Plain stop-and-wait then becomes inadequate.)

Sliding-window flow control

- Allow multiple frames in transit: the receiver has a buffer of W frames; the sender may send up to W frames before an ACK.
- Each frame is numbered; the ACK carries the number of the next frame expected. Sequence numbers are bounded by the field size k and wrap modulo 2^k.
- Sender window grows when an ACK arrives; receiver window is constant and advances when the low-edge sequence number arrives; frames outside the window are discarded.
- The three sliding-window protocols: one-bit (stop-and-wait + error control, piggybacked ACKs), Go-Back-N, and Selective Repeat.

Goal	Window	ACK style	On error
Go-Back-N	Send up to N frames.	Cumulative ACK (ACK N if all < N received); out-of-order packets discarded, no receiver buffering.	Retransmit all frames from the earliest unacked one — wastes bandwidth if error rate is high.
Selective Repeat	Retransmit only the erroneous/lost frame; good frames buffered.	Not cumulative (individual ACKs); out-of-order arrival handled.	Only the bad frame resent on NAK; loses time if frames are lost in bursts.

```
Selective Repeat max window = (Sequence Number Space)/2 with out-of-order buffering, else Sequence Number Space − 1
```

Sliding-window enhancements & timers

- Receiver can send **Receive-Not-Ready** to stop flow without a data ACK, then a normal ACK to resume.
- **Piggybacking** with an auxiliary ACK timer: set the timer when a packet needing an ACK arrives; if no reverse data appears, generate an ACK when it fires.
- **Negative ACK (NAK)** on an out-of-order frame triggers retransmission of that frame; NAKs are not cumulative and duplicate NAKs are not sent.
- Selective Repeat may use N timers or a single timer simulated in software (queued timeouts).
- Pipelining: with a receiver window of 1 a single error forces retransmission of the whole window (Go-Back-N); a large receiver window lets good frames be buffered (selective repeat).

13. SDLC & HDLC

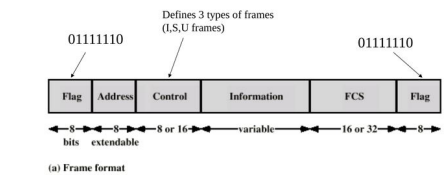
Source deck: 5SDLC & HDLC

SDLC

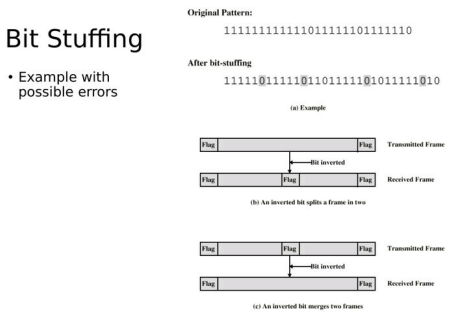
- Synchronous Data Link Communication (IBM): a synchronous **bit-oriented** protocol (continuous stream of bits); simplex/half/full duplex; sliding window; EBCDIC language; 256-character data frame; all link control in a single control field; master-slave (primary/secondary) only.
- Primary issues commands; secondary returns responses.
- States: **transient** (before/after transmission and line turnaround), **idle** (15+ consecutive 1s), **active** (transmitting information or control).

```
General frame: Flag(8) | Address(8) | Control(8) | Data (multiple of 8) | FCS(16) | Flag(8)
```

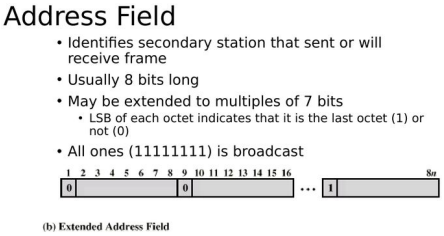
Field	Purpose
Flag	Marks beginning/end of a frame; helps detect frame boundaries and synchronisation.
Address	Identifies the secondary station involved; supports individual, group and broadcast addressing.
Control	Defines the function/type: Unnumbered (U) — link setup, disconnect, control; Supervisory (S) — acknowledgements, flow control, error reporting; Information (I) — data transfer and sequence control.
Information/Data	Optional user data.
FCS	Frame Check Sequence for error detection; ensures integrity.



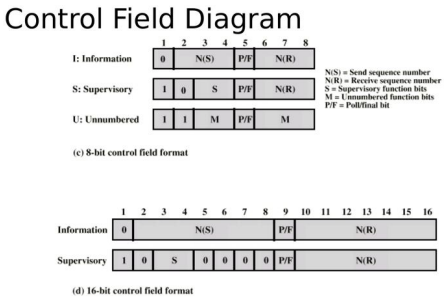
Frame Structure



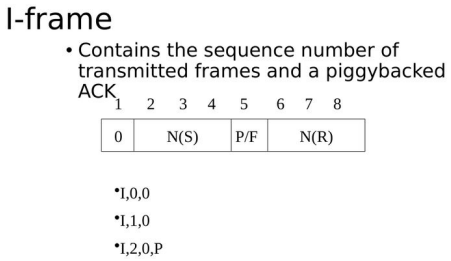
Bit Stuffing



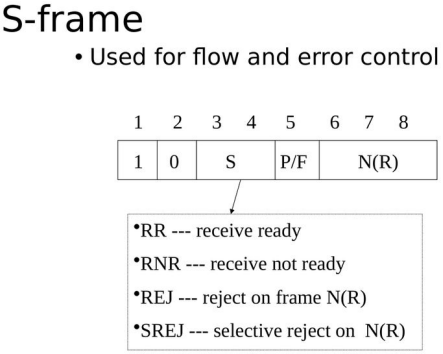
Address Field



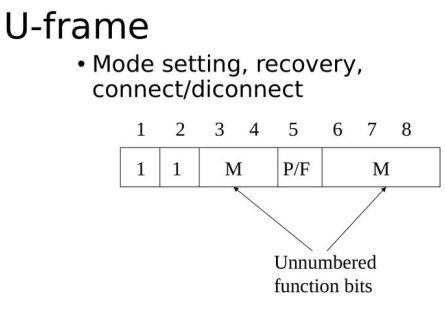
Control Field Diagram



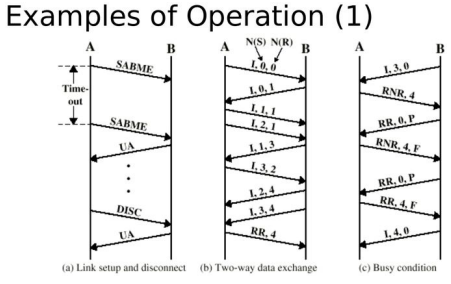
I-frame



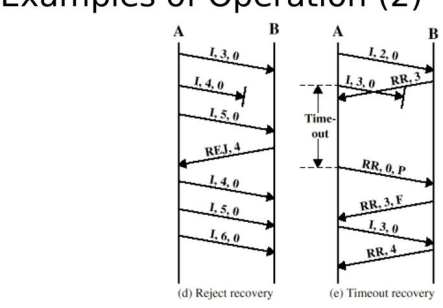
S-frame



U-frame



Examples of Operation (1)



Examples of Operation (2)

HDLC

- High-level Data Link Control — a discipline for managing information transfer over a data communication channel; defines primary/secondary responsibilities, information grouping for control/checking, and the frame format.
- The primary is responsible for organising data flow and link-level error recovery; a frame from primary = **command**, from secondary = **response**. Normally a command expects a response.

Station types and link configurations

Type	Role
Primary station	Controls operation of the link; issues commands; maintains a separate logical link to each secondary.
Secondary station	Under control of the primary; issues responses.
Combined station	May issue both commands and responses.

- **Unbalanced** configuration: one primary and one or more secondaries. **Balanced**: two combined stations. Both support full and half duplex.

Frame types

Frame	Purpose
I-frame (Information)	Carries user data; flow and error control piggybacked on information frames.
S-frame (Supervisory)	Flow and error control: RR (Receive Ready), RNR (Receive Not Ready), REJ (Reject on N(R)), SREJ (Selective Reject on N(R)).
U-frame (Unnumbered)	Link management and supplementary control: mode setting, connect/disconnect.

Frame fields

- **Flag** = 01111110; bit stuffing inserts a 0 after every five 1s. On five 1s the receiver checks the next bit: 0 → delete it; 10 → accept as flag; 11 → the sender is indicating **abort**.
- **Address**: identifies the secondary that sent or will receive the frame; usually 8 bits, extensible to multiples of 7 bits; the LSB of each octet says whether it is the last octet (1) or not (0); all-ones = broadcast.
- **Control**: first one or two bits identify the frame type (I/S/U).
- **P/F bit**: in a command frame P = poll (solicit a response from the peer); in a response frame F = final (response to a soliciting command).
- **Information**: only in I-frames and some U-frames; integral number of octets; variable length.
- **FCS**: error detection; 16-bit CRC (optional 32-bit).

Unnumbered frames

- SNRM (Set Normal Response Mode), SARM (Set Asynchronous Response Mode), SABM (Set Asynchronous Balanced Mode), DISC (Disconnect), UA (Unnumbered Acknowledgement), DM (Disconnect Mode), RD (Request Disconnect), UP (Unnumbered Poll), RSET (Reset), XID (Exchange Identification), TEST, FRMR (Frame Reject).

Data link channel states

State	Meaning
Active	The primary or secondary is actively transmitting a frame, a single abort sequence or interframe time fill; the right to continue transmission is reserved.
Abort	A station aborts a frame by transmitting at least seven contiguous ones; the receiver ignores the frame.
Interframe time fill	Continuous flag bytes are sent between frames.
Idle	15 or more contiguous 1 bits detected; a primary must repoll a secondary before transmitting an I-frame.

Modes of operation

Mode	Behaviour
NRM (Normal Response Mode)	A secondary transmits only in response to a command from the primary; the last frame is marked with the P/F bit; the secondary cannot transmit again until polled.

Mode	Behaviour
ARM (Asynchronous Response Mode)	A secondary may initiate transmission without explicit permission; it is responsible for time-out and retransmission.
ABM (Asynchronous Balanced Mode)	Stations have identical protocols; both may initiate at any time and send commands and responses.

- HDLC operation has three phases: **initialization**, **data transfer**, **disconnect**.

14. Network Layer, Internet Protocol & Fragmentation

Source decks: 1Network Layer, 2INTERNET PROTOCOL (IP)

What the network layer does

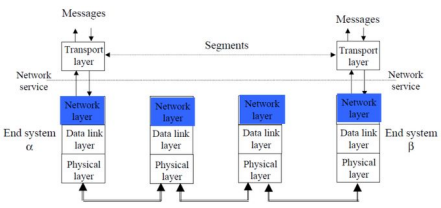
- Gets packets from source all the way to destination, possibly across many hops through intermediate routers — the lowest layer that handles end-to-end transmission.
- **Routing / path determination**: choose the route packets take (routing algorithms).
- **Inter-networking**: the same network layer spans differently-physical networks; headers are stripped and the packet is passed between dissimilar links.
- **Congestion control**: when incoming rate at a router exceeds outgoing rate. Causes: many inputs competing for one output; infinite memory (queued packets time out and their duplicates add load); slow CPUs; low-bandwidth lines.
- **Switching**: move a packet from the router input to the correct output.
- **Call setup**: some architectures set up a path before data flows.

Design issues & services to the transport layer

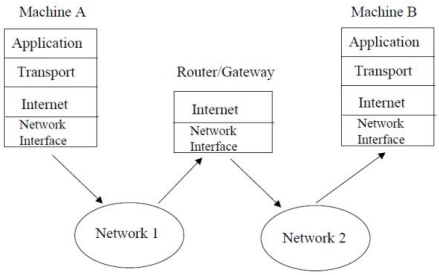
Service should be…	Meaning
Independent of router technology	The transport layer must not depend on how routers are built.
Shielded from router details	Hide the number, type and topology of routers.
Uniform numbering plan	Addresses must be consistent across LANs and WANs.

Store-and-forward packet switching

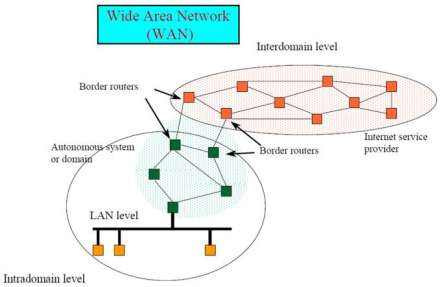
- A host sends a packet to its nearest router (on its LAN or over a point-to-point link). The router stores the packet until it has fully arrived (so the checksum can be verified), then forwards it toward the destination. Data is split into small packets transmitted node to node, processed and forwarded.



1Network Layer — slide 3



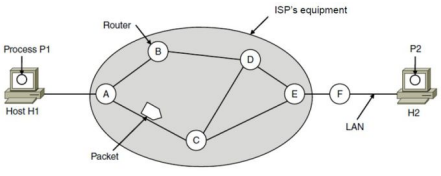
1Network Layer — slide 4



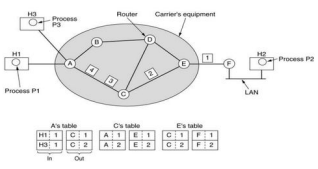
1Network Layer — slide 5

Routing within Virtual Circuit subnet

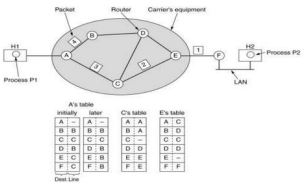
Routing within a datagram subnet



1Network Layer — slide 13



Routing within Virtual Circuit subnet



Routing within a datagram subnet

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Connectionless (datagram) vs connection-oriented (virtual circuit)

Aspect	Datagram subnet (connectionless)	Virtual-circuit subnet (connection-oriented)
Setup	None; each packet routed independently.	A path is established before data; a route is chosen and stored in router tables at setup.
Packet addressing	Each packet carries the full destination address.	Each packet carries a short virtual-circuit identifier.
Ordering	Packets may take different paths and arrive out of order.	Arrive in order along the reserved path.
Release	—	Releasing the connection terminates the circuit.
Trade-off	Address parsing time and per-packet overhead.	Setup time and resources, but reserves resources up front for QoS and congestion avoidance.

Routers

- On each arriving packet a router strips the frame header/trailer, passes the payload to routing software, which uses an internal table (per destination) to pick an output line. The algorithm that builds and maintains these tables is the routing algorithm.

Internet Protocol (IP)

- One of the major TCP/IP protocols, working at the network layer (OSI) / Internet layer (TCP/IP).
- Provides a **connectionless (datagram)** service: each packet treated separately. Packets at this layer are **datagrams** — header + data.
- Header is 20–60 bytes, customarily drawn in 4-byte rows. IP gives **best-effort delivery**: no guarantee, but it does its best.
- Common to all routers; identifies hosts by logical addresses (IPv4 = 32-bit) and routes data among them.

IP design issues

Issue	Key points
Routing	Static tables (may hold alternates) or dynamic tables updated for congestion/errors; source routing (source lists trusted routers) and route recording (routers append their address, for tracing/debugging).
Datagram lifetime	TTL field prevents indefinite looping: hop count (decrement per router) or time count (needs a global clock); on expiry the datagram is discarded, not forwarded.
Fragmentation / reassembly	Networks have different max packet sizes; reassembly in IP is at the destination only.
Error control	Delivery not guaranteed; a router may notify the source of a drop (TTL expiry, congestion, bad checksum). Notifications are not guaranteed.
Flow control	Limited in connectionless systems — e.g. the ICMP source quench packet asks for reduced flow.
Addressing	Each end and intermediate system gets a network-level address (IP address; OSI calls it NSAP). Each process gets a port (SAP in OSI).

IPv4 datagram header

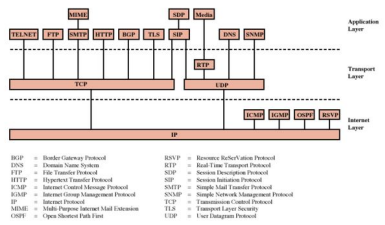
```
Ver | IHL | DSCP/ECN | Total Length | Identification | Flags | Frag Offset | TTL | Protocol
                | Header Checksum | Source IP | Destination IP | Options+Pad
```

Field	Meaning
Version	IP version used (IPv4 = 4).
IHL	Internet header length in 32-bit words, including options; min 5 (20 octets).
DSCP / ECN	Differentiated services code point (was Type of Service) for QoS; ECN signals congestion seen on the route.
Total Length	Whole IP packet (header + payload) in octets; max datagram = 65,535.

Field	Meaning
Identification	16-bit value identical across all fragments of one original packet, so the receiver can group them.
Flags	More-fragments (1 if more follow) and Don't-Fragment.
Fragment Offset	Position of the fragment's data in the original datagram, in multiples of 8 octets (64 bits).
TTL	Hops the packet may still cross; decremented each hop, discarded at 0.
Protocol	Next higher-layer protocol (ICMP = 1, TCP = 6, UDP = 17).
Header Checksum	Checksum of the header only; verified and recomputed at each router.
Source / Destination Address	32-bit sender / receiver addresses.
Options & Padding	Present when IHL > 5 (Security, Record Route, Timestamp, …); padding brings the header to a multiple of 32 bits.

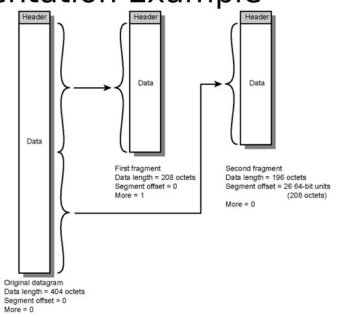
- Data field: any octet length, bounded by the 65,535-octet datagram limit.

Protocols of TCP/IP Protocol Suite



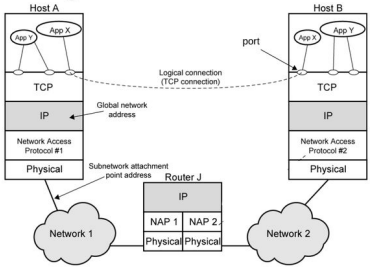
Protocols of TCP/IP Protocol Suite

Fragmentation Example

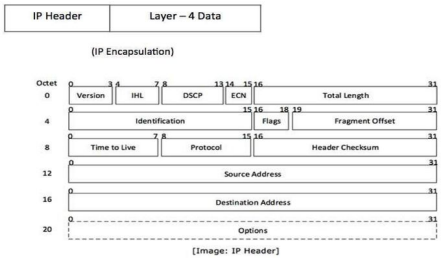


Fragmentation Example

Addressing in TCP/IP



Addressing in TCP/IP



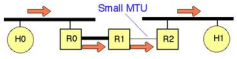
2INTERNET PROTOCOL (IP) — slide 18

- Options
  - This is optional field, which is used if the value of IHL is greater than 5. These options may contain values for options such as Security, Record Route, Time Stamp, etc.

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

- Padding
  - To fill to multiple of 32 bits long

Options This is optional field, which is used if the value of IHL is greater than 5. These o...

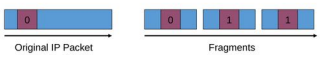


- Two Approaches
- IP Router Segmentation - performing the fragmentation in the routers
- IP Path MTU Discovery - forcing the sender to perform the fragmentation

Two Approaches IP Router Segmentation - performing the fragmentation in the routers IP Path...

Fragmentation and IP Fields

- *More Fragments* field (1 bit)
  - 1 if more fragments
  - 0 if not
  - Source host internet process sets to 0
  - If router fragments, sets More Fragments field in last fragment to 0
  - In all other fragments, sets to 1



Fragmentation and IP Fields

Identification Field

- IP packet has a 16-bit *Identification* field

Version (4)	Hdr Len (4)	TOS (8)	Total Length in bytes (16)
Indication (16 bits)		Flags (3)	Fragment Offset (13)
Time to Live (8)	Protocol (8)	Header Checksum (16)	
Source IP Address			
Destination IP Address			
Options (if any)			PAD
Data Field			

2INTERNET PROTOCOL (IP) — slide 34

Identification Field

- IP packet has a 16-bit *Identification* field
  - Source host internet process places a random number in the Identification field
  - Different for each IP packet

Version (4)	Hdr Len (4)	TOS (8)	Total Length in bytes (16)
Indication (16 bits)		Flags (3)	Fragment Offset (13)
Time to Live (8)	Protocol (8)	Header Checksum (16)	

Identification Field

Identification Field

- IP packet has a 16-bit Identification field
  - If router fragments, places the original Identification field value in the Identification field of each fragment

Identification Field

Identification Field

- Purpose
  - Allows receiving host's internet layer process know what fragments belong to each original packet
  - Works even if an IP packet is fragmented several times

Identification Field

Fragment Offset Field

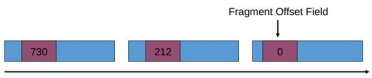
- Fragment offset field (13 bits) is used to reorder fragments with the same Identification field
- Contains the data field's starting point (in octets) from the start of the data field in the original IP packet

Version (4)	Hdr Len (4)	TOS (8)	Total Length in bytes (16)
Indication (16 bits)		Flags (3)	Fragment Offset (13)

2INTERNET PROTOCOL (IP) — slide 38

Fragment Offset Field

- Receiving host's internet layer process assembles fragments in order of increasing fragment offset field value
- This works even if fragments arrive out of order!
- Works even if fragmentation occurs multiple times



Fragment Offset Field

IP fragmentation and reassembly

- A packet larger than a link's **Maximum Transmission Unit (MTU)** must be fragmented or discarded (with an ICMP error to the source). Fragments are smaller than the smallest MTU on the path.
- Two approaches: **IP router segmentation** (routers fragment) or **IP Path MTU Discovery** (the sender fragments). Router fragmentation offloads work onto routers and can cause cascaded, odd fragmentation — not recommended in the modern Internet; Path MTU Discovery is recommended.
- Reassembly happens only at the destination. Fragments with the same 16-bit Identification are stored in one buffer at the offset given by the fragment offset; incomplete packets wait until all fragments arrive or a timer expires, then complete packets go up a layer.
- The sender places a fresh random value in the Identification field per packet; if a router fragments, every fragment copies the original value.
- Offset values let the receiver order fragments even if they arrive out of order or are re-fragmented; the More-Fragments bit is 0 on the last fragment only.
- Reassembly failure is handled with a reassembly lifetime timer on the first fragment; if it expires, partial data is discarded to free buffers.

```
Fragments of one packet share ID; offsets 0, 212, 730 (×8 octets); MF = 1 for all but the last
```

15. Routing Algorithms

Source deck: 3Routing Algorithms

Forwarding vs routing

- Two processes live inside a router: **forwarding** handles each arriving packet by looking up its outgoing line in the routing table; **routing** fills in and updates those tables — that is where the routing algorithm works.

Desirable properties

Property	Meaning
Correctness	Packets actually reach their proper destination.
Simplicity	Low overhead; complexity raises overhead.
Robustness	Copes with topology/traffic changes and router crashes without a reboot.
Stability	Reaches equilibrium and stays there under all conditions.
Fairness	Every node gets a fair chance (often first-come-first-served).
Optimality	Optimal throughput and minimum mean packet delay. Fairness and optimality often conflict.

Classification

Adaptive (dynamic)	Non-adaptive (static)
Based on current traffic/topology measurements. Sub-types: centralized (one node holds all info — single point of failure), isolated (node decides alone), distributed (info from neighbours).	Route computed off-line in advance and loaded at boot. Includes flooding, random walk, and static shortest-path routing.

Adaptive sub-types in detail

- **Centralized**: one central node collects the whole topology/traffic picture and distributes it; only one node needs the information, but its failure downs the network.
- **Isolated**: the node decides without outside information. Example **hot potato** — drop the packet on the shortest output queue regardless of where it leads; a variant combines static link weights with queue lengths.
- **Backward learning**: routing tables adapt from incoming packets carrying source identity + a hop counter. A node records the hop count seen; it updates only if the new value is better. Weakness: when the best route fails it cannot recall the second best, so nodes must periodically forget and restart.
- **Delta routing**: hybrid of centralized and isolated. Nodes send line costs to a central node, which computes the k best i→j paths. Path n is treated as equivalent to the best if $C_{ijn} - C_{ij1} <$ delta. As delta → 0 this becomes centralized; as delta → ∞ all paths become equivalent.
- **Distributed**: node uses information from neighbours; risk is that the network changes between receiving info and sending the packet.

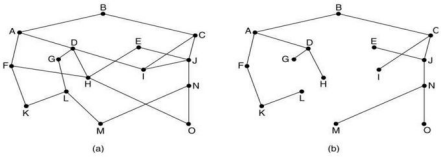
Static strategies

- **Flooding**: every incoming packet is sent on every outgoing line except the one it arrived on. Robust (good for military use) but wastes bandwidth and can loop. Loop controls: sequence numbers (discard already-seen source+seq), hop count (drop at max), spanning tree (send only toward the destination); a practical variant is **selective flooding** (send only on roughly-correct-direction lines).
- **Random walk**: send to a randomly chosen neighbour; highly robust and makes good use of alternative routes when the network is highly connected; usually implemented by choosing the least-queued link.
- **Multipath / bifurcated routing**: each node keeps a table row per destination listing best, second-best, … outgoing lines with relative weights; the packet picks one at random using the weights as probabilities. Tables are computed manually and fixed.

The optimality principle and sink trees

- If router J lies on the optimal path from I to K, then the optimal path from J to K follows the same route. Consequently, the set of optimal routes from all sources to one destination forms a tree rooted at the

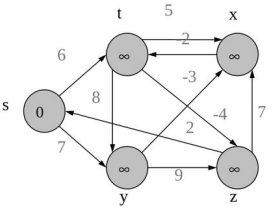
destination — a **sink tree**. Every routing algorithm ultimately aims to discover and use sink trees.



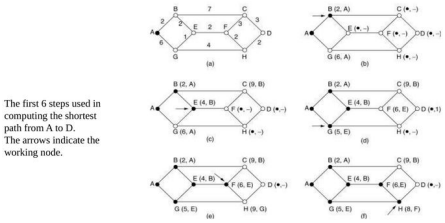
(a) A subnet. (b) A sink tree for router B.

(a) A subnet. (b) A sink tree for router B.

Bellman-Ford Algorithm Example

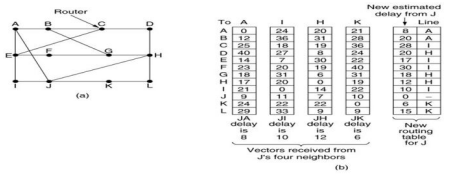


3Routing Algorithms — slide 29



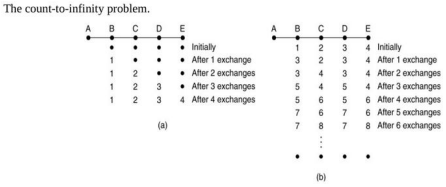
The first 6 steps used in computing the shortest path from A to D. The arrows indicate the working node.

The first 6 steps used in computing the shortest path from A to D. The arrows indicate the w...



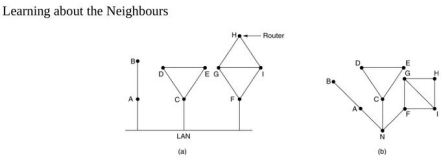
(a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.

(a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.



The count-to-infinity problem.

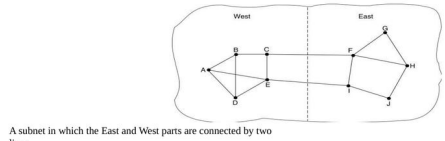
The count-to-infinity problem.



(a) Nine routers and a LAN. (b) A graph model of (a).

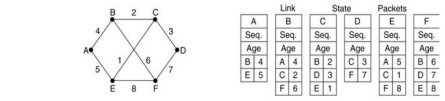
Learning about the Neighbours

Distributing the Link State Packets



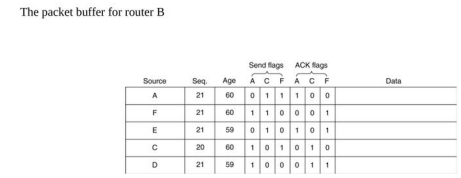
A subnet in which the East and West parts are connected by two lines.

Measuring Line Cost



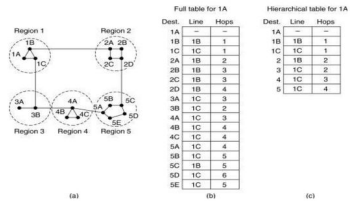
(a) A subnet. (b) The link state packets for this subnet.

Building Link State Packets



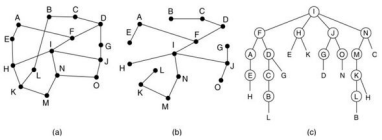
Distributing the Link State Packets

Hierarchical Routing



Hierarchical Routing

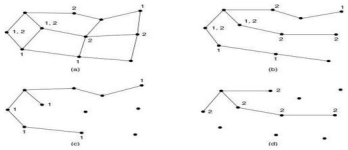
Broadcast Routing



Reverse path forwarding. (a) A subnet. (b) a Sink tree. (c) The tree built by reverse path forwarding.

Broadcast Routing

Multicast Routing



(a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

Multicast Routing

Shortest-path routing

- Model the subnet as a graph: routers = nodes, links = arcs, each arc labelled with a cost (lower is better). Find the shortest path on the graph.
- Cost metrics: minimum hops (unit cost, breadth-first search — ignores load); transmission + propagation delay (reflects capacity and distance but is static); queuing delay (reflects load but not propagation); or a blend of distance, bandwidth, traffic, communication cost, mean queue length and measured delay.
- Assumptions: nodes cost zero w.r.t. themselves; links symmetric ($d_{i,j} = d_{j,i}$); graph complete (missing edge = infinite cost).

Bellman-Ford (distance-vector style, iterates on hop count)

```
D[i, h] = shortest path length from i to node 1 using at most h hops
```

```
Init: D[i,0] = ∞ for i ≠ 1, D[1,h] = 0.   Iterate: D[i,h+1] = min_j{ d_i,j + D[j,h] }.   Stop
when D[i,h] = D[i,h+1] for all i.
```

- Uses the optimality principle; terminates because at most N−1 hops are needed. Complexity **O(N³)**.

Dijkstra (link-state style, iterates on path length)

$$D_j = \min[\ D_j,\ D_i + d_{j,i}\ ] \quad \text{(i just moved to P)}$$

- Set P = nodes with known shortest distance; Q = the rest. Initially P = {1}. Repeatedly move the minimum-cost node of Q into P, then relax. Each moved node's shortest path is then final; N−1 iterations; complexity **O(N²)**.

Floyd-Warshall (all-pairs, iterates on allowed intermediates)

$$\text{Init } D_{i,j}[0] = d_{i,j}. \quad D_{i,j}[n+1] = \min\{\ D_{i,j}[n],\ D_{i,n+1}[n] + D_{n+1,j}[n]\ \}$$

- Grow the intermediate set from {1} to all nodes; gives all-pairs shortest paths. Complexity **O(N³)**.

Distance-vector routing

- Dynamic algorithm: each router keeps a vector of best known distance to every destination and the line to use; tables are updated by exchanging information with neighbours.
- With delay as metric and known neighbour delays, every T ms each router sends its estimated delay list to each neighbour and receives theirs.
- Example: router J measures delays to A, I, H, K of 8, 10, 12, 6 ms. To reach G, if A claims 18 ms, J can reach G via A in 26 ms; via I 41; via H 18; via K 37 — so J picks H with 18 ms. Repeat for all destinations.
- **Count-to-infinity problem**: DV reacts fast to good news but slowly to bad news, because when X tells Y it has a path, Y cannot tell whether it itself is on that path.

Link-state routing

- Replaced distance-vector routing, whose problems were (1) ignoring line bandwidth (delay metric = queue length) and (2) slow convergence (count-to-infinity).
- Each router: discovers its neighbours and learns their network addresses; measures the delay/cost to each neighbour; builds a packet telling everything it has learned; sends that packet to all other routers (link-state packets, with a buffer); computes the shortest path to every other router (e.g. Dijkstra).
- Because every link appears twice (once per direction), a full set of link-state packets lets a router rebuild the entire subnet graph.

Other routing topics

- **Hierarchical routing**: group routers into regions to shrink tables; a router knows full detail inside its region but only a summary of others.
- **Broadcast routing**: reverse path forwarding — a router forwards a broadcast only if it arrived on the link that is on the router's own shortest path back to the source; builds the broadcast tree implicitly from the sink tree.
- **Multicast routing**: one spanning/multicast tree per group, so a packet reaches only group members.

16. IP Addressing — Classful, Classless & Subnetting

Source deck: 4IP Addressing

Addressing modes

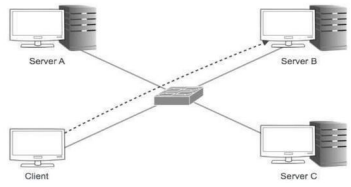
Mode	Destination	Note
Unicast	One host (32-bit destination = that host).	Client → one server.
Broadcast	All hosts on the segment; address 255.255.255.255.	Every host must process it.
Multicast	A group; addresses start 224.x.x.x.	Entertained by more than one host.
Anycast	Nearest host offering a service.	Connection goes to the first responder — best connection.

- Every network reserves one address for the network number and one for broadcast.

Interpreting a 32-bit address

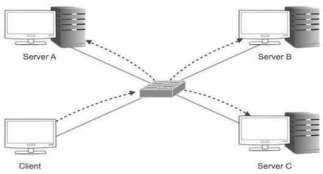
- 10000000 00111101 00010111 11011000 → 128.61.23.216. Writing it with a /24 says the first 24 bits are network, last 8 are host (a classless address). IPv4 uses a hierarchical scheme: network → sub-network → host.

Unicast Addressing Mode



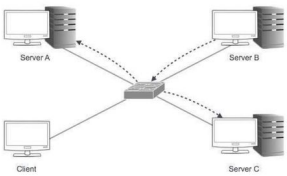
Unicast Addressing Mode

Broadcast Addressing Mode



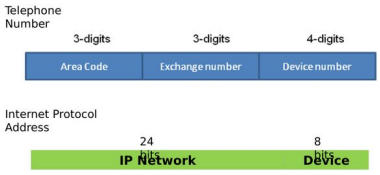
Broadcast Addressing Mode

Multicast Addressing Mode



Multicast Addressing Mode

Internet Addressing



Internet Addressing

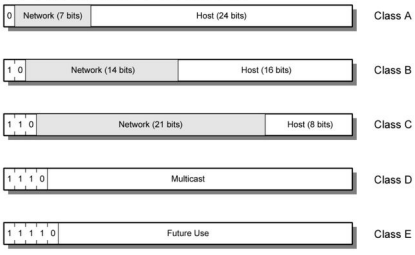
Hierarchical Addressing Scheme

- IPv4 uses hierarchical addressing scheme. An IP address, which is 32-bits in length, is divided into two or three parts as depicted –

- A single IP address can contain information about the network and its sub-network and ultimately the host.
- This scheme enables the IP Address to be hierarchical where a network can have many sub-networks which in turn can have many hosts.

4IP Addressing — slide 12

IPv4 Address Formats



IPv4 Address Formats

Class A Address

- 32 bit global internet address
- Network part (8)and host part (24)
- The first bit of the first octet is always set t
- Range 0.0.0.0-to-127.255.255.255 ; All 0 re
- Class A addresses only include IP starting from 1.x.x.x to 126.x.x.x only.
- The IP range 127.x.x.x is reserved for loopback IP addresses.
- The default subnet mask for Class A IP address is 255.0.0.0 which implies that Class A addressing can have 126 networks (2^7-2) and 16777214 hosts (2^{24}-2).
- Class A IP address format is thus: **0NNNNNNN**.HHHHHHHH.HHHHHHHH.HHHHHHHH
- All allocated

4IP Addressing — slide 15

Class B Address

- An IP address which belongs to class B has the first two bits in the first octet set to 10
- Network part (16)and host part (16)
- Range 128.x.x.x to 191.x.x.x
- Second Octet also included in network address
- 2^{14} = 16,384 class B addresses
- Class B IP Addresses range from 128.0.x.x to 191.255.x.x. The default subnet mask for Class B is 255.255.x.x.
- Class B has 16384 (2^{14}) Network addresses and 65534 (2^{16}-2) Host addresses.
- Class B IP address format is: **10NNNNNN.NNNNNNNN**.HHHHHHHH.HHHHHHHH
- All allocated

4IP Addressing — slide 16

Class C Address

- The first octet of Class C IP address has its first 3 bits set to 110,
- Network part (24)and host part (8)
- Range 192.x.x.x to 223.x.x.x
- Second and third octet also part of network address
- 2^{21} = 2,097,152 addresses
- Nearly all allocated
- Class C IP addresses range from 192.0.0.x to 223.255.255.x. The default subnet mask for Class C is 255.255.255.x.
- Class C gives 2097152 (2^{21}) Network addresses and 254 (2^8-2) Host addresses.
- Class C IP address format is: **110NNNNN.NNNNNNNN.NNNNNNNN**.HHHHHHHH

4IP Addressing — slide 17

Class D Address

- Very first four bits of the first octet in Class D IP addresses are set to 1110, giving a range of –

11100000 – 11101111
224 – 239

- Class D has IP address range from 224.0.0.0 to 239.255.255.255.
- Class D is reserved for Multicasting. In multicasting data is not destined for a particular host, that is why there is no need to extract host address from the IP address, and Class D does not have any subnet mask.

IP Address Class	Format	Purpose	High-Order Bit(s)	Address Range	No. Bits Network/Host	Max. Hosts
A	N.H.H.H[1]	Few large organizations	0	1.0.0.0 to 126.0.0.0	7/24	16,777,214[2] (2^{24} – 2)
B	N.N.H.H	Medium-size organizations	1, 0	128.1.0.0 to 191.254.0.0	14/16	65, 543 (2^{16} – 2)
C	N.N.N.H	Relatively small organizations	1, 1, 0	192.0.1.0 to 223.255.254.0	22/8	245 (2^8 – 2)
D	N/A	Multicast groups (RFC 1112)	1, 1, 1, 0	224.0.0.0 to 239.255.255.255	N/A (not for commercial use)	N/A
E	N/A	Experimental	1, 1, 1, 1	240.0.0.0 to 254.255.255.255	N/A	N/A

1 N = Network number, H = Host number.
2 One address is reserved for the broadcast address, and one address is reserved for the network.

- *Number of blocks and block size in Classful IPv4 addressing*

Class	*Number of Blocks*	*Block Size*	*Application*
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

4IP Addressing — slide 18

Class	Binary	Dotted-Decimal
A	11111111 00000000 00000000 00000000	255.0.0.0
B	11111111 11111111 00000000 00000000	255.255.0.0
C	11111111 11111111 11111111 00000000	255.255.255.0

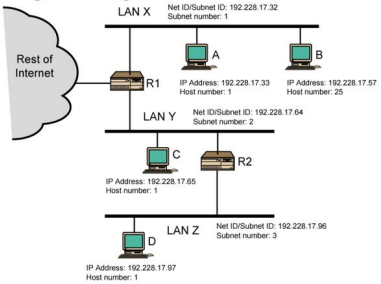
4IP Addressing — slide 20

Subnetting

- When a bigger network is divided into smaller networks, to maintain security, then that is known as Subnetting.
- A subnet is used to divide a large network into a number of smaller, linked networks, which helps to minimize traffic. It reduces the need for traffic to use unnecessary routes, which speeds up the network.

Number of blocks and block size in Classful IPv4 addressing

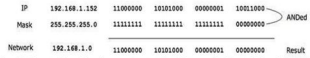
Routing Using Subnets



4IP Addressing — slide 26

Subnet Mask

- The 32-bit IP address contains information about the host and its network. It is very necessary to distinguish both.
- For this, routers use Subnet Mask, which is as long as the size of the network address in the IP address.
- Subnet Mask is also 32 bits long. If the IP address in binary is ANDed with its Subnet Mask, the result yields the Network address.
- For example, say the IP Address is 192.168.1.152 and the Subnet Mask is 255.255.255.0 then –

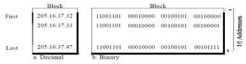


- This way the Subnet Mask helps extract the Network ID and the Host from an IP Address. It can be identified now that 192.168.1.0 is the Network number and 192.168.1.152 is the host on that network.

Subnet Mask

Subnetting

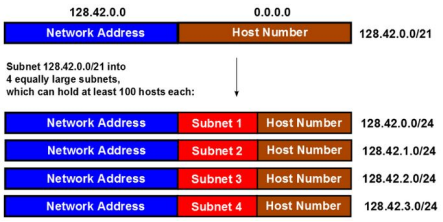
- *A block of 16 addresses granted to a small organization*



- The addresses are contiguous.
- The number of addresses is a power of 2 (16 = 2^4), and the first address is divisible by 16.
- The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

A block of 16 addresses granted to a small organization The addresses are contiguous. The nu...

Routing Using Subnets



4IP Addressing — slide 54

Classful addressing

Class	Leading bits	Range	Net/host split	Networks	Hosts per net	Mask
A	0	1.0.0.0 – 126.255.255.255	8/24	126 (2^7-2)	16,777,214 ($2^{24}-2$)	255.0.0.0
B	10	128.0.0.0 – 191.255.255.255	16/16	16,384 (2^{14})	65,534 ($2^{16}-2$)	255.255.0.0
C	110	192.0.0.0 – 223.255.255.255	24/8	2,097,152 (2^{21})	254 (2^8-2)	255.255.255.0
D	1110	224.0.0.0 – 239.255.255.255	—	multicast	—	none
E	1111	240.0.0.0 – 255.255.255.255	—	experimental	—	none

- 127.x.x.x is reserved for loopback; 0.0.0.0 means "this computer" (bootstrap).
- Masks are 32-bit numbers: contiguous 1s (network) then 0s (host); ANDing an address with its mask yields the network address. In classes A/B/C the address splits into **netid** and **hostid** (1/3, 2/2, 3/1 bytes); this does not apply to D/E.

Special address forms

Prefix (network)	Suffix (host)	Meaning
all 0s	all 0s	this computer (used during bootstrap)
network address	all 0s	identifies the network
network address	all 1s	broadcast on the specified network
all 1s	all 1s	broadcast on the local network
127	any	loopback (testing)

Why classful failed

- Each class has a fixed number of fixed-size blocks; a class A block is far too large for almost any organisation, class B often too large, class C too small — so addresses were wasted.
- Class D was sized for 268,435,456 multicast groups (never needed) and class E went largely unused.
- Rapid Internet growth then caused address depletion. Remedies: subnetting (split a large block) and supernetting (combine class C blocks). Drawback: routing becomes more difficult.

Subnetting, supernetting and masks

- **Subnetting**: divide a class A/B block into contiguous subnets and assign each to a smaller network. It increases the number of 1s in the mask (e.g. /24 → /26). Each LAN gets a subnet number; the host portion splits into subnet + host. A site still looks like one network to the outside.
- **Supernetting**: combine several contiguous class C blocks into a supernet; it decreases the number of 1s (e.g. four class Cs: /24 → /22).
- **CIDR** (Classless Inter-Domain Routing): use addresses independent of class; slashed notation /n. Used on addresses advertised to the Internet (ISP side).
- **VLSM** (Variable-Length Subnet Mask): different masks for different subnets — "subnets of subnets" — used inside companies.

Classless addressing

- No classes, but addresses are still granted in blocks whose size fits the entity (a household may get 2, a large ISP hundreds of thousands). Extends the subnet idea to the whole Internet and guards against exhaustion.
- Three restrictions on a block: addresses must be contiguous; the count must be a power of 2; the first address must be divisible by the block size.
- A block is defined as x.y.z.t/n. From the address and /n you can derive everything.

First address: set the 32−n rightmost bits to 0 | Last address: set the 32−n rightmost bits to 1 | Number of addresses = 2^{32-n}

Worked example — 128.42.5.4/21 (netmask 255.255.248.0)

Quantity	Method	Result
Prefix	Count contiguous 1s in the netmask.	/21 (twenty-one 1s)
Network address	AND address with mask.	128.42.0.0
Broadcast address	Force all host bits to 1 (OR with inverted mask).	128.42.7.255
Host count	$2^{32-21} - 2$	2046

- CIDR notation example: 128.140.168.0/21 → mask 255.255.248.0, lowest host 128.140.168.1, highest host 128.140.175.254.

Subnetting calculations you may be asked for

- Prefix for at least H hosts: host bits = ceil($\log_2 H$); prefix = 32 − host bits. (H = 100 needs 7 host bits → /25.)
- Longest mask containing two addresses: convert both to binary and find the longest common leading bit string (e.g. 128.42.5.17 and 128.42.5.67 → /25).
- Subnet example: 205.16.37.39/28 → first 205.16.37.32, last 205.16.37.47, 16 addresses (2^{32-28}).
- First address = address AND mask; last address = address OR (complement of mask); number of addresses = complement of mask + 1.

Maximum hosts per subnet = $2^{32 - \text{netmask_length}} - 2$ (subtract all-zeros network and all-ones broadcast)

17. IP Multicast

Source deck: 6IP Multicast

Why multicast

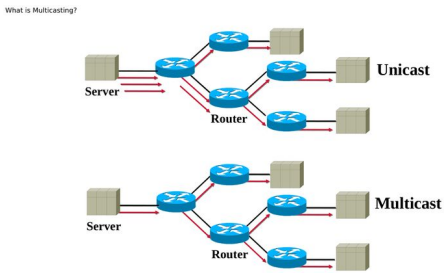
- When sending the same data to multiple receivers: better bandwidth utilisation, less host/router processing. Receivers' addresses are unknown in advance.
- Applications: video/audio conferencing, resource discovery / service advertisement, stock distribution.
- Contrast unicast (one server → one receiver) with multicast (one server → a group).

The MBone and protocols

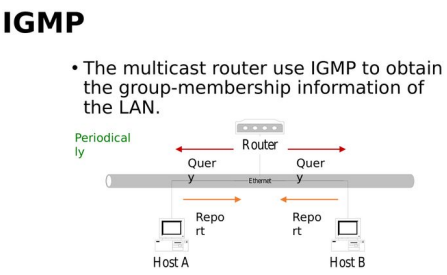
- In 1995 the first multicast network was born: the **MBone** (Internet's Multicast Backbone) — over 2800 subnets in 25+ countries by March 1996; a semi-permanent IP multicast testbed using IP tunnelling.
- Group management protocol: **IGMP** (Internet Group Management Protocol).
- Multicast routing protocols: **source-based trees** (one tree per (S,G)) — DVMRP, MOSPF, PIM-DM; and **shared trees** (one tree per (*,G)) — CBT, PIM-SM.
- IGMP: the multicast router uses IGMP to obtain the group-membership information of the LAN (periodic query/report).

(S,G) notation

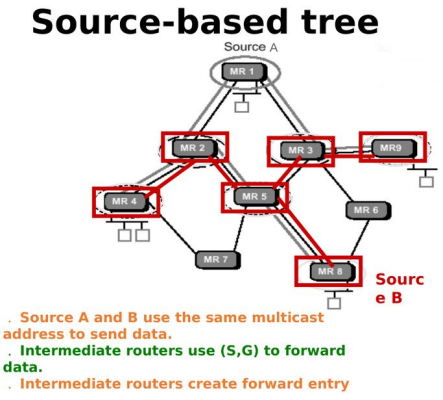
- For every multicast source there are two pieces of information: the source address S and the group address G, written **(S,G)**. Also used is **(*,G)** — every source for a particular group. Routers create tables with (*,G) and (S,G) entries.



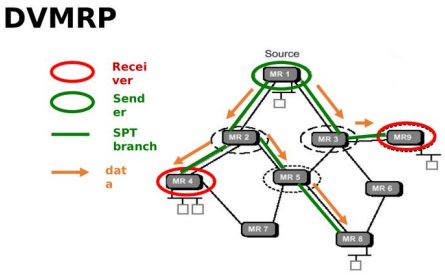
What is Multicasting?



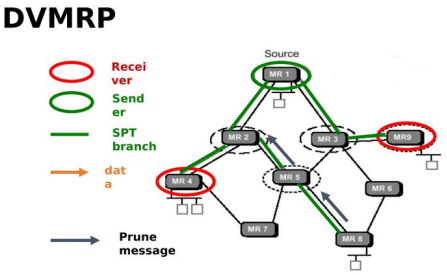
IGMP



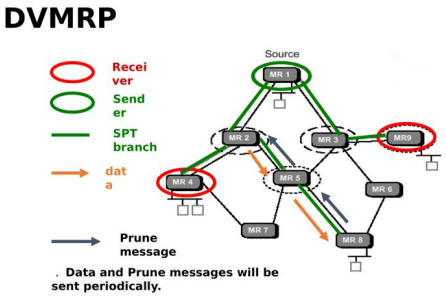
Source-based tree



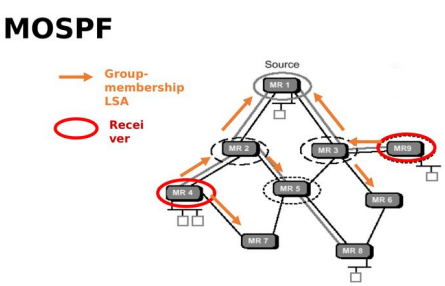
DVMRP

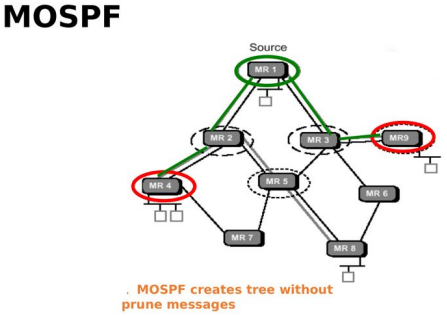


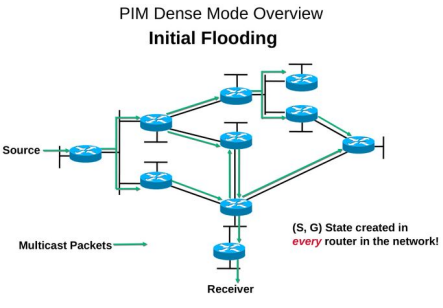
DVMRP



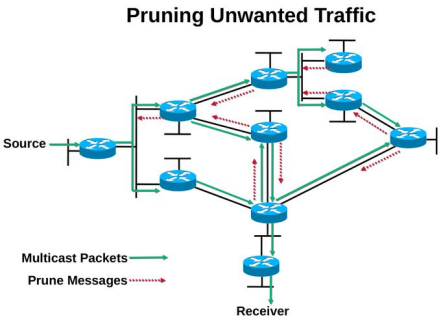
DVMRP

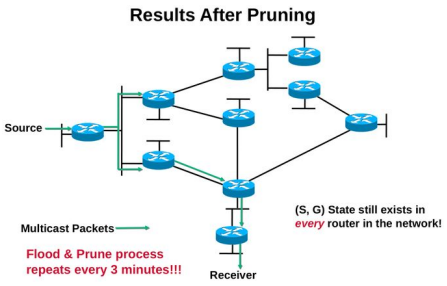






MOSPF *MOSPF* *PIM Dense Mode Overview*





Source *Results After Pruning*

Source-based trees and DVMRP

- Sources A and B may use the same multicast address; intermediate routers use (S,G) to forward data and create forwarding entries as they receive data.
- **DVMRP** (Distance Vector Multicast Routing Protocol): relies on a distance-vector unicast protocol (RIP); for each (S,G) grows a shortest-path tree rooted at the source; uses reverse path multicasting (RPM). DVMRP subnetworks were interconnected through the unicast Internet with tunnels. Flood and Prune technology; very successful in academic circles.
- DVMRP operation: data and Prune messages are sent periodically; a receiver sends a **graft** message toward the source to join the group quickly.
- Problem: DVMRP cannot scale to Internet sizes — distance-vector routing, periodic updates, full table refresh every 60 s, table sizes (Internet > 40,000 prefixes), too many tunnels, hop count up to 32.

MOSPF and PIM

- **MOSPF** (Multicast OSPF): relies on link-state unicast routing (OSPF); for each (S,G) grows a shortest-path tree rooted at the source; uses Dijkstra and group-membership LSAs to compute the shortest path tree. Creates trees without prune messages.
- **PIM-DM** (Protocol Independent Multicast, Dense Mode): flood-and-prune is inefficient; creates (S,G) state in *every* router even with no receivers; complex Assert mechanism (to determine which router on a LAN forwards); no support for shared trees. Initial flooding sends traffic to all PIM-DM neighbours; non-RPF traffic is discarded; prunes are sent on the RPF interface (no downstream members) and on non-RPF interfaces. Flood-and-prune repeats every 3 minutes; (S,G) state remains in all routers.
- **PIM-SM** (Sparse Mode): must configure a **Rendezvous Point (RP)** (statically on every router, or via Auto-RP/Bootstrap Router). Very efficient; explicit join model; traffic flows only where needed; router state only along flow paths; scales better than dense mode; works for sparsely or densely populated networks.

18. Routing Protocols (RIP, OSPF, BGP)

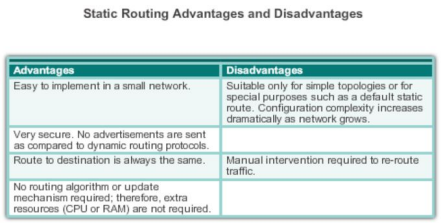
Source deck: 5Routing Protocols

Routing and forwarding

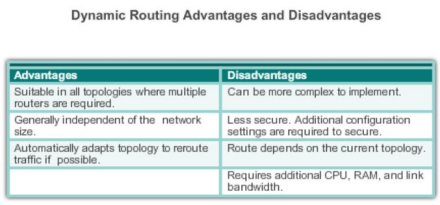
- A **router** determines the next network point to which a packet should be forwarded toward its destination; it creates and maintains a table of routes and their conditions and uses it to pick the best route. Multiple routes may be defined; the route with the lower weight/metric is tried first.
- Routing protocol software in the network layer fills and updates routing tables (finding shortest paths — this is **routing**) and decides the output interface for an incoming packet by referring to the table (this is **forwarding**).
- Static routing: routes change slowly; shortest paths pre-computed offline and entered manually; cannot update automatically on failure; used in small networks. Dynamic routing: each router learns the network state by communicating with neighbours and fills its own table.

Reducing routing-table entries

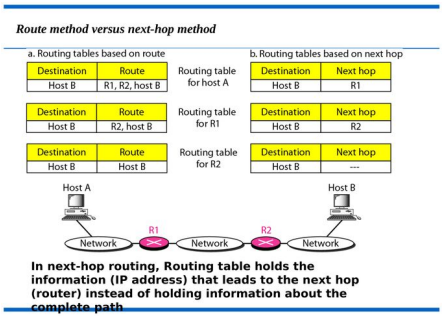
- **Network-specific** routing: list the destination network IP address in the table.
- **Host-specific** routing: list the destination host IP address.
- **Default** routing: a default router is used if the destination network is not found in the table.
- **Next-hop** method: the table holds the IP address of the next hop (router) rather than the complete path.



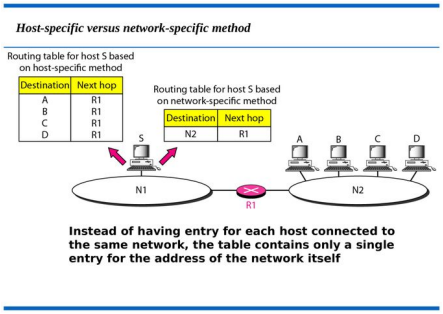
5Routing Protocols — slide 6



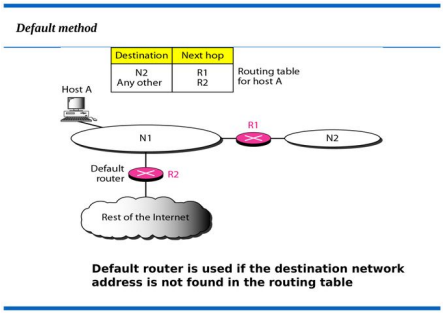
5Routing Protocols — slide 11



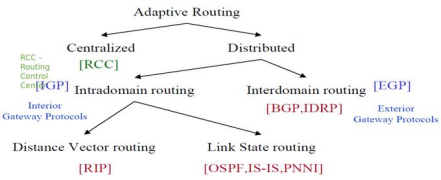
Route method versus next-hop method



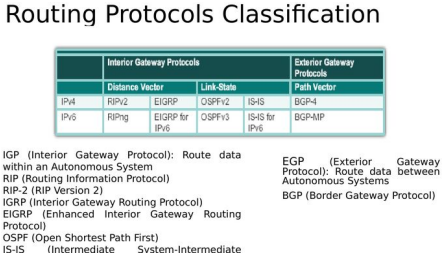
Host-specific versus network-specific method



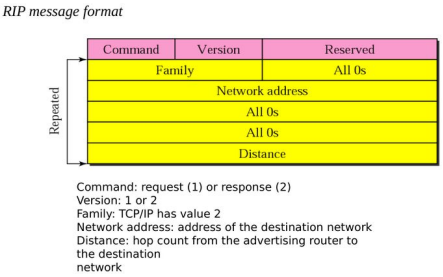
Default method



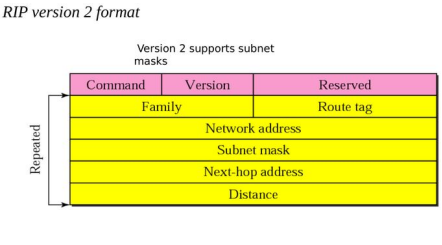
RCC – Routing Control Center



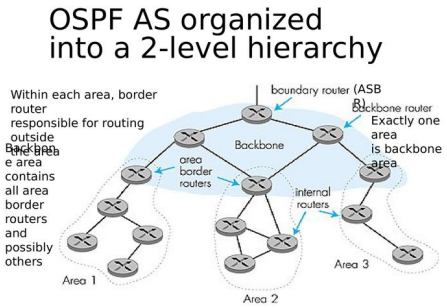
Routing Protocols Classification



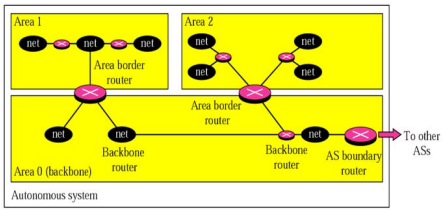
RIP message format



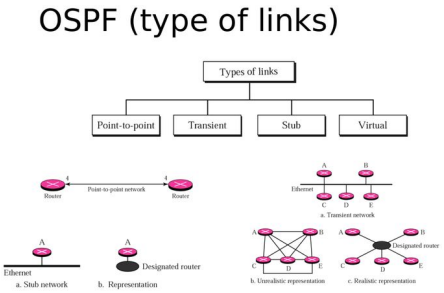
RIP version 2 format



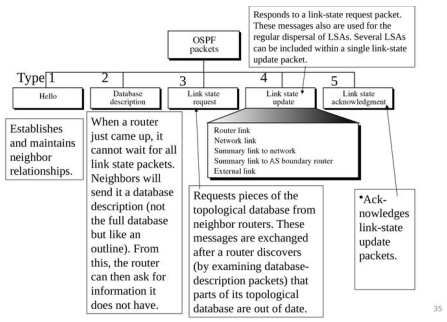
OSPF AS organized into a 2-level hierarchy



5Routing Protocols — slide 31



OSPF (type of links)

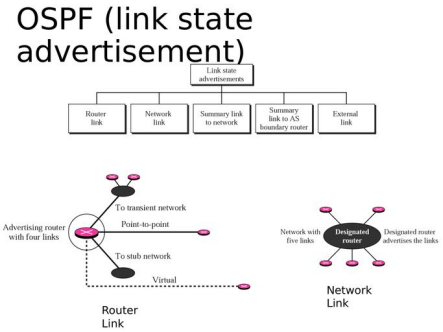


35

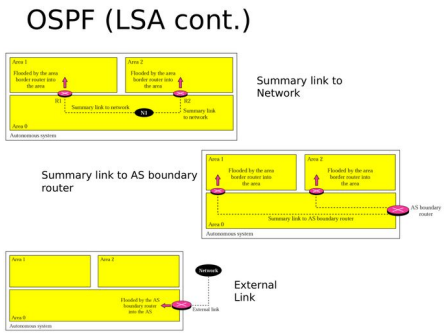
OSPF common header

Version	Type	Message length
Source router IP address		
Area Identification		
Checksum	Authentication type	
Authentication		

OSPF common header



OSPF (link state advertisement)



OSPF (LSA cont.)

BGP Messages

- Messages are sent over TCP connections on port 179.
- Functional procedures
  - Neighbor acquisition (open message, acceptance through Keepalive message)
  - Neighbor reachability (periodic Keepalive messages)
  - Network reachability (broadcast an update message)
    - Each routers maintains a database of networks that can be reached
    - + preferred route to this network.
- RFC does not address
  - How a router knows the address of another router.
  - Up to network admin.

BGP Messages

Classification: IGP vs EGP

Type	Scope	Examples
IGP (Interior Gateway Protocol)	Routes data within an Autonomous System.	RIP, RIP-2, IGRP, EIGRP, OSPF, IS-IS
EGP (Exterior Gateway Protocol)	Routes data between Autonomous Systems.	BGP

Routing Information Protocol (RIP)

- A **distance-vector** protocol using **hop count** as its primary metric; the path with the lowest hop count is the best route.
- Simplest dynamic routing protocol: each router broadcasts its entire routing table frequently (to hosts as well as routers), which interrupts hosts and makes RIP unsuitable for large networks. Slow convergence.
- Prevents routing loops by limiting hops: maximum allowed is 15; a hop count of 16 means unreachable (poison route / infinity metric), so the maximum network diameter is 15 hops.
- Features: updates exchanged periodically; updates always broadcast; full routing tables sent; routers always trust neighbour information ("routing on rumours").
- Sends periodic updates every 30 s; uses the Bellman-Ford distance-vector algorithm; supports IP and IPX; uses UDP port 520; administrative distance 120.
- If multiple equal-cost paths exist, RIP load-balances (default up to 4) by round-robin — which can cause pinhole congestion (e.g. a 9600-baud link and a T1 get equal traffic).

Version	Update	Address	Authentication	Class
RIP v1	Broadcast	255.255.255.255	No	Classful

Version	Update	Address	Authentication	Class
RIP v2	Multicast	224.0.0.9	Yes	Classless (supports classful too)
RIPng	Multicast	FF02::9 (IPv6 only)	—	Classless updates

- RIP message format: Command (1 = request, 2 = response), Version (1 or 2), Family (TCP/IP = 2), Network address, Distance (hop count from the advertising router). Version 2 supports subnet masks.

RIP timers

Timer	Value / purpose
Update timer	Frequency of routing updates — every 30 s IP RIP sends a complete copy of its routing table, subject to split horizon (IPX RIP: 60 s).
Invalid timer	Absence of refreshed content; RIP waits 180 s to mark a route invalid and immediately puts it into hold-down.
Hold-down timer	Ensures regular updates do not inappropriately cause a loop; the router does not act on non-superior new information for a period. RIP hold-down = 180 s.
Flush timer	RIP waits an additional 240 s after hold-down before removing the route from the table.

Open Shortest Path First (OSPF, RFC 1247)

- A **link-state** interior routing protocol; has a value in the IP header (8-bit protocol field). Its domain is an Autonomous System (AS) — a collection of IP routing prefixes under one administrative entity with a common routing policy.
- Divides an AS into **areas** and routes hierarchically within the AS; special/backbone (AS boundary) routers dissipate information about other ASes into the current system.
- Metric based on type of service (minimum delay/RTT, maximum throughput, reliability); complex; link-state packet (LSP) databases must be protected; uses designated routers to reduce endpoints.
- OSPF organises an AS into a **2-level hierarchy**: within each area a border router handles routing outside the area; exactly one area is the **backbone area**, containing all area border routers (and possibly others).

OSPF link types

Type	Meaning
Point-to-Point	A link between two routers with no hosts or intermediate routers.
Transient	Many routers attached to a single network (LAN/Wi-Fi).
Stub Network	A network connected to the OSPF domain through a single router; all traffic goes through that router.
Virtual Link	When a direct link is broken or impossible, a virtual link connects two routers indirectly.

Five types of OSPF packets

Type	Name	Purpose
1	Hello	Discovers neighbours, establishes and maintains neighbour relationships; elects the Designated Router (DR) on broadcast networks.
2	Database description	An outline of the database sent to a newly-booted router so it can request what it lacks.
3	Link-State Request	Requests pieces of the topological database from neighbours (after discovering out-of-date parts).
4	Link-State Update	Responds to a request and carries the regular dispersal of LSAs (several LSAs per packet).
5	Link-State Acknowledgement	Acknowledges link-state update packets.

- All OSPF packets share a header: Version (8-bit), Type (8-bit), Message length (16-bit), Source router IP (32-bit), Area identification (32-bit), Checksum (16-bit), Authentication type (16-bit: 0 = none, 1 = password), Authentication (64-bit).

Link-state updates and LSAs

- Each Link State Update packet carries a set of new link state advertisements (LSAs) one hop further from their origin; a single packet may carry LSAs of several routers; each LSA is tagged with the originating

router's ID and a checksum.
- LSA types: Router Link, Network Link, Summary link to Network, Summary link to AS boundary router, External Link.

OSPF operations

- **Hello protocol**: Hello packets are transmitted to all interfaces periodically to discover neighbours, establish/maintain adjacency, and elect the DR.
- **Database synchronization**: two neighbours exchange database description packets (LSA headers only); newer LSAs are requested later; packets sent by the master are acknowledged by the slave.
- **Propagation of link-state information**: a link-state request is sent to update the database; a link-state update is sent when requested, when a link-state changes, or periodically; a link-state acknowledgement is sent in response; updates are retransmitted until acknowledged.
- **Flooding LSAs**: a node installs each LSA then decides on which links to flood it (it may not flood an interface if neighbours likely already have it).
- **Building the routing table**: with the complete link-state database, Dijkstra produces a shortest-path tree rooted at the router; only the next hop is used in the table.

Border Gateway Protocol (BGP, RFC 1771)

- Why DV and link-state are unsuitable for inter-AS routing: DV assumes all routers share a common distance metric (different meanings → unstable, loop-prone routes), gives no information about the ASes visited, and the smallest hop count is not always preferred (e.g. an insecure AS); link-state is too big for the Internet, needs flooding, huge databases and long Dijkstra runs.
- BGP is an inter-AS routing protocol based on **path vector** routing: no metrics; information about which networks a router can reach and the ASes to be crossed (a distance vector annotated with the entire path plus policy attributes); guaranteed loop-free.
- Path vector differs from DVA: it does not include a distance/cost estimate; it lists all ASes visited; it can use non-tree backbone topologies; it uses TCP to disseminate DVs (reliable but subject to TCP flow control); policies are complex to set up.
- Messages are sent over TCP connections on **port 179**. Procedures: neighbour acquisition (open message, accepted via Keepalive), neighbour reachability (periodic Keepalive), network reachability (broadcast an update). Each router keeps a database of reachable networks and the preferred route to each. The RFC does not address how a router knows another router's address (up to the network admin).

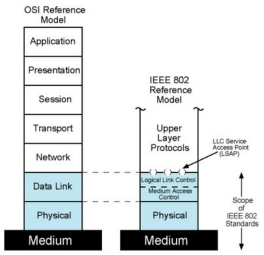
19. LAN Protocols: MAC, ALOHA, CSMA, IEEE 802

Source decks: 6LANMAC, 7IEEE802

IEEE 802 LAN architecture

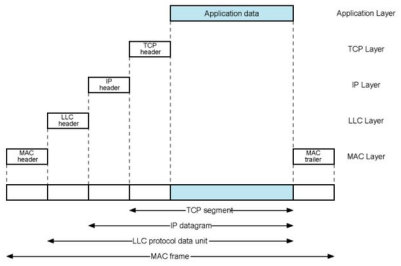
- Corresponds to the lower two OSI layers; most LANs follow IEEE 802. Layer 2 is split into **LLC** (interface to higher levels, flow control, based on classical DLC) and **MAC** (framing, error detection, address recognition, access to medium).
- Topologies: bus, ring, star. MAC coordinates access to the shared broadcast medium so simultaneous transmissions do not collide.
- The MAC layer prepares data for transmission, detects errors, recognises addresses and governs access to the medium; it discards erroneous frames, and LLC optionally retransmits unsuccessful frames.

IEEE 802 Protocol Layers vs. OSI Model



IEEE 802 Protocol Layers vs. OSI Model

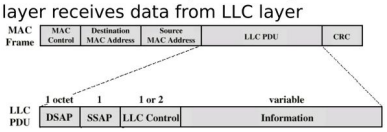
LAN Protocols in Context



LAN Protocols in Context

Generic MAC & LLC Format

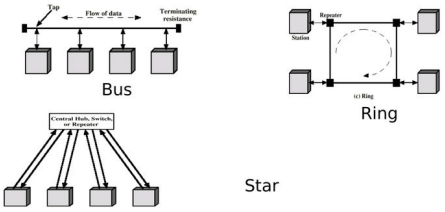
- Actual format differs from protocol to protocol
- MAC layer receives data from LLC layer



- MAC layer detects errors and discards frames
- LLC optionally retransmits unsuccessful frames

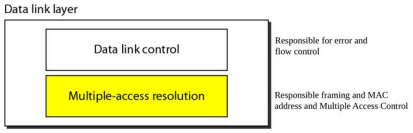
Generic MAC & LLC Format

LAN Topologies



LAN Topologies

Data link layer divided into two functionality-oriented sublayers

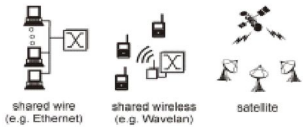


Data link layer divided into two functionality-oriented sublayers

Multiple Access

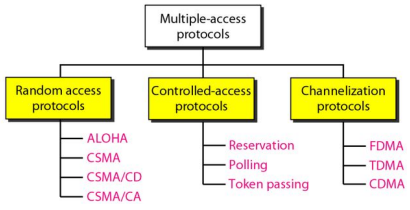
- **Broadcast link** used in LAN consists of multiple sending and receiving nodes connected to or use a single shared link

Broadcast links Examples



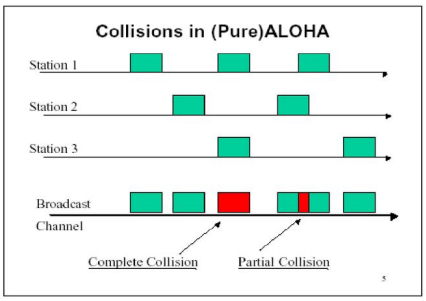
Multiple Access

Taxonomy of multiple-access protocols



Taxonomy of multiple-access protocols

Pure ALOHA

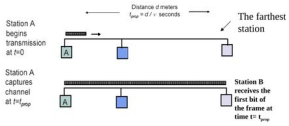


In pure ALOHA, frames are transmitted at completely arbitrary times.

Pure ALOHA

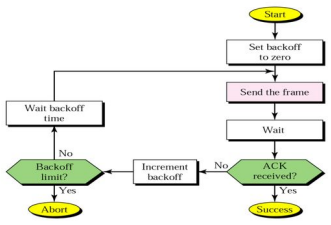
Maximum Propagation Delay

- **Maximum propagation delay**(t_{prop}): time it takes for a bit of a frame to travel between the two most widely separated stations.



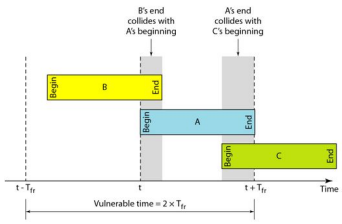
Maximum Propagation Delay

Procedure for ALOHA protocol



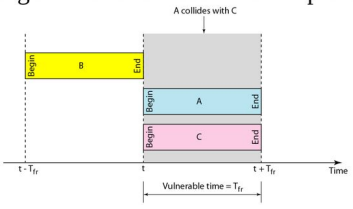
Procedure for ALOHA protocol

Critical time for pure ALOHA protocol



Critical time for pure ALOHA protocol

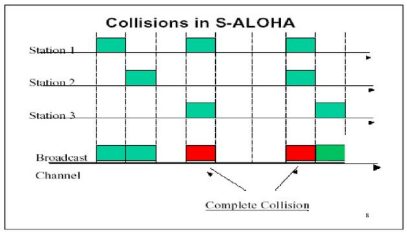
In danger time for slotted ALOHA protocol



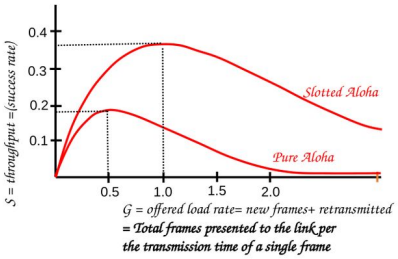
In danger time for slotted ALOHA protocol

Efficiency of Aloha

Random Access – Slotted ALOHA



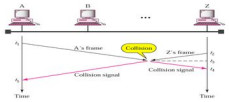
6LANMAC — slide 25



Efficiency of Aloha

Random Access – Carrier Sense Multiple Access (CSMA)

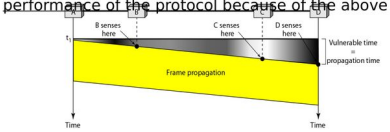
- To improve performance, avoid transmissions that are certain to cause collisions
- Based on the fact that in LAN propagation time is **very small**
- → If a frame was sent by a station, All stations knows immediately so they can **wait before start sending**
  - → A station with frames to be sent, should **sense the medium** for the presence of another transmission (carrier) before it starts its own transmission (Listen Before Talk)
- This can **reduce** the possibility of collision but it cannot eliminate it.
  - Collision can only happen when more than one station begin transmitting within a short time (the **propagation time** period)



Random Access – Carrier Sense Multiple Access (CSMA)

Random Access – Carrier Sense Multiple Access (CSMA)

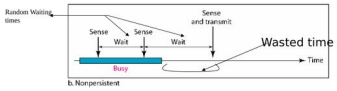
- Vulnerable time for CSMA is the **maximum propagation time**
- The longer the propagation delay, the worse the performance of the protocol because of the above case.



Random Access – Carrier Sense Multiple Access (CSMA)

Nonpersistent CSMA

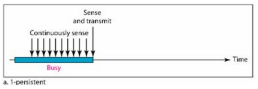
- A station with frames to be sent, should sense the medium
  1. If medium is idle, **transmit**; otherwise, go to 2
  2. If medium is busy, **(backoff)** wait a ***random* amount of time** and repeat **1**
- Non-persistent Stations are **deferential (respect others)**
- Performance:
  - Random delays reduces probability of collisions because two stations with data to be transmitted will wait for different amount of times.
  - Bandwidth is **wasted** if waiting time (backoff) is large because medium will remain idle following end of transmission even if one or more stations have frames to send



Nonpersistent CSMA

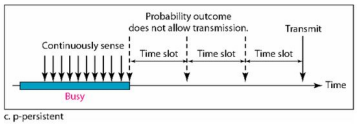
1-persistent CSMA

- To avoid idle channel time, 1-persistent protocol used
- Station wishing to transmit listens to the medium:
  1. If medium idle, **transmit** immediately;
  2. If medium busy, **continuously listen** until medium becomes idle; then transmit immediately with probability 1
- Performance
  - 1-persistent stations are **selfish**
  - If two or more stations becomes ready at the same time, **collision guaranteed**



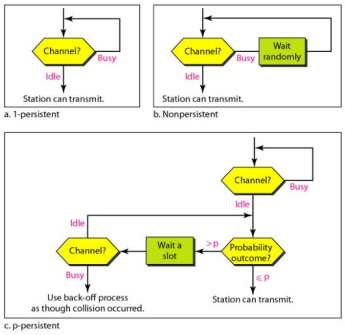
1-persistent CSMA

P-persistent CSMA



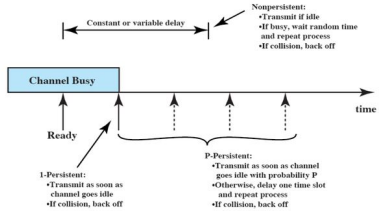
P-persistent CSMA

Flow diagram for three persistence methods



Flow diagram for three persistence methods

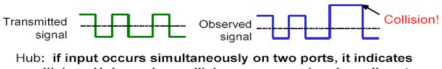
All CSMA Persistence schemes altogether



All CSMA Persistence schemes altogether

How does a node detect a collision?

Transceiver: **A node monitors the media while transmitting. If the observed power is more than transmitted power + attenuated reflection of its own signal, it indicates a collision.**

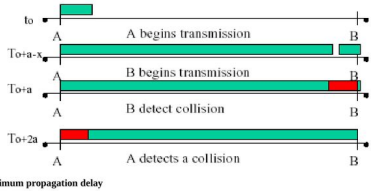


Hub: **if input occurs simultaneously on two ports, it indicates a collision. Hub sends a collision presence signal on all ports.**

6LANMAC — slide 40

CSMA/CD

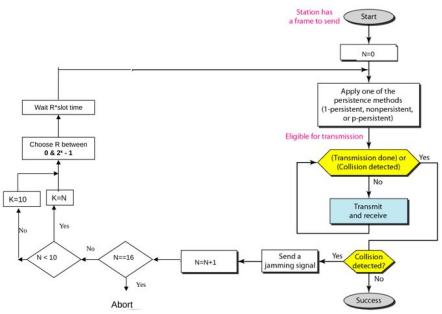
- ***Question:*** How long does it take to detect a collision?
- ***Answer:*** *In the* ***worst case*****, twice the maximum propagation delay of the medium**



Note: **a = maximum propagation delay**

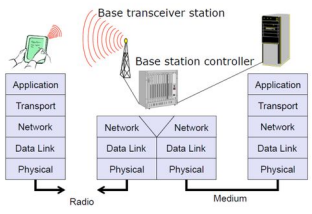
CSMA/CD

Flow diagram for the CSMA/CD



Flow diagram for the CSMA/CD

Reference Model



Reference Model

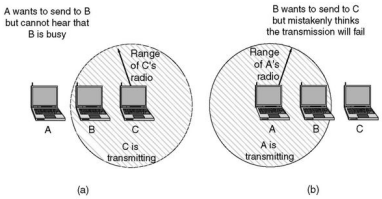
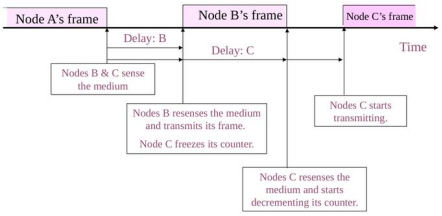


Figure 4-26.(a)The hidden station problem. (b) The exposed station problem.

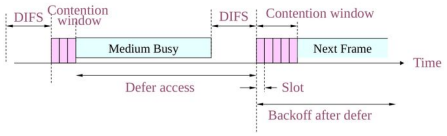
Figure 4-26.(a)The hidden station problem. (b) The exposed station problem.

CSMA/CA



CSMA/CA

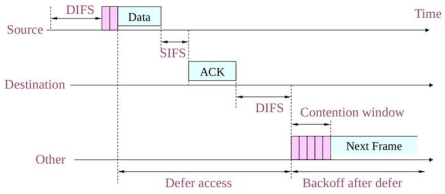
CSMA/CA



DIFS – Distributed Inter Frame Spacing

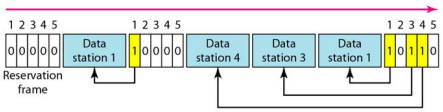
CSMA/CA

CSMA/CA/ACK



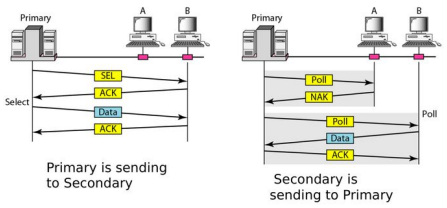
SIFS – Short Inter Frame Spacing

CSMA/CA/ACK



6LANMAC — slide 64

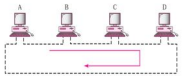
Select and poll functions in polling access method



Select and poll functions in polling access method

Token-Passing network

- On a local area network, **token passing** is a channel access method where a signal called a *token* is passed between nodes to authorize that node to communicate
- In contrast to polling access methods, there is no pre-defined "master" node
- Implements Distributed Polling System

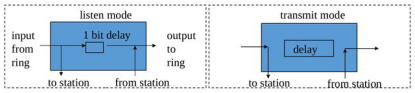


Token-Passing network

Token-Passing network

bits are copied to the output bits with a one bit delay

Bits are inserted by the station

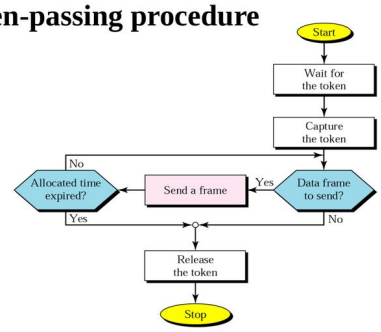


Station Interface is in two states:

- **Listen state**: Listen to the arriving bits and check the destination address to see if it is its own address. If yes the frame is copied to the station otherwise it is passed through the output port to the next station.
- **Transmit state**: station captures a special frame called **free token** and transmits its frames. **Sending** station is responsible for **reinserting** the free token into the ring medium and for **removing** the transmitted frame from the medium.

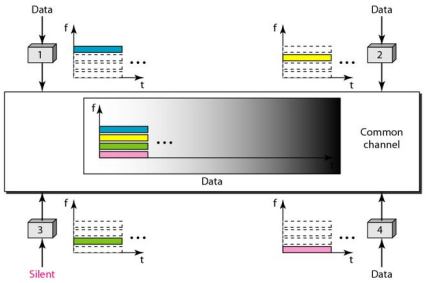
6LANMAC — slide 69

Token-passing procedure



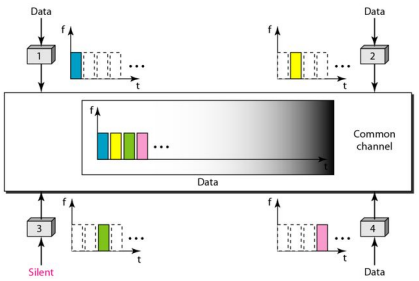
Token-passing procedure

Frequency-division multiple access (FDMA)



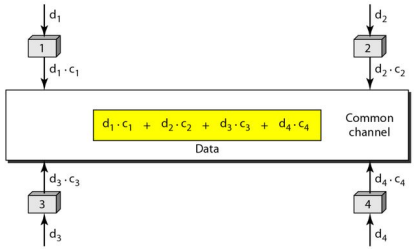
Frequency-division multiple access (FDMA)

Time-division multiple access (TDMA)



Time-division multiple access (TDMA)

Simple idea of communication with code



Simple idea of communication with code

Multiple-access taxonomy & random access

Protocol	Rule	Max throughput
Pure ALOHA	Transmit any time; after a frame wait up to $2 \cdot t_{prop}$ for an ACK, else wait a random time and resend. Vulnerable time = 2T.	$S = G \cdot e^{-2G}$, $S_{max} = 0.184$ at $G = ½$ (≈18%)
Slotted ALOHA	Time in slots = one frame time; transmit only at a slot start; a central clock marks slots. Vulnerable time = T.	$S = G \cdot e^{-G}$, $S_{max} = 0.368$ at $G = 1$ (37%)
Reservation ALOHA	Group N slots; each station owns one slot; unused slots reusable.	TDM-like, adapts to load

`Pure ALOHA: S = G·e^{-2G}, S_max = 0.184 at G = ½ | Slotted ALOHA: S = G·e^{-G}, S_max = 0.368 at G = 1`

- ALOHA advantages: a lone node can use the full rate R; simple; no master. Disadvantages: collisions when M nodes transmit give each far below R/M and low utilisation.

Carrier-sense multiple access (CSMA)

- Sense the medium before transmitting (listen before talk); propagation time is small, so collisions happen only when stations start within the propagation window. Vulnerable time = maximum propagation time.
- **Non-persistent**: if idle transmit; if busy wait a random time and retry — reduces collisions but wastes idle bandwidth during backoff.
- **1-persistent**: if busy, keep listening and transmit immediately when idle — avoids idle time but two ready stations collide (selfish).
- **p-persistent**: for slotted channels — if idle send with probability p, defer to the next slot with 1−p; larger p → smaller utilisation and less waiting.

CSMA/CD, CSMA/CA and controlled access

- **CSMA/CD** (Ethernet, bus topology): listen while talking. On detecting a collision the sender aborts and sends a 48-bit **jam signal**, backs off a random time, then retransmits. Detect a collision in the worst case in 2× maximum propagation delay. The frame must be long enough to detect the collision, else CSMA/CD is pointless.
- **Binary exponential backoff**: after the k-th collision pick R ∈ [0, 2^k−1] and wait R × slot time (Ethernet slot = 51.2 µs at 10 Mbps); cap k at 10 (max 1023); give up after 16 attempts. Gives a last-in-first-out effect among waiting stations.
- **CSMA/CA** (IEEE 802.11 WLAN): collision detection is not used because a node cannot listen and transmit at once. Distributed Coordination Function (DCF): sense the channel; if free for a DIFS transmit; else defer and use a random backoff in [0, W−1], decrementing only while idle and freezing when busy; ACK sent after a SIFS; if no ACK, retransmit after EIFS (doubling CW to a maximum).
- **Hidden terminal**: A cannot hear C's transmission to B and wrongly starts, causing a collision at B. **Exposed station**: B hears A's transmission and wrongly refrains from sending to C. RTS/CTS optionally mitigates these.
- Random-access performance: simple, decentralised, low delay at low load; but limited throughput, unbounded delay under heavy load, and potentially unfair.

```
CSMA/CD worst-case collision detection = 2 × max propagation delay  |  retry R ∈ [0, 2^k−1];
                                                                       give up after 16
```

Controlled access

Mechanism	Idea
Reservation	Cycles begin with a reservation interval of N minislots (one per station); a station reserves in its own minislot; everyone learns the order; reserved stations then send at full rate.
Polling	Centralized (primary polls secondaries via select/poll frames; NAK if nothing; round-robin or predetermined order) or distributed (known polling order, highest priority gets access then passes it on).
Token passing	A token authorises a node to transmit; no master; station interface has a listen state (copy if addressed, else forward) and a transmit state (capture free token, send frames, reinsert token, remove own frame).

- **Channelization**: share bandwidth by frequency (FDMA), time (TDMA) or code (CDMA). In FDMA/TDMA a node is limited to R/M even when alone; in CDMA all transmissions share one channel, each coded so receivers can separate them.

IEEE 802.3 (Ethernet)

- CSMA/CD with slotted channels: if idle transmit; if busy wait for an idle slot then transmit. Stations listen while transmitting; on collision they cease and wait a random (exponentially backed-off) time.
- Contention detail: no ACKs, so a station must ensure all stations see its transmission and no collision occurred — it transmits for the worst-case round-trip time (for 2500 m of coax at 10 Mbps, ≈ 50 µs).
- Minimum frame: 50 µs at 10 Mbps → 500 bits, rounded up to 512 bits (64 bytes); maximum frame 1518 bytes. (RFC 894 encapsulates IP in Ethernet; RFC 1042 in IEEE 802.)
- Frame: Preamble (alternating 1/0 for clock sync) | SFD 10101011 | Destination MAC | Source MAC | Length (LLC data) | Data | FCS (32-bit CRC, excludes preamble and SFD). Addresses are IEEE-assigned.
- 48-bit MAC address burned into the NIC; first 3 bytes assigned by IEEE. Least-significant bit of the first byte: 0 = unicast, 1 = multicast; broadcast is FF:FF:FF:FF:FF:FF.

- 10 Mbps variants: 10Base2/10Base5 (bus coax), 10BaseT with hub (one collision domain, star, twisted pair), 10BaseT with switch (one collision domain per port). Half-duplex send-xor-receive; full duplex when attached to a switch port. Manchester encoding.
- Speeds: Fast Ethernet 802.3u (100 Mbps; 100BASE-TX/FX/T4, TX most popular), Gigabit 802.3z/ab (1000 Mbps, enterprise backbone), 10 Gigabit 802.3ae (10 Gbps, 64b/66b line code).
- Performance: U = T_{trans} / (T_{trans} + T_{prop}) with no collisions; add $T_{contention}$ when collisions occur.

> Ethernet utilization (no collisions): U = T_{trans} / (T_{trans} + T_{prop}) | with collisions add $T_{contention}$

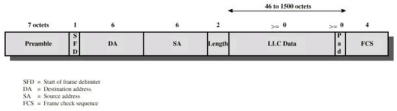
IEEE 802 Standards Working Groups

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

The important ones are marked with *. The ones marked with ↓ are hibernating. The one marked with † gave up.

IEEE 802 Standards Working Groups

IEEE 802.3 Frame Format



Preamble is alternating 1's and 0's (for clock synchronization)

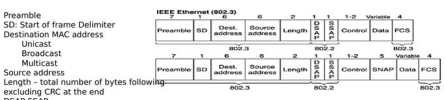
SFD is 10101011

Length is of the LLC data

FCS is 32-bit CRC (Cyclic Redundancy Check) code and excludes Preamble and SFD

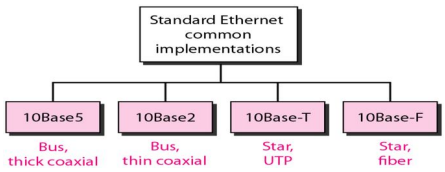
Addresses are uniquely assigned by IEEE to manufacturers.

IEEE 802.3 Frame Format



Preamble
SD: Start of frame Delimiter
Destination MAC address
Unicast
Broadcast
Multicast
Source address
Length – total number of bytes following excluding CRC at the end
DSAP SSAP,
Control
SNAP (Subnet Network Access Protocol)
Data
Frame Check Sequence (FCS)

Preamble SD: Start of frame Delimiter Destination MAC address Unicast Broadcast Multicast So...



7IEEE802 — slide 9

Ethernet Addressing

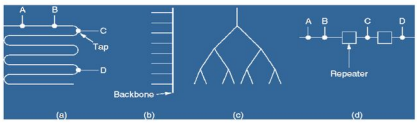
- Ethernet uses MAC address burned into each NIC.
- 48 bits (6 bytes)
- First 3 bytes assigned by IEEE
- Unicast MAC Address
- Broadcast MAC Address (FFFF.FFFF
- Multicast Address
- The least significant bit of the first byte defines the type of address.
  If the bit is 0, the address is unicast; otherwise, it is multicast.
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.

06:01:02:01:2C:4B

6 bytes = 12 hex digits = 48 bits

Ethernet Addressing

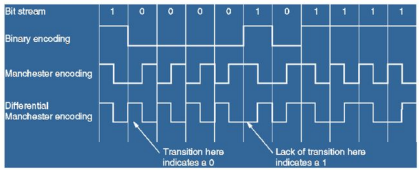
Cable topologies



Linear Spine (bus) Tree Segmented

Cable topologies

Manchester encoding



Manchester encoding

Summary of Standard Ethernet implementations

Characteristics	10Base5	10Base2	10Base-T	10Base-F
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

Summary of Standard Ethernet implementations

Summary of Fast Ethernet implementations

Characteristics	100Base-TX	100Base-FX	100Base-T4
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T

7IEEE802 — slide 17

Summary of Gigabit Ethernet implementations

Characteristics	1000Base-SX	1000Base-LX	1000Base-CX	1000Base-T
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

Summary of Gigabit Ethernet implementations

Summary of Ten-Gigabit Ethernet implementations

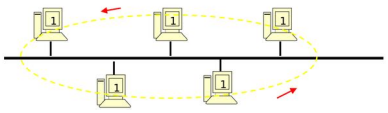
Characteristics	10GBase-S	10GBase-L	10GBase-E
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-nm single mode
Maximum length	300 m	10 km	40 km

Encoding used **64b/66b** - line code that transforms 64-bit data to 66-bit line code

Summary of Ten-Gigabit Ethernet implementations

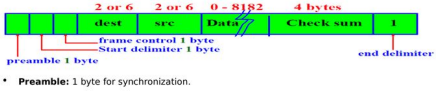
Token Bus – IEEE 802.4

- A network which implements the modified **Token Ring** protocol over a **"virtual ring"** on a coaxial cable with a bus topology.
- It is mainly used for industrial applications.



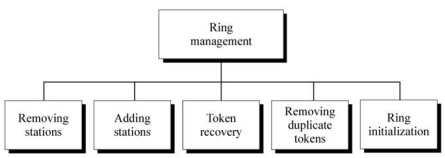
Token Bus – IEEE 802.4

Token Bus Frame Format



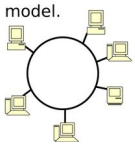
- **Preamble:** 1 byte for synchronization.
- **Start Delimiter:** 1 byte that marks the beginning of the frame.
- **Frame Control:** 1 byte that specifies whether this is a data frame or control frame.
- **Destination Address:** 2-6 bytes that specifies address of destination station.
- **Source Address:** 2-6 bytes that specifies address of source station.
- **Payload:** A variable length field that carries the data from the network layer.
- **Checksum:** 4 bytes frame check sequence for error detection.
- **End Delimiter:** 1 byte that marks the end of the frame.

Ring Management

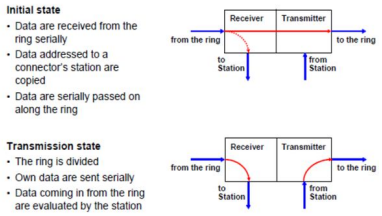


Token Ring – IEEE 802.5

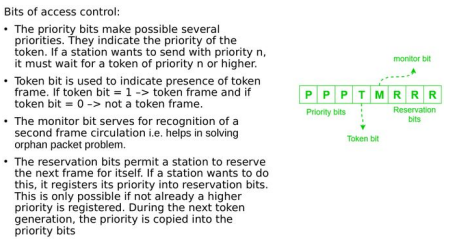
- A **ring** topology network developed in the late 1960s. Supported mainly by IBM. pushed into the background by **Ethernet** in the 1990s.
- a LAN protocol which resides at the data link layer (**DLL**) of the **OSI** model.



Sending and receiving



Sending and receiving



Bits of access control: The priority bits make possible several priorities. They indicate th...

Token bus (802.4) & token ring (802.5)

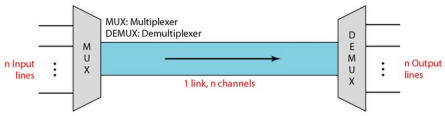
- **Token Bus**: token ring over a virtual ring on a bus/tree coax (industrial use). Each station knows its left/right neighbours; a logical ring is formed in address order (a coordinator initialises). Token passes high→low address; the holder transmits for a limited time then hands over; priorities via a queue and token-holding-time fractions. Frame: Preamble | Start Delimiter | Frame Control | Dest Addr | Source Addr | Payload | Checksum | End Delimiter. Messages: token T(ID, nextID) and data M(ID). Ring maintenance handles a station leaving (tell predecessor the new successor) and joining (open a window; conflict resolved by sending a request whose length encodes the last 2 ID bits). High token overhead (512 bit times).
- **Token Ring** (802.5, IBM): ring topology, unidirectional, Differential Manchester encoding, token passing, 4/16 Mbps. Frame: SD | AC | FC | DA | SA | PDU | CRC | ED | FS. AC bits: P (priority), T (token), M (monitor), R (reservation). Status bits A (address recognised) and C (frame copied). A station converts a free token to a data frame and back when its frame returns. Active monitor (chosen by CLAIM_TOKEN election) acts as master clock, inserts a 24-bit delay, ensures exactly one token, detects a broken ring and removes circulating frames; standby monitors take over.

20. Multiplexing & Switching

Source deck: 7Multiplexing

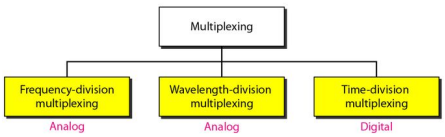
- **Multiplexing**: how links are shared between sessions. **Switching**: how traffic is switched from input to output ports at nodes.
- Multiplexing is similar to medium access control, but with MAC users are geographically separate (coordination and contention arise); with multiplexing the sharing is centralised in a switch, so coordination is easier.
- Static multiplexing gives each user a fixed portion; dynamic (statistical) multiplexing varies by demand.

Dividing a link into channels



Dividing a link into channels

Categories of multiplexing

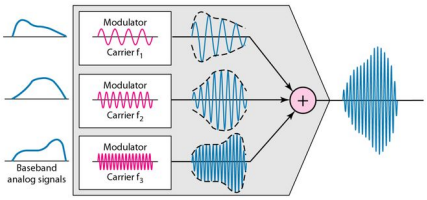


Categories of multiplexing

Frequency-division multiplexing (FDM)

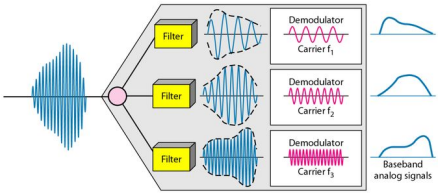
Frequency-division multiplexing (FDM)

FDM process



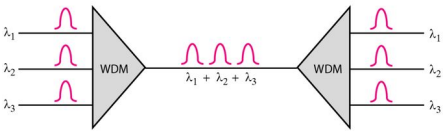
FDM process

FDM demultiplexing example



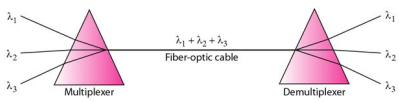
FDM demultiplexing example

Wavelength-division multiplexing (WDM)



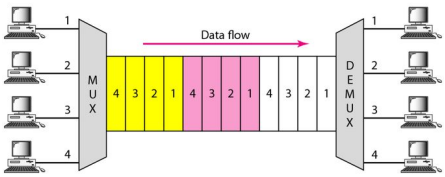
Wavelength-division multiplexing (WDM)

• *Prisms in wavelength-division multiplexing and demultiplexing*



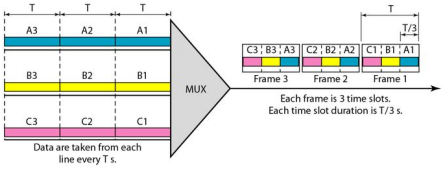
Prisms in wavelength-division multiplexing and demultiplexing

Time Division Multiplexing (*TDM*)

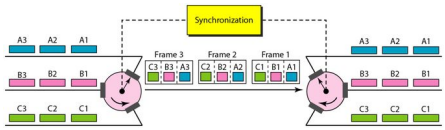


Time Division Multiplexing (TDM)

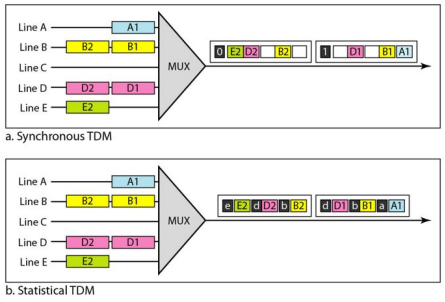
Synchronous time-division multiplexing



Synchronous time-division multiplexing

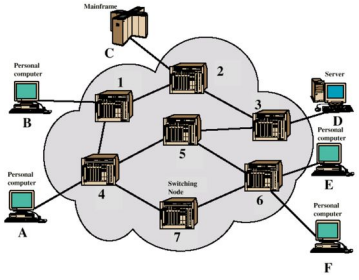


7Multiplexing — slide 22



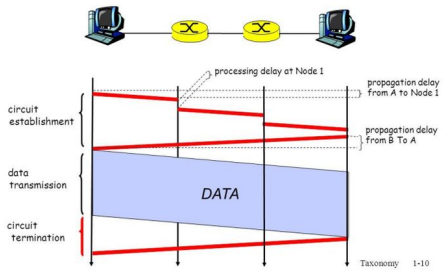
7Multiplexing — slide 24

Simple Switched Network



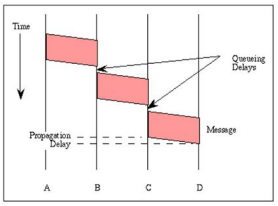
Simple Switched Network

Circuit Switching in MultiHop Route



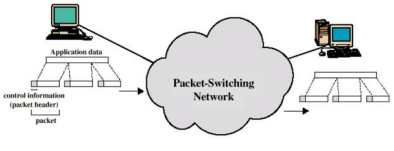
7Multiplexing — slide 33

Message Switching

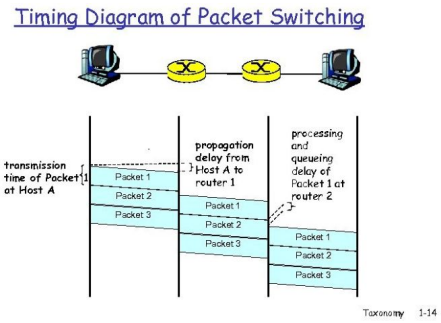


Message Switching

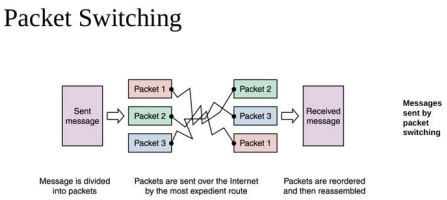
Use of Packets



Use of Packets



7Multiplexing — slide 38



Packet Switching

Frequency-Division Multiplexing (FDM)

- Divide the frequency spectrum of a link into disjoint bands, one per session; each signal is modulated onto a different carrier; carriers are separated by **guard bands** (e.g. broadcast radio). A channel is allocated even if it has no data.
- Advantages: users can be added by adding a transmitter/receiver pair; supports full duplex. Disadvantages: high initial cost (cable and connectors); a problem with one user can affect others; each user needs a precise carrier frequency.

Wavelength-Division Multiplexing (WDM)

- FDM for optical fibre: different sessions modulate light sources (LEDs or lasers) at different wavelengths over the same fibre; separated with optical components such as diffraction gratings.
- Current systems multiplex 40–128 channels onto a single fibre at 2.5–10 Gbps each. Low-rate optical streams can be multiplexed "all-optically" (passive components). Each wavelength can be used independently for analog or digital data; WDM is often combined with TDM.

Time-Division Multiplexing (TDM)

- A digital technique combining several low-rate channels into one high-rate channel: time is divided into slots and each session gets a set of slots in cyclical order. One slot may carry a bit, a byte, or a fixed-size packet; the sequence of slots (one per user) is a **frame**.
- **Synchronous TDM**: the data rate of the link is n times faster and the unit duration n times shorter; slots are preassigned and fixed; slots are allocated even if there is no data. **Interleaving** = taking a group of bits from each input onto one output.
- **Statistical TDM**: allocates slots dynamically on demand; the multiplexer scans inputs and collects data until the frame is full; the line data rate can be lower than the aggregate of the inputs.
- Advantages: single link; no precise carrier matching; high channel utilisation; easy expansion; no need to identify each traffic stream. Disadvantages: high sensitivity to other users; high initial cost; more technical complexity.

Switching

- A network is a collection of nodes and connections; long-distance transmission is done over switched nodes; nodes are not concerned with the content of data; end devices are stations.
- Nodes may connect to other nodes or to stations; node-to-node links are usually multiplexed; the network is usually partially connected with some redundancy for reliability.
- Types: **circuit switching** (used with static multiplexing) and **packet switching** (used primarily with statistical multiplexing).

Type	How it works	Note
Circuit switching	A dedicated communication path between two stations; three phases: establish, transfer, disconnect.	Inefficient for data: resources dedicated to a call, much idle time, fixed data rate (both ends must match), setup takes time; but once connected, transfer is transparent.
Message switching	The whole message is stored and forwarded.	Store-and-forward of an entire message.

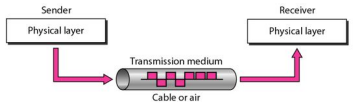
Type	How it works	Note
Packet switching	Messages split into fixed-sized, numbered packets (typically ~1000 octets), each with user data + control (routing) info; routers direct packets; packets are received, buffered and passed on (store-and-forward).	Circuit switching pairs with static mux; packet switching with statistical mux.

21. Transmission Media

Source deck: 7Transmission Media

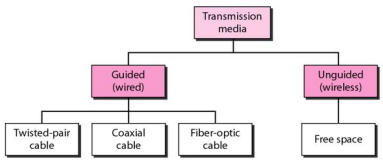
- The transmission medium is the physical path between transmitter and receiver. In one type, transmission occurs through a solid medium (twisted pair, coax, fibre); in the other, wirelessly through the atmosphere, space or water.
- Design factors: **bandwidth** (greater bandwidth → higher data rate), **transmission impairments** (attenuation, noise, distortion), **interference** (competing signals in overlapping bands), and **number of receivers** (point-to-point vs shared link).

Transmission medium and physical layer



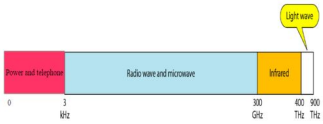
Transmission medium and physical layer

Classes of transmission media



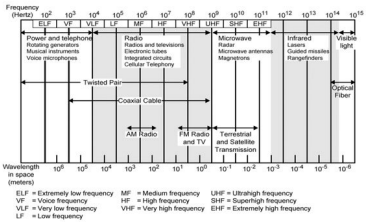
Classes of transmission media

Electromagnetic spectrum for Transmission Media



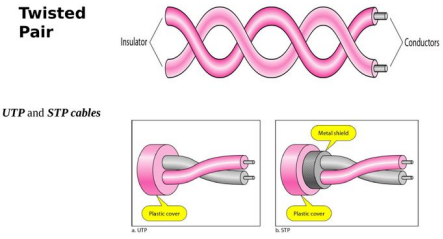
Electromagnetic spectrum for Transmission Media

Electromagnetic Spectrum



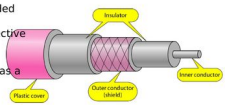
Electromagnetic Spectrum

UTP and STP cables



UTP and STP cables

A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material.
The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh.
The outer conductor is covered in a protective plastic
sheath.
The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.

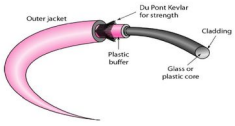


A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating mat...

Fiber-Optic Cable

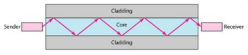
- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.

An optical fiber has an cylindrical shape and consists of 3 concentric section –
(i) Core
(ii) Cladding
(iii) Jacket



Fiber-Optic Cable

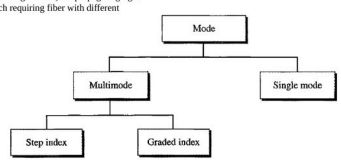
- Optical fibers use reflection to guide light through a channel.
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected refracted into it.



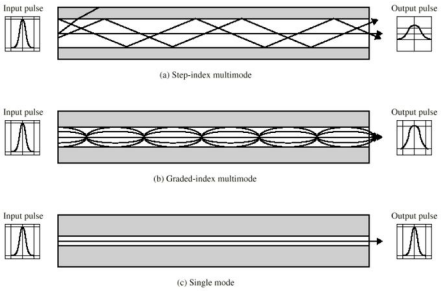
Optical fibers use reflection to guide light through a channel. A glass or plastic core is s...

Propagation Modes

Two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics.



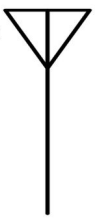
Propagation Modes



7Transmission Media — slide 22

Antennas

- electrical conductor used to radiat collect electromagnetic energy
  1)transmission antenna
  2)reception antenna



Antennas electrical conductor used to radiate or collect electromagnetic energy 1)transmissi...

- Antennas Types

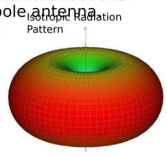
1)Omnidirectional Antenna

2)Unidirectional Antenna or Directional Antenna or Parabolic antenna

Antennas Types 1)Omnidirectional Antenna 2)Unidirectional Antenna or Directional Antenna or...

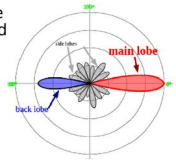
Isotropic antenna

- The radiation pattern of a simple omnidirectional antenna , a vertical half-wave dipole antenna
- In this graph the antenna is at th
center of the "donut," or torus.



Isotropic antenna

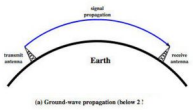
- In a directive antenna, the largest lobe, in the desired direction of
propagation, is called the "main lobe".
The other lobes are called "side
represent radiation in unwanted



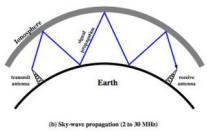
In a directive antenna, the largest lobe, in the desired direction of propagation, is called...

Wireless Propagation

- Ground Wave

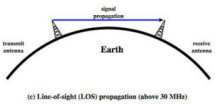


- Sky Wave

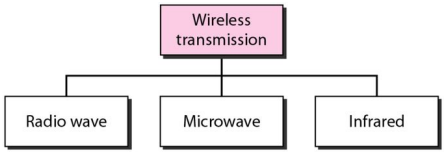


Wireless Propagation

- Line of Sight

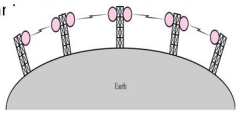


Line of Sight



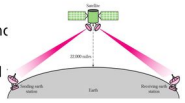
7Transmission Media — slide 37

Terrestrial
- used for long distance telecommunications and short point-to-point links,
- use a parabolic dish to focus a narrow beam onto a receiver ar



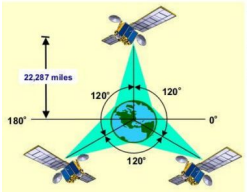
Terrestrial used for long distance telecommunications and short point-to-point links, use a...

Satellite is relay station
- receives on one frequency,
- repeats signal and transmits on anc
frequency
- Separate frequencies are assigned
- upward transmission (uplink)
- downward transmission(downlink)
- eg. uplink 5.925-6.425 GHz & downlink 3.7-4.2 GHz



Satellite is relay station receives on one frequency, repeats signal and transmits on anothe...

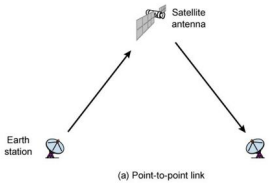
Geostationary Satellite



- **Remains in a fixed position relative to ground station.**
- **Used for communication purposed.**
- **Used 3 satellites to cover all over the earth except the polar extreme (latitudes > 81° north or south).**

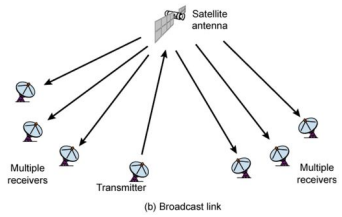
7Transmission Media — slide 42

Satellite Point to Point Link



Satellite Point to Point Link

Satellite Broadcast Link



Satellite Broadcast Link

Guided media

Twisted pair

- Two insulated copper wires (~1 mm) twisted in a helix. Twisting prevents the pair acting as an antenna: waves from different twists cancel, so it radiates less. One wire carries the signal, the other is a ground reference; the signal is the voltage difference, giving immunity to external noise. Twisting reduces crosstalk between adjacent pairs; neighbouring pairs use different twist lengths.
- Carries analog or digital information; bandwidth depends on wire thickness and distance (several Mbps over a few km); widely used due to adequate performance and low cost.
- **UTP** (unshielded) is most common; **STP** (shielded, IBM) adds a metal foil/braid around each pair — better against noise/crosstalk but bulkier and more expensive.
- Transmission characteristics: analog needs amplifiers every 5–6 km; digital needs repeaters every 2–3 km; limited distance, bandwidth (1 MHz) and data rate (100 MHz); susceptible to interference and noise.
- **Near-end crosstalk**: coupling of a signal from one pair to another — the near transmitted signal is picked up by the near receiving pair.

Coaxial cable

- Better shielding and greater bandwidth than UTP, so it spans longer distances at higher speeds. **Baseband** 50-ohm coax is for digital transmission; **Broadband** 75-ohm coax is for analog transmission (cable TV).
- Construction: a stiff copper core, insulation, a cylindrical braided-mesh conductor, and a protective plastic sheath. The outer metallic wrapping both shields against noise and acts as the second conductor completing the circuit.
- Transmission characteristics: analog needs amplifiers every few km (closer if higher frequencies; up to 500 MHz); digital needs repeaters every 1 km (closer for higher data rates).

Fibre-optic cable

- Made of glass or plastic; transmits signals as light. Three concentric sections: **core**, **cladding**, **jacket**. Uses reflection to guide light: a denser core is surrounded by less-dense cladding so a beam is reflected rather than refracted.
- Benefits: greater capacity (hundreds of Gbps), smaller size and weight, lower attenuation, electromagnetic isolation, greater repeater spacing (tens of km).
- Act as a waveguide for 10^{14}–10^{15} Hz (infrared and visible spectrum). Sources: LED (cheaper, wider temperature range, lasts longer) and injection laser diode (more efficient, greater data rate). Supports WDM.

Propagation mode	Characteristics
Multimode step-index	Multiple beams travel different paths; core density is constant, so light travels straight until the core/cladding interface where it changes angle abruptly. The suddenness (step) distorts the signal.
Multimode graded-index	Index of refraction (density) is highest at the centre and decreases gradually to the edge; this reduces signal distortion.
Single-mode step-index	A highly focused light source limits beams to a small range of near-horizontal angles; much smaller diameter and lower density → critical angle near 90° so beams are almost identical with negligible delays and little distortion.

Unguided media: wireless

- Unguided media transport electromagnetic waves without a physical conductor — through air (sometimes water); available to anyone who can receive them. Types: radio waves, microwaves, infrared.
- Bands: 30 MHz–1 GHz (radio, omnidirectional, broadcast); 2–40 GHz (microwave, highly directional, point-to-point); 3×10^{11}–2×10^{14} (infrared, local point-to-point/multipoint in confined areas such as a room).
- An **antenna** is an electrical conductor that radiates or collects electromagnetic energy (transmission or reception).

Radiation pattern and antenna types

- An antenna does not radiate equally in all directions; the **radiation pattern** is a graphical representation of the radiated power. An **isotropic antenna** is an ideal reference that radiates equal energy in all directions.
- **Omnidirectional** (e.g. a vertical half-wave dipole): the pattern looks like a "donut"/torus with the antenna at the centre. **Unidirectional / directional / parabolic**: the largest lobe in the desired direction is the **main lobe**; other lobes are **side lobes** (unwanted radiation).
- **Antenna gain** measures directionality — the power output in a particular direction compared with an isotropic antenna; measured in decibels (dB).

Propagation and bands

Medium	Range / nature	Use
Radio waves	3 kHz–300 GHz; broadcast radio 30 MHz–1 GHz; omnidirectional; sky-wave propagation; penetrates walls (line of sight gives better results); suffers multipath interference (reflections).	FM radio, UHF/VHF television
Microwaves	2–40 GHz; used for unicast; higher frequencies cannot penetrate walls. Terrestrial (parabolic dish, point-to-point) and satellite.	Cellular telephones, satellite networks, wireless LANs, long-distance links
Infrared	Transceivers modulate infrared light; must be in line of sight (directly or via reflection); does not penetrate walls.	Local point-to-point / multipoint in a room

- Wireless propagation modes: **ground wave**, **sky wave**, and **line of sight**.
- A **satellite** is a relay station: it receives on one frequency, repeats the signal and transmits on another. Separate frequencies are assigned for uplink and downlink (e.g. uplink 5.925–6.425 GHz, downlink 3.7–4.2 GHz).

22. Network Components

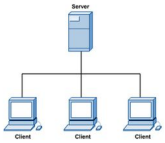
Source deck: 3NETWORK COMPONENTS

Basic components & functions

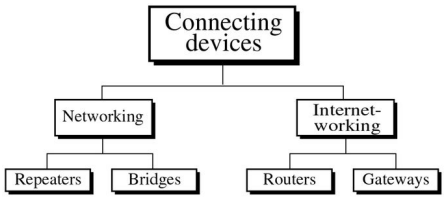
- Basic components: client, server, network interface card, cables, modems, hub, switch, router.
- Functions of network devices: **separating/connecting networks or expanding the network** (repeaters, hubs, bridges, routers, brouters, switches, gateways) and **remote access** (56K and ADSL modems).
- Client/server: network services are located in a dedicated computer whose only function is to respond to client requests; the server holds file, print, application and security services and is continuously available.

Clients and Servers

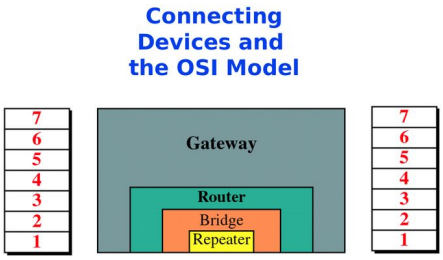
- In a client/server network arrangement, network services are located in a dedicated computer whose only function is to respond to the requests of clients.
- The server contains the file, print, application, security, and other services in a central computer that is continuously available to respond to client requests.



Clients and Servers

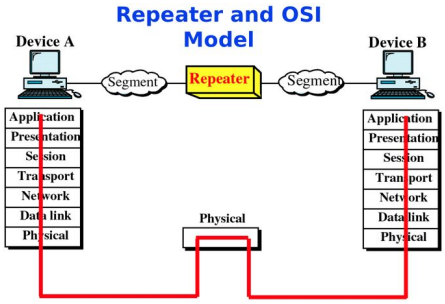


3NETWORK COMPONENTS — slide 10

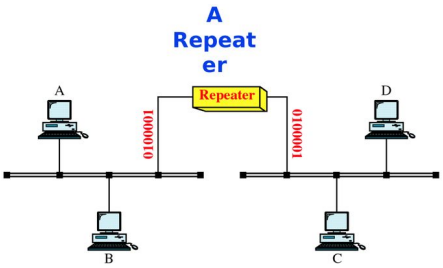


Connecting Devices and the OSI Model

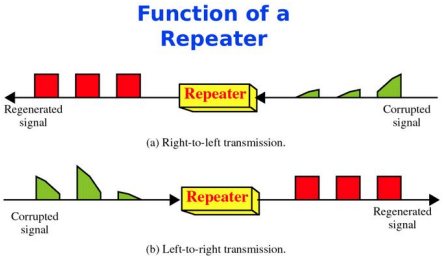
3NETWORK COMPONENTS — slide 13



Repeater and OSI Model

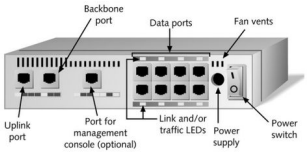


A Repeater



Function of a Repeater

Hubs

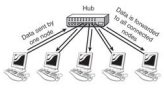


Hubs

The basic function of a hub is to take data from one of the connected devices and forward it to all the other ports on the hub.

This method of operation is inefficient because, in most cases, the data is intended for only one of the connected devices.

Due to the inefficiencies of the hub system and the constantly increasing demand for more bandwidth, hubs are slowly but surely being replaced with switches.

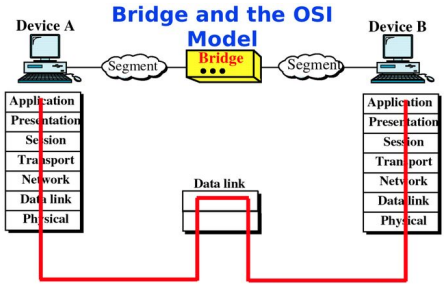


The basic function of a hub is to take data from one of the connected devices and forward it...

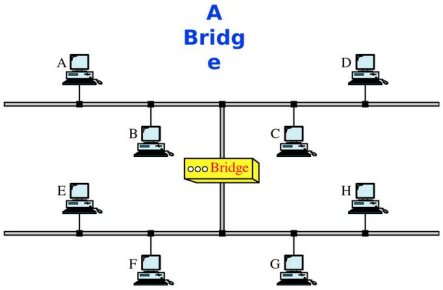
Bridges

- Like a repeater, a bridge has a single input and single output port
- Unlike a repeater, it can interpret the data it retransmits

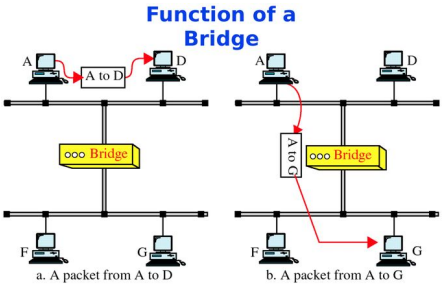
Bridges



Bridge and the OSI Model



A Bridge

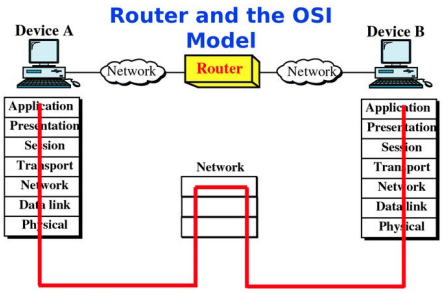


Function of a Bridge

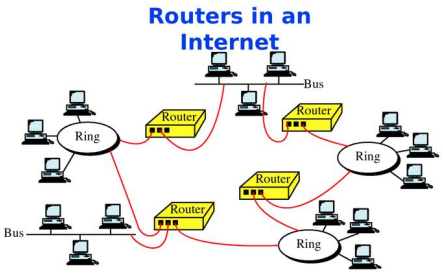
Switches

- Subdivide a network into smaller logical pieces
- A switch is a multiport bridge with a buffer designed that can boost its efficiency(a large number of ports imply less traffic) and performance.
- A switch is a data link layer device.

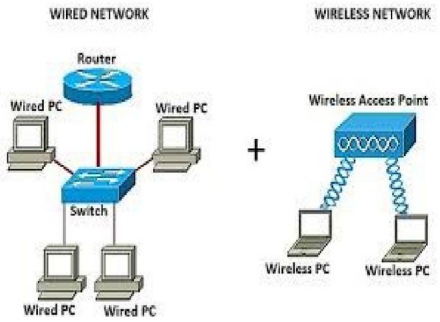
Switches



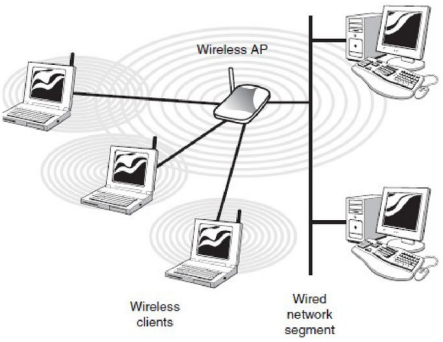
Router and the OSI Model



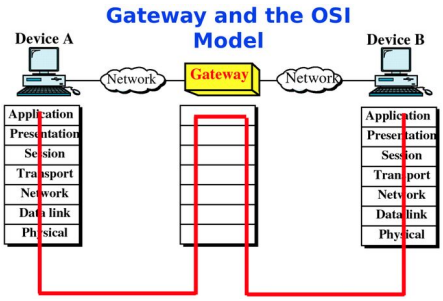
Routers in an Internet



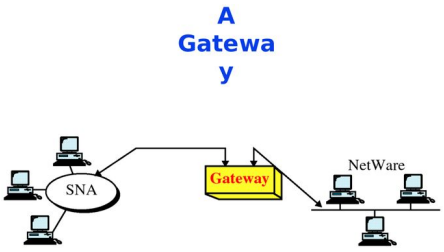
3NETWORK COMPONENTS — slide 51



3NETWORK COMPONENTS — slide 52



Gateway and the OSI Model



A Gateway

Network adapters, cables and modems

- **Network adapters** (NICs) connect a node to the media and contain the data transceiver (format, send, receive); they carry a unique MAC address. Types: expansion boards (via the computer bus), PCMCIA/PC Cards, USB, parallel port, wireless, and various Ethernet adapters.
- **Cables** are network linking devices that connect nodes; a crossover cable connects two computers directly.
- **Modems** (modulator/demodulator) are provided by the ISP to gain Internet access; they convert between analog and digital signals on the landline.

Connecting devices and the OSI model

Device	Layer	Function
Repeater	Physical	Receives a weak/corrupted signal, cleans and regenerates it, and retransmits. Consists of a receiver, signal processor/filter, amplifier and transmitter. A 2-port regenerator (not an amplifier); copies bit by bit at original strength; uses no logic, so data and noise are both regenerated; extends a single LAN's range. Now largely built into hubs/switches.
Hub	Physical	A multiport repeater with multiple interfaces: accepts an incoming signal and repeats it onto all other interfaces. "Unintelligent" — broadcasts to all. Active hubs regenerate (need power); passive hubs do not regenerate (no power). Intelligent hubs add processing; standalone and stackable variants exist. Inefficient (data usually intended for one device); being replaced by switches.
Bridge	Data link	A single-input/single-output device that can interpret the data it retransmits. Connects two similar (or closely similar) LANs; examines the destination address to forward or not, and the source address to build a routing table (backward learning).

Device	Layer	Function
Switch	Data link	A multiport bridge with a buffer; forwards only to the port where the destination is connected; error-checks before forwarding. Components: I/O ports, switching fabric (data plane), routing processor (control plane).
Router	Network	A multiport connectivity device that integrates LANs/WANs at different speeds and protocols; removes LAN headers/trailers and encapsulates WAN headers/trailers; digs into the network layer to read the destination address.
Brouter / routing switch	Data link + network	A brouter combines bridge and router features (filters LAN traffic as a bridge, routes across networks as a router). A routing switch combines a router and a switch.
Gateway	Any layer	Connects two dissimilar networks; a protocol converter (combination of router and modem) that can operate at any OSI layer.

Bridge types

Type	Idea
Transparent basic	No programming; observes all traffic and builds routing tables (backward learning). Two ports, a routing table per port; extracts the source address of each arriving frame. Found with CSMA/CD LANs.
Transparent learning	A bridge processor decides destination ports via a routing database; if the destination is unknown it broadcasts to all ports except the incoming one; it also stores the source address.
Transparent spanning tree	Uses a subnet of the full topology for loop-free operation; stores port state information (block/forward/disabled) to decide whether a port can be used for a destination.
Translational	Converts from one networking system to another (e.g. Ethernet to Token Ring), adding/removing fields as needed.
Source-routing	Found with token ring; does not learn routing tables — the entire route is embedded in the frame. A workstation must know the exact path, or send a discovery frame to record it.
Remote	Passes frames between two LANs separated by a WAN; encapsulates WAN headers/trailers at the source bridge and removes them at the destination bridge.

Switch forwarding methods & full-duplex

Method	Behaviour
Cut-through	Begins forwarding as soon as the header is read; no error checking, so errors can be propagated.
Store-and-forward	Waits for the entire frame, checks it for accuracy, then forwards.
Fragment-free	Reads only the part of the packet needed to identify transmission fragments.

- A **full-duplex switch** allows simultaneous transmission and reception, eliminating collisions; two sets of wires are needed (one for receive, one for transmit).

Router functions and wireless access points

- Router features: filter broadcast traffic, prevent certain traffic reaching a network, support simultaneous local/remote activity, provide fault tolerance via redundant components, monitor traffic and report to a MIB, diagnose problems and trigger alarms; often include firewall functions.
- Router functions: forwarding, routing, NAT (translate private IPs to a public IP), security (firewalls), VPN connectivity, bandwidth management, monitoring/diagnostics.
- **Wireless Access Point (WAP)**: allows wireless devices to connect to a wired network; a transceiver (transmitter + receiver) for WLAN radio signals with a built-in antenna, transmitter and adapter. It bridges WLANs and a wired Ethernet LAN, may have extra ports, and performs authentication, encryption and channel selection. Range and number of clients depend on the wireless standard and interference; multiple WAPs extend range/coverage.
- Gateway types: web application firewalls, cloud storage gateways, API/SOA/XML gateways, IoT gateways, media gateways, email security gateways, VoIP trunk gateways. Gateways differ from routers in that they communicate using more than one protocol and can operate at any of the seven OSI layers; a router joins two similar networks while a gateway joins two dissimilar networks.

23. Application Layer (HTTP, DNS, Email)

Source deck: 5Application Layer

Application-layer protocols

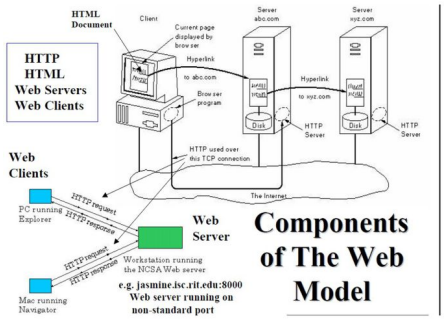
- An application-layer protocol defines how processes (clients and servers) on different end systems pass messages: the types of messages (request/response), their syntax (fields and delimiters), the semantics of fields, and the rules for when and how a process sends and responds.
- Most widely known TCP/IP application protocols: DNS (resolve names to IP addresses), HTTP (transfer web-page files), SMTP (mail transfer), Telnet (remote terminal access), FTP (interactive file transfer). They are defined by RFCs maintained by the IETF.

The World Wide Web and HTTP

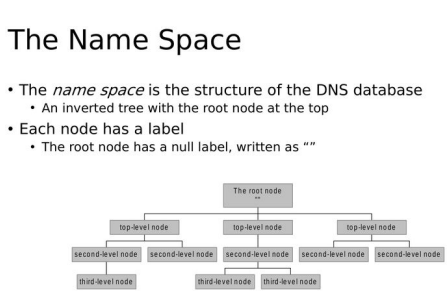
- The WWW was invented at CERN by Tim Berners-Lee and associates (1989–1991): HTML, HTTP, a Web server and a text-based browser. Marc Andreessen released the Mosaic GUI browser (1993) and formed Mosaic Communications, later Netscape.
- HTTP is the Web's main application-layer protocol, implemented as a client (browser) and a server that exchange HTTP messages; it defines the message structure and how they are exchanged, and uses TCP connections.
- When a URL is typed, the browser connects to the web service using HTTP. Example: **http://www.nitt.edu/web-server.htm** — the browser interprets the protocol (http), server name (www.nitt.edu) and filename (web-server.htm), checks a name server to convert the name to a numeric address, sends a GET request, receives the HTML, and formats the page.
- HTTP is a request/response protocol. Three common message types: **GET** (client request for data), **POST** and **PUT** (upload data to the server).
- HTTP is not secure: POST data is sent in plain text and responses (HTML) are unencrypted. **HTTPS** (Secure HTTP) uses authentication and encryption and specifies additional rules between the application and transport layers.

URLs

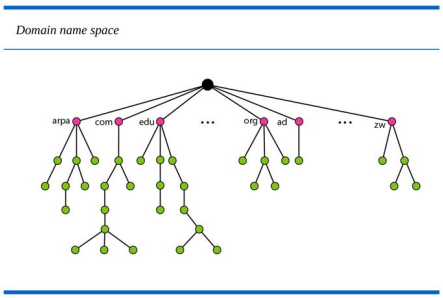
- A URL specifies the protocol used, the host name/alias/IP address, the port number (if not default), the path to the data, and the resource requested. Form: **http://hostname:port/path/resource**.



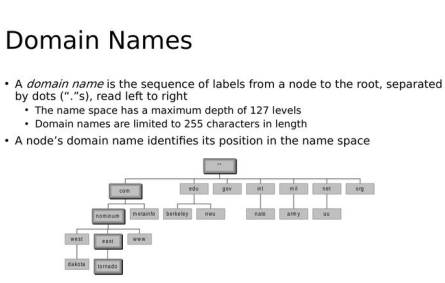
5Application Layer — slide 8



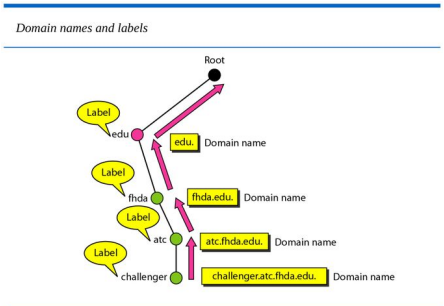
The Name Space



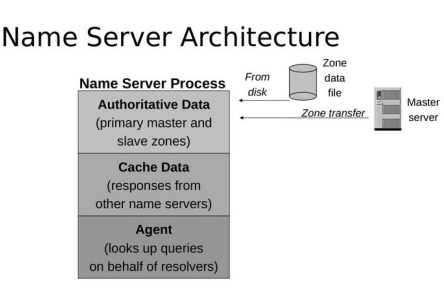
Domain name space



Domain Names

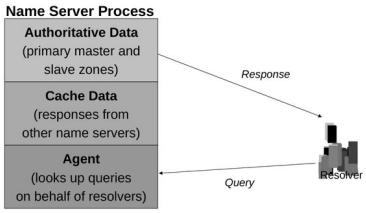


Domain names and labels



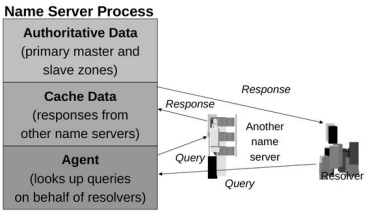
Name Server Architecture

Authoritative Data



Authoritative Data

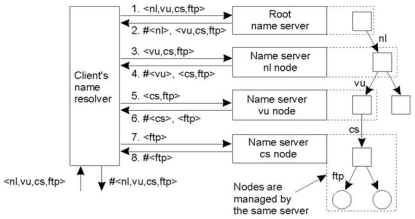
Using Other Name Servers



Using Other Name Servers

Iterative Name Resolution

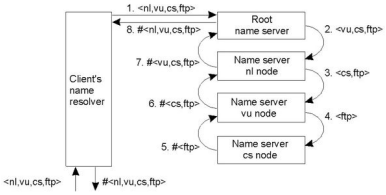
- The principle of iterative name resolution.



Iterative Name Resolution

Recursive Name Resolution

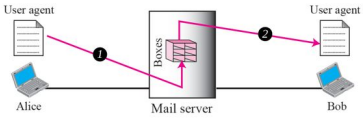
- The principle of recursive name resolution.



Recursive Name Resolution

Email Architecture

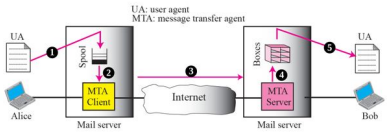
First scenario



When the sender and the receiver of an e-mail are on the same mail server, we need only two user agents.

Email Architecture

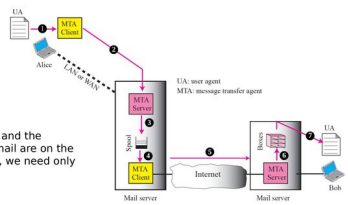
Second scenario



When the sender and the receiver of an e-mail are on different mail servers, we need two UAs and a pair of MTAs (client and server).

Second scenario

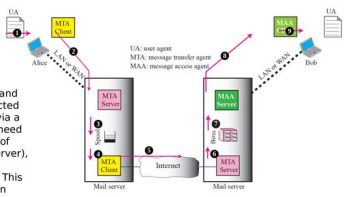
Third scenario



When the sender and the receiver of an e-mail are on the same mail server, we need only two user agents.

Third scenario

Fourth scenario



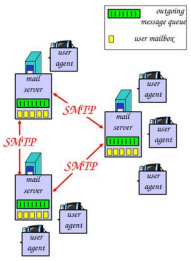
When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server). This is the most common situation today

Fourth scenario

Electronic Mail

Three major components:

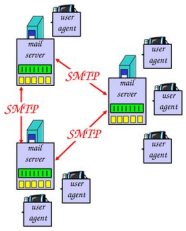
- user agents
- mail servers
- Simple Mail Transfer Protocol: SMTP used to exchange electronic mail.
- User Agent (Email Reader)
- a.k.a. "mail reader"
- composing, editing, reading mail messages
- e.g., Eudora, Outlook, elm, Netscape Messenger
- outgoing, incoming messages stored on server



5Application Layer — slide 53

Electronic Mail: mail servers

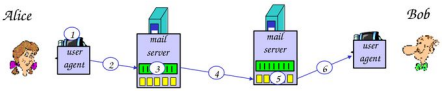
- Mail Servers
- mailbox contains incoming messages for user
- message queue of outgoing (to be sent) mail messages
- SMTP protocol between mail servers to send email messages
  - client: sending mail server
  - "server": receiving mail server



5Application Layer — slide 54

Scenario: Alice sends message to Bob

1) Alice uses Email Reader to compose message and "to" bob@bilkent.edu.tr
2) Alice's UA sends message to her mail server; message placed in message queue
3) Client side of SMTP opens TCP connection with Bob's mail server
4) SMTP client sends Alice's message over the TCP connection
5) Bob's mail server places the message in Bob's mailbox
6) Bob invokes his user agent to read message

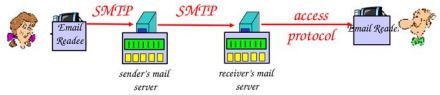


5Application Layer — slide 55

Push versus pull

Push versus pull

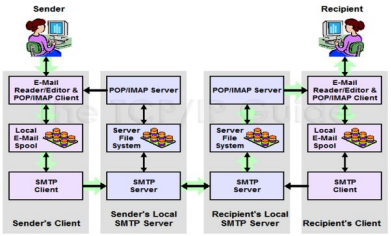
Mail access protocols



- SMTP: delivery/storage to receiver's server
- Mail access protocol: retrieval from server
  - POP: Post Office Protocol [RFC 1939]
    - authorization (agent <-->server) and download
  - IMAP: Internet Mail Access Protocol [RFC 1730]
    - more features (more complex)
    - manipulation of stored messages on server

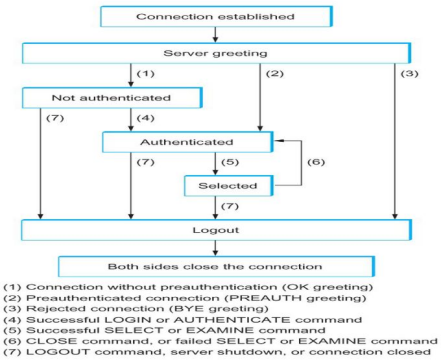
5Application Layer — slide 57

SMTP Communication Model



SMTP Communication Model

IMAP State Transition Diagram



(1) Connection without preauthentication (OK greeting)
(2) Preauthenticated connection (PREAUTH greeting)
(3) Rejected connection (BYE greeting)
(4) Successful LOGIN or AUTHENTICATE command
(5) Successful SELECT or EXAMINE command
(6) CLOSE command, or failed SELECT or EXAMINE command
(7) LOGOUT command, server shutdown, or connection closed

IMAP State Transition Diagram

Domain Name System (DNS)

- Created in 1983 by Paul Mockapetris (RFCs 1034 and 1035); the mechanism by which Internet software translates names to attributes (and vice versa). A **name** is how an endpoint is referenced; an **address** is how you get to it.
- A hierarchical, distributed database using a hierarchy of name servers to resolve host names into IP addresses (usually Unix machines running BIND; gethostbyname() issues a DNS query on Unix).
- Properties: keys are domain names; each name has resource records, each individually retrievable. Data is maintained locally but retrievable globally; remote data is locally cacheable. Loose coherency (each zone has a serial number incremented on change). Scalable (no limit on size/queries) and reliable (data replicated to slaves). Dynamic (the master can be updated dynamically — a single point of failure).
- Components: the **name space**, the **servers**, and the **resolvers**.

The name space and domains

- The name space is an inverted tree with the root at the top (null label ""). Each node has a label (up to 63 bytes); sibling labels must be unique; hostnames use A–Z, 0–9 and "-".
- A **domain name** is the sequence of labels from a node to the root, separated by dots, read left to right; maximum depth 127 levels, length ≤ 255 characters.
- A domain is a subdomain of another if its name ends in the other's name (cse.nitt.edu is a subdomain of nitt.edu and edu).
- **Delegation**: an administrator can delegate a subdomain to someone else; the parent keeps links. Each delegation creates a new unit of administration — a **zone**; the boundary between zones is a point of delegation. Delegation is the key to scalability.

Servers, resolvers and name resolution

- Each host has a **resolver** (a library applications link to; local name servers are configured in /etc/resolv.conf). **Name servers** are responsible for a zone or act as local servers doing lookups for local hosts.
- Name servers store the name space in **zones**; servers that load a complete zone are **authoritative** for it. Usually more than one server is authoritative for a zone (redundancy, load spreading); one server may be authoritative for many zones.
- Types: **Authoritative** (maintains the data; master where data is edited, slave where it is replicated to) and **Caching** (stores data obtained from an authoritative server). No special hardware is needed.
- A DNS query has three parameters: the domain name (e.g. www.nitt.edu), the class (e.g. IN) and the type (e.g. A). On a query a name server first looks in authoritative data and cache; if that fails, the answer must be looked up — **iterative** or **recursive** name resolution.
- The DNS imposes no constraints on the hierarchy except a single root and label restrictions; a site not connected to the Internet may use any hierarchy, but connecting to the Internet implies using the existing DNS hierarchy.
- There are about 13 **root name servers** worldwide ({a–m}.root-servers.net), operated by various organisations; local name servers contact root servers when they cannot resolve a name.

Electronic mail: SMTP, POP, IMAP

- Email is one of the oldest network applications. Distinguish the user interface (mail reader) from the message transfer protocols (SMTP/IMAP) and from the message-format companion protocol (RFC 822 and MIME).
- **SMTP** (RFC 821/2821) transfers messages from senders' mail servers to recipients' mail servers using TCP connections (port 25). It follows the client/server model: the client runs on the sender's mail server, the server on the recipient's; both run on every mail server. It uses persistent connections, direct transfer, three phases (handshaking/greeting, transfer, closure), and command/response interaction (ASCII commands such as HELO, MAIL FROM; responses are a status code and phrase). Messages must be 7-bit ASCII; the end of the message is CRLF.CRLF.
- The process of sending: a user agent (mail reader) creates the message; it goes to the outgoing queue in the local mail server (acting as SMTP client); the client opens a TCP connection to the destination SMTP server; after handshaking it sends the 7-bit ASCII message; the remote server places it in the recipient's mailbox.

Message format and MIME

- Mail is a text file. The **envelope** contains sender/receiver addresses and other information; the **message** has a header (sender, receiver, subject...) and a body (the actual information).
- RFC 822 defines a header and a body in ASCII text; the header is a series of CRLF-terminated lines, separated from the body by a blank line; each header line has a type and value separated by a colon.

- **MIME** (Multipurpose Internet Mail Extensions) extends RFC 822 so the body can carry many data types (audio, video, images, PDF). Binary data is first encoded to ASCII (e.g. base64) avoiding certain characters (including "."). MIME has three pieces: header lines (MIME-Version, Content-Description, Content-Type, Content-Transfer-Encoding), definitions of content types/subtypes (e.g. image/gif, image/jpeg), and encoding methods.

Mail access protocols and architecture

- SMTP is for delivery/storage to the receiver's server; a **mail access protocol** retrieves from the server: POP (Post Office Protocol, RFC 1939 — authorization and download), IMAP (Internet Mail Access Protocol, RFC 1730 — more features, manipulation of stored messages on the server), and HTTP (Hotmail, Yahoo! Mail).
- Key players: the user interacts with a mail reader; a mail daemon (message transfer agent, MTA) runs on each host with a mailbox and plays the role of a post office; mail may traverse mail gateways, each running an MTA; finally the user retrieves messages with the mail reader (locally, or via POP/IMAP).
- IMAP is a client/server protocol over TCP; the client issues CRLF-terminated ASCII commands and the server responds in kind; the exchange begins with authentication and mailbox identification.
- Architecture scenarios: same mail server → two user agents; different servers → two UAs and a pair of MTAs; both connected via a LAN/WAN → two UAs, two pairs of MTAs and a pair of MAAs (most common today).

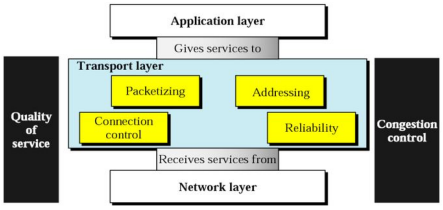
24. Transport Layer: TCP

Source decks: 1Transmission Control Protocol (TCP), 2TCP Data Flow

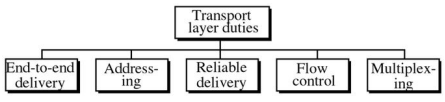
The transport layer

- Provides **process-to-process** delivery (vs the data link layer's node-to-node). Addressing: data link = MAC address, network = IP address, transport = **port number** (to choose among processes on the destination host).
- Transport and data-link error control differ in scope; the transport layer handles sequence control, loss control, duplication control, flow control (sliding window) and multiplexing.

Transport Layer

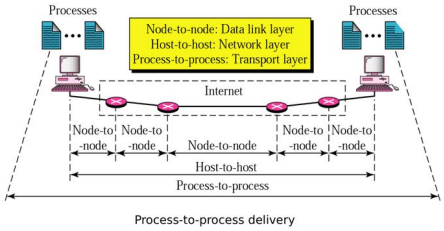


Transport Layer



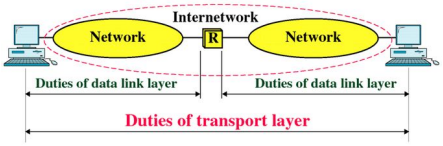
1Transmission Control Protocol (TCP) — slide 3

Transport Layer



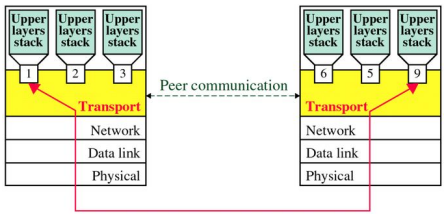
Transport Layer

Transport Layer and Data Link Layer



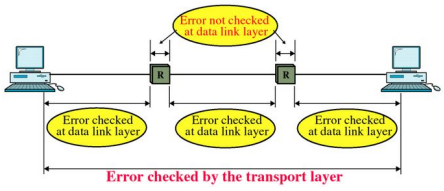
Transport Layer and Data Link Layer

Service Points



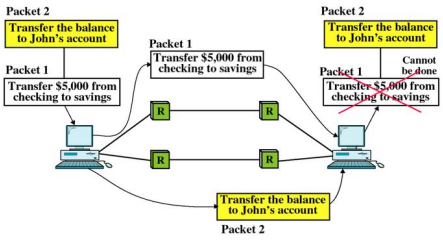
Service Points

Transport and Data Link Error Control



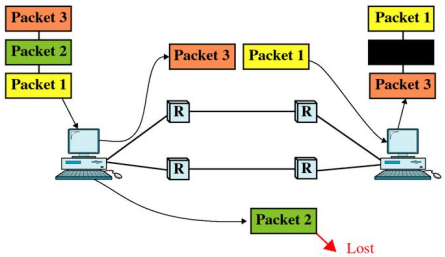
Transport and Data Link Error Control

Sequence Control



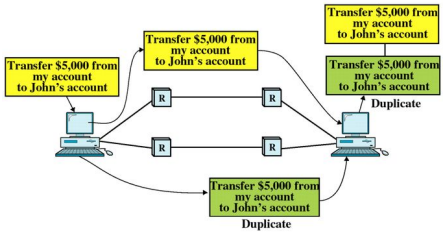
Sequence Control

Loss Control



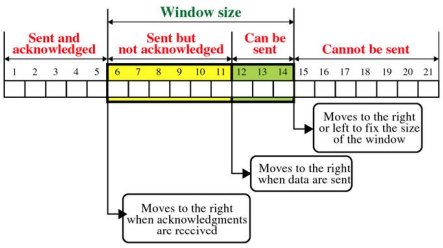
Loss Control

Duplication Control



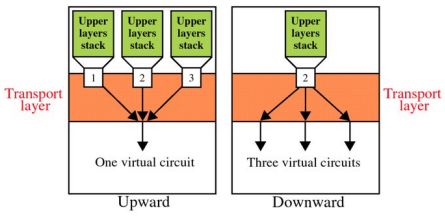
Duplication Control

Flow Control Sliding Window



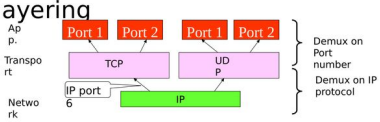
Flow Control Sliding Window

Multiplexing



Multiplexing

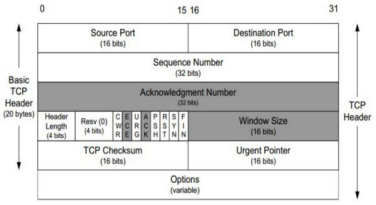
TCP layering



- A connection needs:
  - Source: (address, port) AND Destination: (address, port)
  - Only need one port on host to allow multiple connections, since each connection will have different (host, port) at other end
    - E.g. single host can serve multiple telnet connections
- Passive open: application contacts OS & indicates it will accept incoming connection, OS assigns port and listens
- Active open: application requests OS to connect to an (host, port)

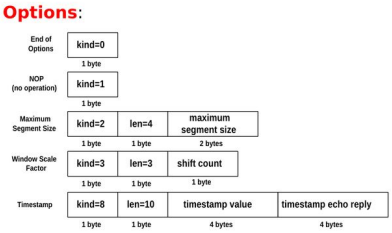
1Transmission Control Protocol (TCP) — slide 22

The TCP Protocol – The Header



The TCP Protocol – The Header

TCP header fields



TCP header fields

TCP overview

- TCP was designed to provide a **reliable, end-to-end byte stream** over an unreliable internetwork (parts with widely different topologies, bandwidths, delays, packet sizes). It dynamically adapts to the internetwork and is robust to many failures.
- RFC 793 (1981) is the base protocol; RFC 1122 (clarifications), RFC 1323 (high performance), RFC 2018 (SACK), RFC 2581 (congestion control), RFC 3168 (ECN).
- TCP provides a **connection-oriented** service: two applications establish a connection before exchanging data; exactly two endpoints (no broadcast/multicast).
- **Reliable**: the byte stream is broken into **segments**; the receiver acknowledges segments; if no ACK arrives TCP retransmits and waits longer (it maintains a timer). Errors: TCP has checksums for header and data (invalid segments are discarded) and every byte has a sequence number.
- **Byte stream**: a TCP connection is a byte stream, not a message stream — message boundaries are not preserved (writes of 10, 20 and 50 bytes are indistinguishable at the other end).
- Conceptually: place a call, agree on details, be notified of the connection, verify data continuously, then tear down the connection. Provides buffering and flow control; handles lost packets, out-of-order, duplicates and long delays; isolates the application from network details.
- Round-trip time (RTT) is estimated dynamically to know how long to wait; sequencing associates a sequence number with every byte; TCP reorders out-of-order segments and discards duplicates.

TCP service model and sockets

- A connection needs Source (address, port) and Destination (address, port). One port on a host allows multiple connections (each has a different host/port at the other end) — e.g. a single host serving multiple telnet connections.
- **Passive open**: an application tells the OS it will accept incoming connections; the OS assigns a port and listens. **Active open**: an application asks the OS to connect to a (host, port).
- The socket model: a single daemon (**inetd**) runs at well-known ports and waits for the first incoming connection; on the first connection it forks a new process and starts the corresponding daemon (e.g. httpd on port 80, ftpd on port 21).
- Uses sockets to define an end-to-end connection: (Source IP, Source Port, Source ISN, Destination IP, Destination Port, Destination ISN).

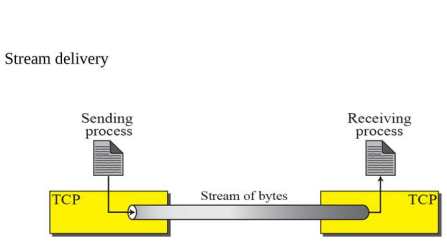
The TCP header

Field	Purpose
Source Port	Identifies the originating application.
Destination Port	Identifies the destination application.
Sequence Number	Sequence number of the first octet in the segment.
Acknowledgment #	Sequence number of the next expected octet (if the ACK flag is set).
Len	Length of the TCP header in 4-octet units.
Flags	SYN, FIN, RST, PSH, ACK, URG, CWR, ECE.
Window	Number of octets from the ACK that the sender will accept.
Checksum	Checksum of IP pseudo-header + TCP header + data.
Urgent Pointer	Pointer to the end of "urgent data".
Options	Special options such as MSS and Window Scale.

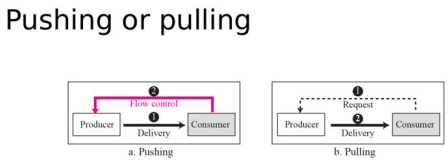
- Flags: **CWR** (Congestion Window Reduced), **ECE** (ECN Echo), **URG** (urgent pointer significant), **ACK** (acknowledgement number significant), **PSH** (push — deliver promptly), **RST** (reset), **SYN** (synchronise sequence numbers), **FIN** (no more data from sender, but it can still receive).
- The source/destination ports plus the source/destination IP addresses uniquely identify each connection; the combination of an IP address and a port is an **endpoint** or **socket**.
- Sequence/acknowledgement numbers are 32-bit; every byte has its own 32-bit sequence number. TCP uses sliding-window flow control: the acknowledgement number contains the next expected byte (cumulative). E.g. ACK 31245 means bytes up to 31244 were received and 31245 is expected.
- Options: NOP (pad the header to a multiple of 4 bytes), Maximum Segment Size, Window Scale (increases the window from 16 to 32 bits; only usable in the SYN segment), Timestamp (round-trip measurement).

Segments and buffers

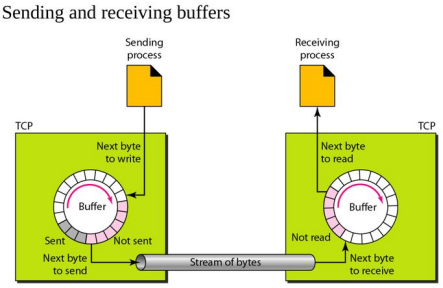
- A TCP segment is a fixed 20-byte header (plus optional part) followed by zero or more data bytes. TCP may accumulate data from several write() calls into one segment or split one write() into multiple segments. Segment size is restricted by the IP payload (65515 bytes) and the link MTU.
- Sending/receiving buffers: at the sender the buffer has empty chambers, bytes sent but not yet acknowledged (kept until acknowledged), and bytes to be sent. At the receiver, empty chambers and received bytes that can be read; a read chamber is recycled.
- How a segment is created: writes go to the sender buffer; the sender maintains a dynamic window; modern TCP uses path MTU discovery (via ICMP) and sets the MSS during connection establishment. After an ACK, if the window is less than the MSS, build one segment, otherwise build multiple MSS-sized segments.
- Challenges: segments are constructed dynamically, so a retransmission need not contain the same data; segments may arrive out of order and the receiver must handle this to minimise data wastage.



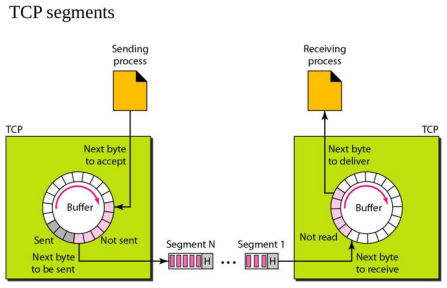
Stream delivery



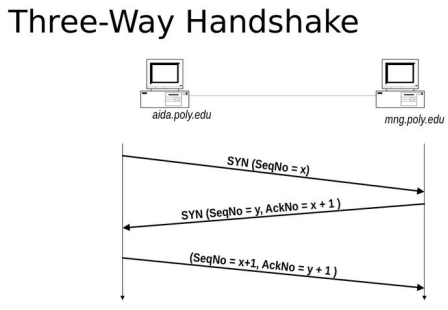
Pushing or pulling



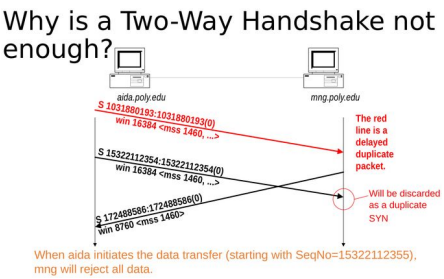
Sending and receiving buffers



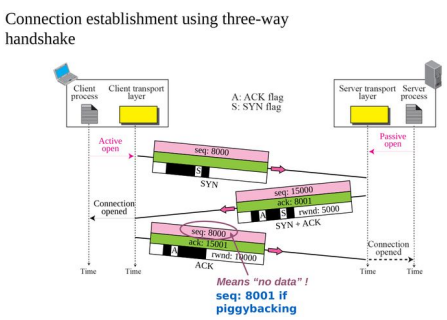
TCP segments



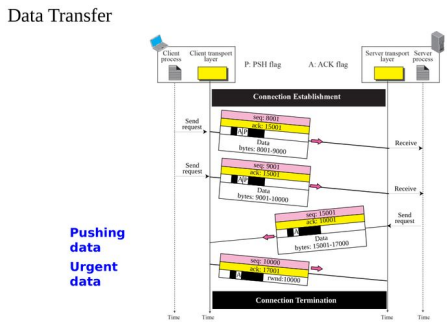
Three-Way Handshake



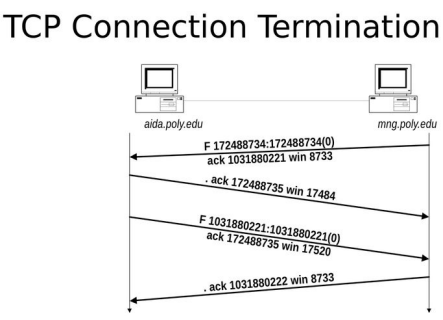
Why is a Two-Way Handshake not enough?



Connection establishment using three-way handshake

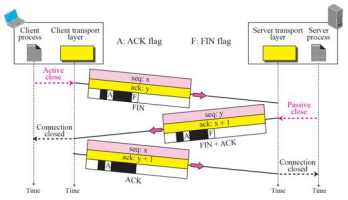


Data Transfer



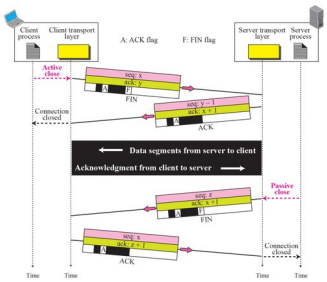
TCP Connection Termination

Connection termination using three-way handshake

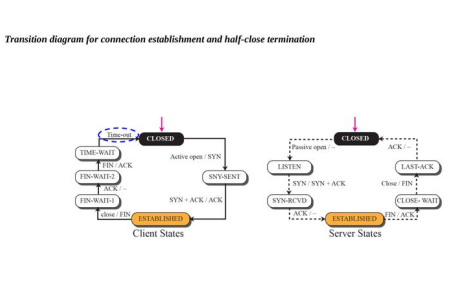


Connection termination using three-way handshake

Half-Close



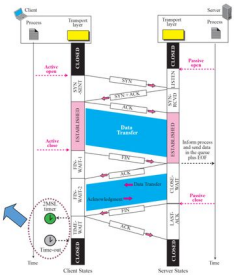
Half-Close



Transition diagram for connection establishment and half-close termination

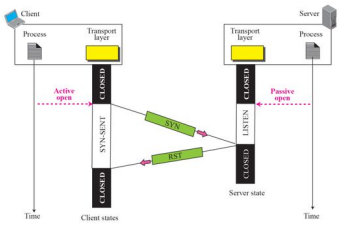
Time-line diagram

1. Enough time for an ACK to be lost and a new FIN to arrive. If during the TIME-WAIT state, a new FIN arrives, the client sends a new ACK and restarts the 2MSL timer
2. To prevent a duplicate segment from one connection appearing in the next one, TCP requires that incarnation cannot take place unless 2MSL amount of time has elapsed.
   Another solution: the ISN of the incarnation is greater than the last seq. # used in the previous connection.



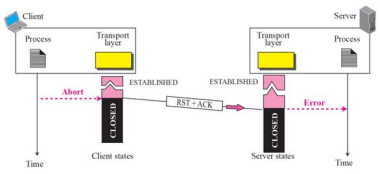
Time-line diagram

Denying a connection



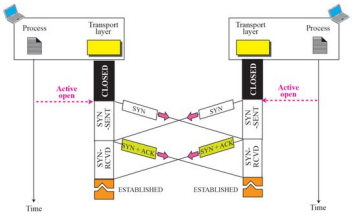
Denying a connection

Aborting a connection



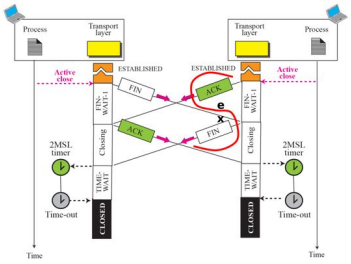
Aborting a connection

Simultaneous open



Simultaneous open

Simultaneous close



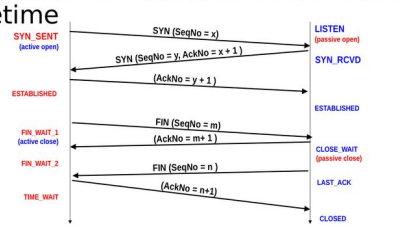
Simultaneous close

TCP States

State	Description
CLOSED	No connection is active or pending
LISTEN	The server is waiting for an incoming call
SYN RCVD	A connection request has arrived; wait for Ack
SYN SENT	The client has started to open a connection
ESTABLISHED	Normal data transfer state
FIN WAIT 1	Client has said it is finished
FIN WAIT 2	Server has agreed to release
TIMED WAIT	Wait for pending packets ("2MSL wait state")
CLOSING	Both Sides have tried to close simultanesously
CLOSE WAIT	Server has initiated a release
LAST ACK	Wait for pending packets

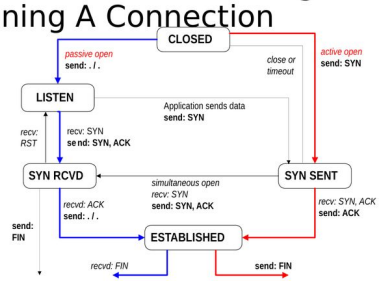
TCP States

TCP States in "Normal" Connection Lifetime



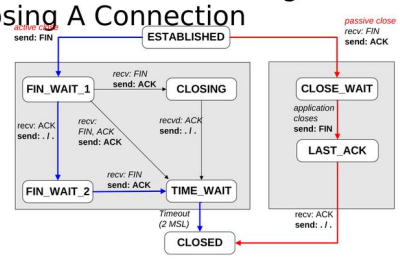
TCP States in "Normal" Connection Lifetime

TCP State Transition Diagram Opening A Connection



TCP State Transition Diagram Opening A Connection

TCP State Transition Diagram Closing A Connection



TCP State Transition Diagram Closing A Connection

Connection management

Three-way handshake (open)

- (1) **Active open**: the client sends a segment with SYN set, its port and its initial sequence number (ISN).
- (2) **Passive open**: the server responds with SYN set (counts as one byte), its ISN, and an ACK for the client's ISN.
- (3) The client acknowledges by sending a segment with an ACK of the server's ISN.
- A SYN segment cannot carry data but consumes one sequence number; a SYN + ACK segment also consumes one sequence number; an ACK segment with no data consumes no sequence number.
- A two-way handshake is not enough: old duplicate SYN segments could be accepted as new connection requests.

Data transfer

- After establishment, bidirectional transfer takes place; acknowledgements are piggybacked on data; the last segment carries only an acknowledgement. Client data segments may set the **PSH** (push) flag so the server delivers data as soon as received.

- **Pushing data**: the sending TCP normally buffers and chooses segment sizes; an application can request a push so the sending TCP sends immediately with PSH set (though most current implementations ignore such requests).
- **Urgent data**: an application can send a segment with **URG** set; the sending TCP places the urgent data at the beginning of the segment and the urgent pointer field marks the end of the urgent data. It is neither a priority nor an expedited service — it merely marks bytes for special treatment by the receiving application.

Connection termination

- Each end shuts down independently ("half-close"). A FIN segment means no more data will be sent. Four steps: (1) X sends FIN to Y (active close); (2) Y ACKs the FIN (Y can still send data); (3) Y sends FIN to X (passive close); (4) X ACKs the FIN. A FIN segment consumes one sequence number if it carries no data; so does a FIN + ACK.
- **Half-close**: one end stops sending while still receiving (e.g. a client half-closes after sending so the server can sort, then the server replies and closes).
- **2MSL wait state (TIME_WAIT)**: after an active close and the final ACK, the connection stays in TIME_WAIT for twice the maximum segment lifetime (2MSL = 2 × MSL; MSL is 2 min, 1 min or 30 s). Reasons: to give TCP a chance to resend the final ACK if it is lost (the server will time out and resend the FIN), and to prevent a duplicate segment from an old connection appearing in a new one (an incarnation cannot occur until 2MSL has elapsed; alternatively, the new ISN is greater than the last sequence number used).

Connection reset

- Reset by setting the **RST** flag. When is RST set? A connection request arrives and no server process is waiting on the destination port; aborting a connection causes the receiver to throw away buffered data (it does not acknowledge the RST); a TCP may deny a connection request, abort an existing connection, or terminate an idle connection.
- Also covered: simultaneous open and simultaneous close.

TCP states and window size

- TCP has a well-defined state machine for a normal connection lifetime, for opening a connection and for closing a connection.
- Flow control uses a variable-sized sliding window: the window-size field tells how many bytes the receiver can receive based on its current free buffer space. A TCP acknowledgement is a combination of the acknowledgement number and the window size.

25. Formula & Quick-Reference Sheet

Every formula in one place.

Signals & bandwidth

Period & frequency: $T = 1/f$

Wavelength: $\lambda = v / f = v \cdot T$

Nyquist bit rate (noiseless): $\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2 L$

Shannon capacity (noisy): $C = \text{Bandwidth} \times \log_2(1 + \text{SNR})$

$\text{SNR}_{dB} = 10 \log_{10}(\text{SNR})$; $dB = 10 \log_{10}(P_2/P_1) = 20 \log_{10}(V_2/V_1)$

SNR_{dB} = signal level (dBm) − noise level (dBm); $dBm = dBm0 + dBr$

Digital data rate vs bandwidth (first harmonic): $B = n/2 \rightarrow n \leq 2B$

Line coding & modulation

Signal (baud) rate: $S = c \cdot N \cdot (1/r)$ bauds (c = case factor)

Multilevel condition: $2^m \leq L^n$

ASK: $B = (1 + d)\ S$, $d \in [0,1]$

FSK: $B = (1 + d)\ S + 2\Delta f$; Multi-level FSK: $B = (1 + d)S + (L-1)/2 \cdot \Delta f = L \cdot S$

PSK: $B = (1 + d)\ S$

AM: $B_{AM} = 2B$

FM: $B_{FM} = 2(1 + \beta)B$, $\beta \approx 4$

PM: $B_{PM} = 2(1 + \beta)B$, $\beta \approx 2$

Analog-to-digital (PCM)

Sampling rate: $f_s = 1/T_s$; Nyquist: $f_s \geq 2\ f_{max}$

Quantization zone height: $\Delta = (\max - \min)/L$

Bits per sample: $n_b = \log_2 L$

PCM bit rate = $n_b \times f_s$

Error control

Hamming redundancy: $2^r \geq m + r + 1$

Detect s errors: $d_{min} \geq s + 1$

Correct d errors: $d_{min} = 2d + 1$

Hamming distance: $d(x,y)$ = number of 1s in $x \oplus y$

CRC: transmitted frame = $x^r M(x) \oplus$ remainder, divisible by $G(x)$

Internet checksum: complement(wrapped one's-complement 16-bit sum); receiver accepts iff result = 0

Flow & link efficiency

Transmission time = frame bits / bit rate

Propagation time = distance / propagation speed

Ethernet utilization (no collisions): $U = T_{trans} / (T_{trans} + T_{prop})$

With collisions: add $T_{contention}$

Stop-and-wait efficiency: $U = T_{trans} / (T_{trans} + 2 \cdot T_{prop})$

Sliding window: max window = $2^k - 1$ (GBN); Selective Repeat = (seq space)/2 (or seq space − 1)

Multiple access

Pure ALOHA: $S = G \cdot e^{-2G}$, $S_{max} = 0.184$ at $G = \frac{1}{2}$ (≈18%)

Slotted ALOHA: $S = G \cdot e^{-G}$, $S_{max} = 0.368$ at $G = 1$ (37%)

CSMA/CD worst-case collision detection = 2 × max propagation delay

Ethernet slot time = 51.2 µs; retry $R \in [0, 2^k - 1]$; give up after 16 attempts

Channelization: FDMA/TDMA limit each node to R/M; CDMA shares one channel by code

IP addressing

Block of /n: first = address AND mask; last = address OR ¬mask; count = 2^{32-n}

Usable hosts = $2^{32-n} - 2$

Prefix for at least H hosts: host bits = $\lceil \log_2 H \rceil$; prefix = 32 − host bits

Subnetting increases the number of 1s; supernetting decreases them

Routing complexity

Bellman-Ford: $O(N^3)$ | Dijkstra: $O(N^2)$ | Floyd-Warshall: $O(N^3)$

Distance-vector update: $D_{i,j} = \min_k \{ c_{i,k} + D_{k,j} \}$

Topology & other

Mesh links for n nodes: $L = n(n-1)/2$

PCM/decoder: hold circuit + low-pass filter with cutoff = highest pre-sampled frequency

2MSL = 2 × Maximum Segment Lifetime

Rapid-fire revision list

Topic	Remember
Network types	LAN, WLAN, MAN, WAN, PAN, SAN; node; five components of data communication
Models	OSI 7 layers; TCP/IP 4 layers; service vs protocol; upper vs lower layers
Topologies	Bus, ring, star, tree, mesh; mesh links = n(n−1)/2; peer-to-peer vs primary-secondary
Physical layer	Analog vs digital; amplitude/frequency/phase; bandwidth; Nyquist & Shannon
Impairments	Attenuation (dB), distortion (delay/intersymbol), noise (thermal, impulse, crosstalk)
Line coding	NRZ, RZ, Manchester, AMI; block coding 4B/5B, 8B/10B; scrambling B8ZS, HDB3
PCM	Sample → quantize → encode; $f_s \geq 2f_{max}$; bit rate = $n_b \cdot f_s$
Modulation	ASK/FSK/PSK/QAM; AM/FM/PM; constellation diagrams
Spread spectrum	FHSS, DSSS, THSS, CSS; $B_{ss} \gg B$; anti-jam, LPI
Data link	Framing (byte count, byte/bit stuffing, coding violations); LLC & MAC
Error detect	Parity (d=2), CRC (divisible by G(x)), checksum
Error correct	Hamming $2^r \geq m+r+1$, convolutional (K=7, rate ½), RS (255,233), LDPC
DLL protocols	Utopia, stop-and-wait, ARQ; sliding window; Go-Back-N vs Selective Repeat
SDLC/HDLC	Flag 01111110; I/S/U frames; NRM/ARM/ABM; P/F bit
Network layer	Datagram vs virtual circuit; store-and-forward; congestion control
IP header	IHL=5→20 B; TTL; protocol 1/6/17; ID+offset+MF for fragments
Fragmentation	Reassemble only at destination; Path MTU Discovery preferred
Routing algorithms	Sink tree; Bellman-Ford, Dijkstra, Floyd-Warshall; DV vs link-state
Routing protocols	RIP (hop ≤ 15, DV), OSPF (link-state, areas), BGP (path vector, port 179)
Addressing	Classes A–E, masks, subnet/supernet, CIDR, VLSM
Multicast	IGMP; (S,G) vs (*,G); DVMRP, MOSPF, PIM-DM/SM
LAN MAC	ALOHA 18%/37%, CSMA variants, CSMA/CD, CSMA/CA, token passing, FDMA/TDMA/CDMA
802.x	802.3 Ethernet, 802.4 token bus, 802.5 token ring, 802.11 WLAN
Multiplexing	FDM, WDM, TDM (sync/statistical); circuit vs message vs packet switching
Transmission media	UTP/STP, coax, fibre (multi/single-mode); radio, microwave, infrared
Components	Repeater, hub, bridge, switch, router, brouter, gateway, WAP
Application	HTTP/HTTPS, URLs, DNS (zones, resolvers, root servers), SMTP/POP/IMAP, MIME
Transport (TCP)	3-way handshake, sliding window, FIN/2MSL, RST, byte stream, ports

Exam tip: nearly every topic reduces to three questions — where in the stack?, what goes wrong?, and which formula sizes it? Anchor each answer to a layer, a failure mode and a number.